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ANALYSIS OF STEREO IMAGE IN VAST REPOSITORIES OF RECORDED MUSIC

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Abstract
Spatial information is crucial to the listener's experience of music, and stereophonic recordings have long been the standard for capturing this spaciousness. Recent advancements in computing, coupled with the availability of new, very large musical corpora, have created opportunities for large-scale quantitative analysis of the stereo image of recorded music. In this licentiate thesis I address a key gap in current research: direct analysis of raw music recordings to describe the stereo image. The work comprises practice-led research, objective experiments, perceptual listening tests, and an extensive literature review. A combination of immersion in the creative aspects of stereo field and thorough mathematical evaluation of the recorded signals created a strong foundation for future research. The studies together revealed that the while there are many existing stereo descriptors, the musical features visible to each one vary considerably and therefore have differing use cases. Moreover, they don't always align with human perception. The combination of research methods was hence a highly effective way to critique the strengths and weakness of objective measurements. Future work should focus on applying and adapting the discovered methods to modern music datasets.

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Special thanks to Rikard Lindell, Andreas Rossholm, and Elaine Åstrand for their advice and support, and to Wolmar Nyberg Åkerström for donating his LaTeX template.

List of papers

This thesis is based on the following papers, which are referred to in the text by their roman numerals.

I A. Franklin et al., 'Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications,' *Journal of The Audio Engineering Society*, vol. 73, no. 12, pp. 872–881, Mar. 2025. DOI: 10.17743/jaes.2022.0245

II R. Mitic and A. Rossholm, 'A Systematic Literature Review of Methods to Describe the Stereo Image of Recorded Music,' *Journal on Audio, Speech, and Music Processing*, vol. 2026, no. 1, p. 29, 9th Jun. 2026, ISSN: 3091-4523. DOI: 10.1186/s13636-026-00463-4

III R. Mitic and A. Rossholm, 'Towards an objective comparison of panning feature algorithms for unsupervised learning,' in *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)*, Ancona, IT, 2nd Sep. 2025. [Online]. Available: https://dafx.de/paper-archive/2025/DAFx25_paper_26.pdf

IV R. Mitic et al., 'Perceptual Evaluation of Binaurally Panned Speech and Music Mixtures,' *Journal of The Audio Engineering Society*, in review, submitted Aug. 2026

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Abbreviations

AAC Advanced Audio Coding 32
ADRess Azimuth Discrimination and Resynthesis 33
AGR Automatic Genre Recognition 32
AMT Auditory Modelling Toolbox 29, 30
API Application Programming Interface 15
AR Augmented Reality 25
ASF Amplitude Stereo Features 36, 37
ASW Apparent Source Width 40, 41

BCC Binaural Cue Coding 32
BMS Bivariate Mixture Space 33

CD Compact Disc 27

DAW Digital Audio Workstation 22–24, 27, 31, 38
DJ Disc Jockey 42
DL Deep Learning 13, 32, 40, 42, 46
DSP Digital Signal Processing 14, 16, 18, 32, 40, 44, 47

ERB Equivalent Rectangular Bandwidth 30, 31, 36, 42

FA Frequency-Azimuth 34
FFT Fast Fourier Transform 28, 36
FMA Free Music Archive 13, 31, 32, 35, 36, 51

HRIR Head-Related Impulse Response 20, 21

IACC Interaural Cross Correlation 18, 33, 35, 40
ICLD Inter-channel Level Difference 42
ILD Interaural Level Difference 18–20, 33, 35, 36, 40, 42
IPD Interaural Phase Difference 18, 19
IRESAP Information Retrieval in Embedded Systems for Audiovisual Artistic Processes iii
ITD Interaural Time Difference 18–20, 35, 40, 42

KKS The Knowledge Foundation iii
KMH Kungliga Musikhögskolan iii

LEF Lateral Energy Fraction 40
LEV Listener Envelopment 40, 41
LFE Low Frequency Effects 25

MDU Mälardalen Universitet iii
MIR Music Information Retrieval 13, 14, 32, 49
ML Machine Learning 13, 31, 32, 42, 46
MP3 MPEG layer 3 32
MPEG Moving Picture Experts Group 32

PC Panning Coefficients 36
PH Panning Histogram 35, 36

PRISMA Preferred Reporting Items for Systematic reviews and Meta-Analyses 46
PS Parametric Stereo 32
PSF Phase Stereo Features 37

RMS Root Mean Squared 19
RQ research question 14, 31, 42–44, 49

SIP Stereo Image Profile 15, 35, 36, 41, 42, 46, 47, 51
SLR Systematic Literature Review 15, 46
SOM Self Organising Map 42
SPF Stereo Panning Features 35, 36, 47
SPL Sound Pressure Level 19
SSPS Spectral Stereo Phase Spread 37
STFT Short-time Fourier Transform 28, 29, 34, 36, 37, 52

TF time-frequency 28, 30, 31, 33, 34

VR Virtual Reality 25

Part I

Thesis

1. Introduction

It is a widely accepted truth in the world of music production and recording that spatial information in audio is crucial to listeners' sense of immersion and connection with the music. To capture that spaciousness, stereophonic recording and reproduction has been the standard format since it gained popularity in the 1950s [1, 2]. The spatial analysis of subsequent decades worth of stereo recordings presents a unique opportunity to study human musicality, as each recording reflects and encodes the creative choices made by the musicians and recordists involved.

Within the music community it is common to speak about spaciousness in a qualitative sense, using terms such as 'stereo image' and 'stereo field' [1] to encompass all audible spatial aspects. In academia however – and specifically in the Music Information Retrieval (MIR) community – there have been several explicit attempts to translate these discussions into a quantitive domain [3–5]. All were performed in an era when musical datasets were hand-curated/-labelled, fairly small (1000 to 2000 tracks), and, most notably, private. In recent years new musical corpora have become available, e.g. MTG-Jamendo [6] and Free Music Archive (FMA) [7], which are both orders of magnitude larger than their predecessors, containing approximately 55 000 and 100 000 tracks respectively. In addition, computing techniques in Machine Learning (ML) and Deep Learning (DL) have progressed to the point where it is now possible to learn extremely high-level features on these vast datasets, thereby making sense of the human creativity represented in the recordings.

A typical use case for spatial descriptors that has emerged both in academia and industry is classification, and in particular, auto-tagging of musical genre/performer [8, 9]. Such features can be used in music similarity and recommendation tasks [10, 11]. It has also been shown that stereo field classification of subsections within a larger work can act as a precursor to other tasks such as segmentation, audio enhancement, upmixing, and source separation [12–14]. The latter is very relevant to the audio distribution industry, which is currently experimenting with newer audio formats such as ambisonic and object-based audio. While these new formats offer exciting opportunities for both distributors and listeners, there remains a huge back-catalogue of stereophonic recordings that need to remain playable on modern reproduction equipment. A detailed understanding of their spatial content would therefore be beneficial.

Historically, binaural audio (which is a subset of stereo) has been of particular interest because its relevance to human hearing. Thus, it is extremely widely studied, and the underlying fundamental techniques are well-established [15]. Yet there remains a gap in current research regarding direct analysis of raw music recordings, and in particular, describing the stereo image of said recordings at a high level. Through the work described in this thesis, I aim to fill that gap.

I firstly demonstrate use cases where the spatial analysis of stereo audio has the opportunity to solve real-world problems in audio transmission, dissemination, and reproduction. Accordingly, I gather existing objective methods to describe stereo image and assess their abilities to detect the spatial aspects of a variety of musical audio sources.[2]

[1] The terms 'stereo image' and 'stereo field' are freely interchangeable and, as is normal in the music community, I use them both throughout this thesis.

[2] Throughout this thesis I use the word 'objective' quite frequently. The reader is directed to chapter 5 for a full discussion of the philosophical implications, but in essence I use the term to infer that the signal processing techniques discussed are deterministic and applicable to any musical recording.

2. Research Questions and Contributions

The underlying purpose of the research documented in this thesis relates to MIR, and in particular to generating an understanding of the links between human musical creativity and musical output. While it might be possible for the creative people involved to explain how they make their music, it is far more objective to analyse their output as a representation of their musical ideas.

2.1 Research Questions

Given the aforementioned context, the overarching research question (RQ) can be formed as:

How do music producers use stereo field?

The basis of the research is access to extremely large musical corpora and modern computing techniques. The number and variety of recordings available for analysis makes them a decent representation of music producers in general. The spatial aspects of each track can be analysed individually using traditional Digital Signal Processing (DSP) techniques and a population-level analysis can be performed on the spatial features of all tracks together. Thus, I have broken down the main RQ into two smaller ones:

RQ_1: *How does the stereo image of a recording relate to real-world musical knowledge?*

RQ_2: *What methods can be used to describe the stereo image of a music recording?*

These concrete questions are addressed through the four papers included in this licentiate thesis. RQ_1 provides an important context of what quantitive spatial aspects receive attention from music practitioners, while RQ_2 establishes a state-of-the-art baseline of relationships between those quantitive descriptions and objective measurements. The overarching research question spans both licentiate and doctoral studies, with future plans being detailed in section 7.1.

2.2 Research contributions

This section presents a summary of the research contributions achieved by the four included papers and explains their relationship to the RQs. Many parts of the research required spatial descriptors to be extracted from raw audio signals. To facilitate this I implemented a library of software tools, allowing all the research tracks to be linked by a common mathematical framework.

2.2.1 An understanding of the musical uses of stereo field

With respect to the overarching research question, the act of personally creating musical output gives an insight into how music producers in general might use stereo field. We initially embarked on such a project, using audio from a remotely-located stereo microphone as the primary sonic material in a set of musical performances. Paper I was a retrospective account of the process, providing technical information to any musicians wishing to make a similar performance in

the future. The paper also provided suggestions for future iterations of similar instruments and the standardized Application Programming Interfaces (APIs) which are built into web browsers, giving a basis for future work.

The software library presented the opportunity to apply a more technical approach to RQ_1. The exploration of stereo field in existing recordings furthered the knowledge gained from an individual musical project by allowing us to analytically compare stereo fields created by many producers. This allowed us to establish an understanding of the range and variation that stereo field can embody.

2.2.2 Signal processing techniques relating to stereo field

The software implementation and the resulting paper III contribute an understanding of how some established techniques for measuring stereo field react to existing recorded music. Moreover, that understanding is not limited to the techniques specified in paper III since I developed the software continuously throughout several literature searches, resulting a full-featured library that includes many other spatial analysis techniques.

In order to fully answer RQ_2, an extensive Systematic Literature Review (SLR) (paper II) was performed with two intentions. First and foremost, the intention was to conclusively find all existing mathematical techniques that describe stereo field. In addition, there was a goal to find 'cross-pollination' opportunities from disparate areas of audio research which nonetheless derive spatial cues from audio signals. In doing so, it was possible to form a 'general model' of Stereo Image Profiles (SIPs), thereby making it easier to for future researchers to find or create a SIP that suits their specific use case.

2.2.3 Psychoacoustics of mixed music and speech

The perceptual listening test described in paper IV provided a new perspective on a classic problem in psychoacoustics research. While there is extensive literature describing how people perceive music or speech that has been intentionally masked by another sound, our experiment used music and speech that were intended to complement each other. The results firstly suggested that while people prefer speech that is panned towards the centre, there was an opportunity to increase the audible pleasantness by using very subtle panning. Contrary to our initial assumptions, the musical content had far less influence that originally expected, and speech became the dominant audial stream which was paramount to pleasantness for a majority of listeners.

2.2.4 Setup for future work on large datasets of audio

In regard to the future work described in section 7.1, the SLR provided both a list of relevant audio datasets, and many standardized implementations of mathematical techniques that extract spatial information from audio. Paper III also presented a good opportunity to assess the computational complexity of the algorithms implemented in the software library, thereby highlighting some opportunities for optimization when they are applied to extremely large corpora.

3. Background

Music production techniques related to stereo field are thoroughly rooted in the human perception of sounds, exploiting well-understood auditory phenomena for creative means. However, that understanding is a fairly recent development. In fact, musical practice involving stereophonic sound predated the scientific knowledge of perception by several decades, with one of the earliest known examples dating back the late 19th century [16]. It wasn't until mid 20th century that the work of Fletcher and Munson [17] and Jeffress [18] outlined the psychoacoustic principles that music production would eventually exploit. In this chapter we draw a line from the basic human physiology of the auditory system, through psychoacoustics, DSP theory, and music production, eventually linking them all to the modern computing techniques used as the basis for the work detailed in this thesis.

3.1 The mechanics of hearing and auditory perception

The physical characteristics of the human auditory system govern what we as humans are able to perceive. In particular, our perception of frequency, loudness, and directionality are key when assessing stereo field.

3.1.1 Physiology of the inner ear

The ear contains three major sections, loosely labelled the 'outer', 'middle', and 'inner' ear (see fig. 3.1). When air vibrations (i.e. sound waves) flow into the outer ear, mechanical coupling in the middle ear transmits vibrations to the cochlea – the major organ of the inner ear. The cochlea is a conical tube which is coiled up into a spiral, filled with an incompressible fluid, and covered on either end by a membrane. Through the centre of the tube runs the basilar membrane, which is a covered in hairs named cilia. When vibrations are mechanically transmitted to the base of the cochlea, the internal fluid responds by transmitting those vibrations to the basilar membrane, and the cilia will bend and twist accordingly. The cilia are linked to nerve fibres, and will initiate electrical pulses when manipulated. These electrical pulses are sent to the brain and interpreted as sounds [19].

The basilar membrane reacts to a pressure wave from an incoming sound in an elastic way, with a bulge travelling from the base (the widest section) to the apex (the narrowest section). That bulge will change amplitude as it travels due to the basilar membrane's non-uniform width and stiffness, and the distance at which is reaches its maximum amplitude depends on the frequency of the incoming sound. Thus, the cilia at different positions along the cochlea have different 'critical frequencies' that they respond to. The critical frequency increases logarithmically as one moves from the base to the apex, as shown in fig. 3.2. The cochlea therefore acts to convert time-domain sound waves into time- and frequency-dependent electrical excitations, enabling us to perceive different frequencies simultaneously.

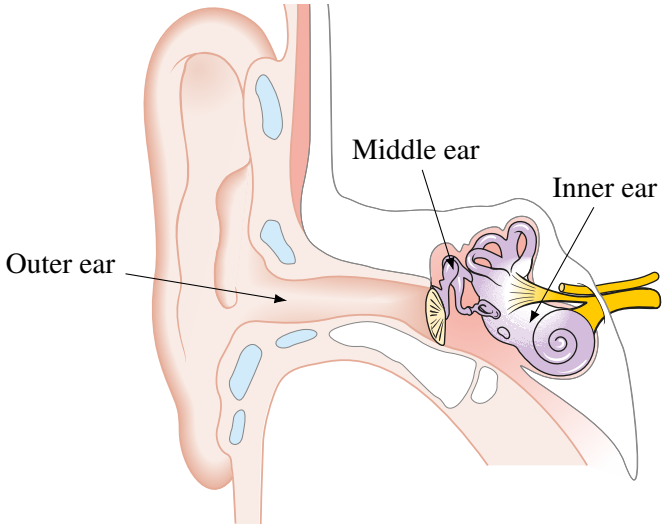


Figure 3.1. Diagram of the human ear*. Air flows through the outer ear and vibrates the ear drum. The middle ear mechanically couples the outer ear to the inner ear. The inner ear converts mechanical movements to electrical signals.

*Original image by Cancer Research UK is licensed under CC BY-SA 4.0. Annotations have been edited.

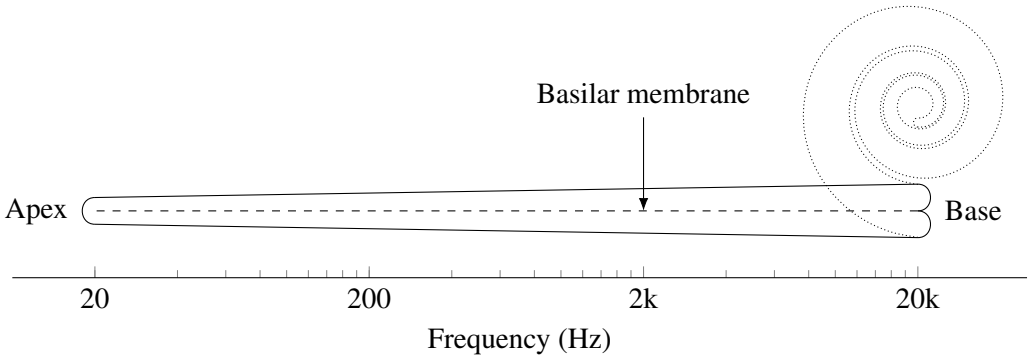


Figure 3.2. Tonotopic diagram of the human cochlea. Dotted lines represent the cochlea in its normal shape while solid lines show a hypothetical unfurled cochlea with the basilar membrane running down the centre shown as a dashed line. The frequency response of the basilar membrane changes logarithmically along its length, with high frequencies detected at the base and low frequencies at the apex.

3.1.2 Directional hearing

Section 3.1.1 explains how sound waves are translated in to electrical impulses inside a single ear. When this mechanism is expanded to both ears, combinations of the two sets of electrical impulses can be combined to determine the direction of an incoming sound.

The psychoacoustic study of spatial hearing has been an active area of research since the mid 20th century, with early examples including the Jeffress model in which time differences between sound events at each ear contribute to a person's ability to locate sounds in space [18]. The work of Lindemann a few decades later expanded on this idea by translating the earlier theory to a DSP context, and establishing the Interaural Level Difference (ILD), Interaural Time Difference (ITD), and Interaural Cross Correlation (IACC) as important measures to describe the spatial aspects of binaural signals in a quantifiable way [20].

Put simply, we are able to perceive the incoming direction of a sound by observing both the amplitude difference and time difference of coincident sound waves arriving at the left and right ears. Figure fig. 3.3 shows how these two properties enable directional hearing. Firstly, the head itself creates a physical barrier that attenuates sound waves at one ear more-so than at the other ear. Secondly, the difference in distance between the sound source and each ear will create a time delay between the received signals. Over time, we learn to associate patterns in ITD/ILD with the physical location of the sound source.

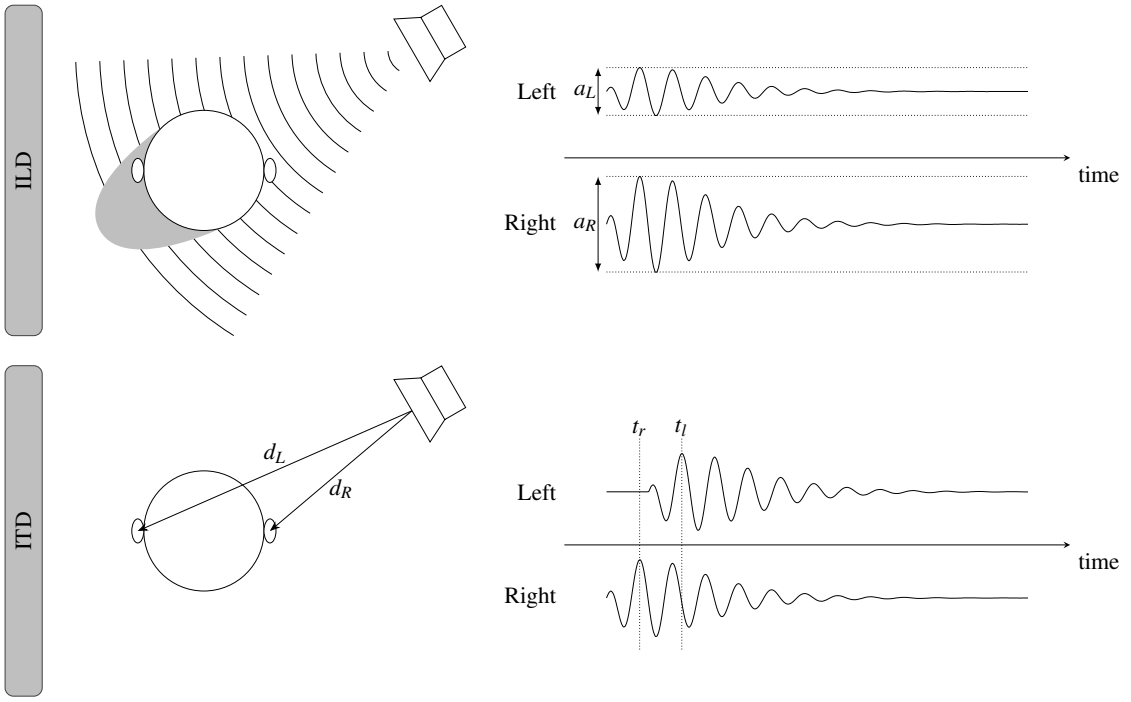


Figure 3.3. Illustration of how humans perceive the direction of an incoming sound. The physical barrier of the head causes an Interaural Level Difference (ILD) between sound waves received at the two ears ($a_R - a_L$). The difference in distance from the source to the ears ($d_R - d_L$) causes an Interaural Time Difference (ITD) between the two ears ($t_l - t_r$).

The relative importance of ILD and ITD differ depending on the frequency of the incoming sound. The reasons for this are twofold. Firstly, a person's interaural distance (the physical distance between their ears) must be longer than the wavelength of the incoming sound wave in order for them to detect the phase difference between wave peaks. In this case, it is often referred to as Interaural Phase Difference (IPD), which accounts for the centre frequency of the critical band. Secondly the cilia that line the basilar membrane inside the cochlea can only detect the fine structure of a sound wave up until roughly 1.6 kHz [20]. Lindemann gives a rule of thumb that time differences of less than 7.5 ms are heard as a single auditory event (for which direction could be inferred), and differences above 21 ms are heard as uncorrelated events [20]. Hence,

at 1.6 kHz and below ITD is more important, whereas above 1.6 kHz ILD is more important. Furthermore, Blauert's subsequent work found firstly that both ILD and ITD are relevant within the frequency band surrounding 1.6 kHz, and secondly that the angle of incidence of the sound wave plays a role in how the cues are perceived [21]. Despite this, Lindemann's rules of thumb are still broadly accepted, even being encoded into international standards such as *ISO/IEC 11172-3* [22]. As an example, fig. 3.4 shows two recordings (a high-frequency hi-hat and a low-frequency kick drum) that originate from the right-hand side of a listener. The high-frequency hi-hat has a large ILD and ambiguous IPD, whereas the low-frequency kick drum has negligible ILD, but an interpretable IPD.

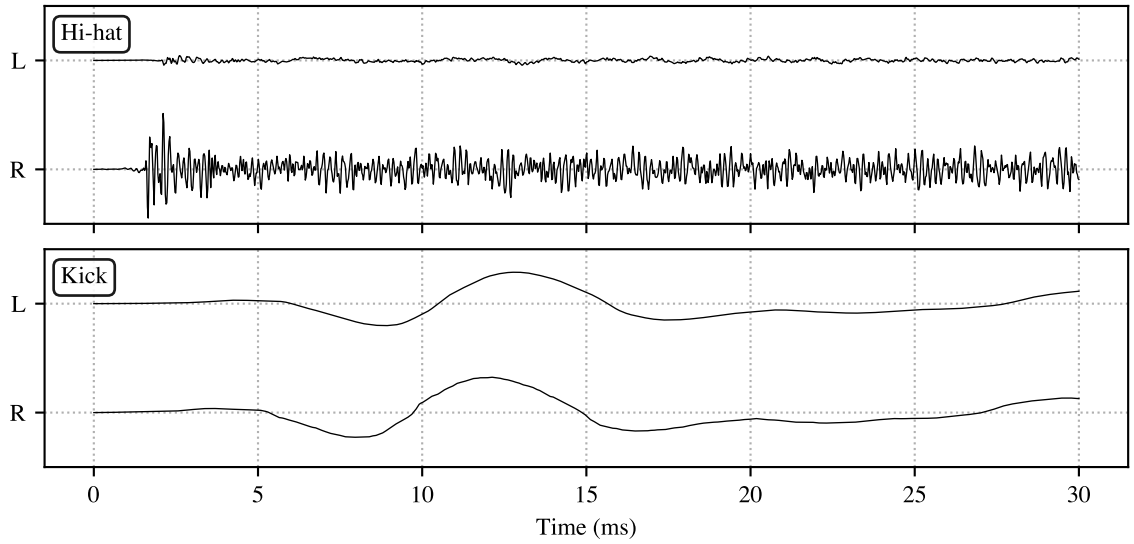


Figure 3.4. Example sounds coming from the right-hand side, recorded binaurally. High-frequency content (above) has a large ILD and ambiguous IPD, whereas low-frequency content (below) has negligible ILD, but an interpretable IPD.

3.1.3 Loudness perception

While we have established that the ILD is a key factor of directional hearing, a more detailed understanding of the link between perceived loudness and sound wave intensity is necessary for musical applications.

We perceive the loudness of sound, not in terms of the absolute amplitude of the sound wave, but the *power* of the sound wave. Mathematically speaking, perceived loudness is proportional to the Root Mean Squared (RMS) Sound Pressure Level (SPL) raised to an exponent [23]. Observing that the calculation of RMS naturally involves taking the square of the original time function (see eqs. (3.1) and (3.2) for the continuous and discrete forms respectively), it follows that multiplication of a signal by $\sqrt{2}$ will result in double perceived loudness. Conversely, multiplying a signal by $\frac{1}{\sqrt{2}}$ will halve the perceived loudness.

$$f_{RMS}(t) = \sqrt{\frac{1}{T_2 - T_1} \int_{T_1}^{T_2} [f(t)^2] dt} \tag{3.1}$$

$$f_{RMS}[n] = \sqrt{\frac{1}{N} \sum_{n=0}^{N} f[n]^2} \tag{3.2}$$

This basic rule is relevant when two sounds are mixed in music production. It defines a precise, mathematical relationship between perceived loudness and the amplification applied by the production equipment or software.

3.2 Music production

The principles of our auditory perception have been exploited for creative use since the turn of the 20th century. Early stereophony came from the world of cinema, with sound technicians adding multiple speakers to enable spatial effects[1]. Eventually migrating into the music industry, stereophonic recording became the norm in the 1950s [2, 24, 25]. While both cinema and music practitioners have long known that using two speakers for reproduction enabled spatial effects, knowledge of how to *control* those effects came with the psychoacoustic understanding detailed in section 3.1.

3.2.1 Panning

The concepts of directional hearing are applied in music production to make sound sources appear as though they originate from a specific direction or position, often referred to as the 'phantom position'. In recording studio vernacular, this is call 'panning'. In accordance with the two signal differences a that humans are able to detect (ILD and ITD), there are also two basic types of panning – amplitude panning and time panning. By way of example, fig. 3.5 shows how the phantom position can be moved anywhere in between the speakers that are reproducing the sound by applying different amplitude gains to the left and right channel of a stereo recording.

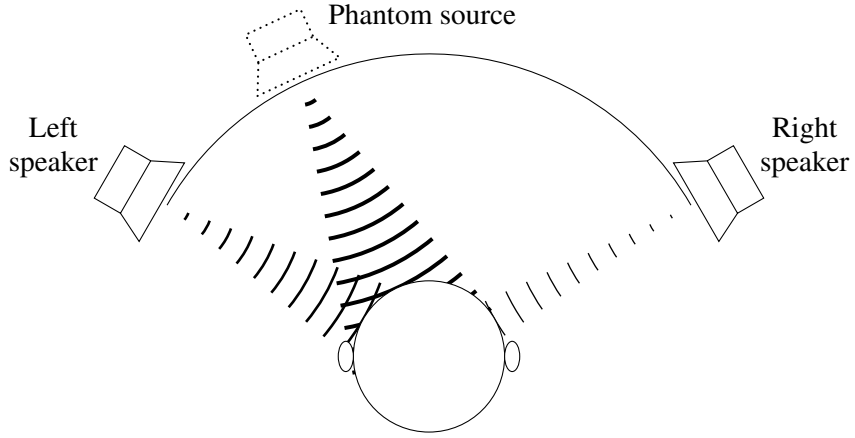


Figure 3.5. Perception of a phantom sound source created by amplitude panning. The sound wave from the left speaker has higher amplitude than that of the right speaker, thereby creating a phantom source which appears to come from the left.

Time panning (otherwise known as the 'Haas effect') exploits the phenomenon that the first ear to hear a sound generally dictates the direction that it appeared to come from. The process is to simply delay the signal in one channel by a time value within the 'Haas window' (typically less than 40ms). The undelayed channel will then dictate the phantom direction of the original source. In music production it is used both for panning and to add spaciousness to a recording [24, 26].

Another frequently used technique is 'binaural panning', which assigns combinations of amplitude and time difference to each stereo channel. This is typically achieved by using a Head-Related Impulse Response (HRIR). An HRIR is a pair of impulse responses recorded with microphones placed in the ears of either a real person or a dummy head. To obtain an HRIR, a known excitation signal is played from a specific position in a room and recorded with said binaural microphones (see fig. 3.6). The HRIR can then be convolved with any other monophonic recording to make it appear as though the recording came from the position of the excitation source. An HRIR can only be used to pan a sound to a single position, so it is typical to record many from different positions [27].

In traditional recording studio setups, the recording engineer or producer will use one or more microphones to capture an original sound source. These singular sound sources are commonly

[1] A typical use case was to match the origin of a voice with the actor's location on the screen.

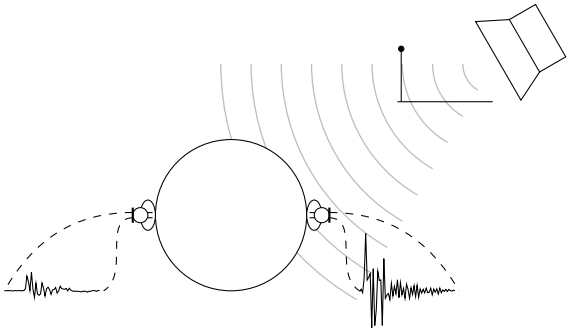


Figure 3.6. Illustration of a binaural HRIR being recorded. An impulse is played back through the loudspeaker and recorded at microphones placed in each ear. The head acts a filter, altering the left and right microphone signals differently.

called 'stems'. During the production process, the stems may be panned to create a sense of space in the final mix. In the case that the stem was recorded using two microphones, the stem would be called a 'stereo stem', and the term 'panning' is usually replaced by 'balance', and only amplitude panning (as opposed to time or binaural panning) would be applied.

Amplitude panning is the most common panning technique in traditional music production [28]. It is associated with so-called 'panning laws', which govern how the amplitude is changed in each channel according the desired pan position [24]. One example is the 'energy-preserving pan law' (sometimes called the '-3dB pan law') in which the squared sum of the channel gains always equals 1. The gains of the left and right channel are calculated by $g_L = \cos(\frac{\pi}{2}a)$ and $g_R = \sin(\frac{\pi}{2}a)$ respectively where a is the panning position in the range $[0, 1]$. Figure 3.7 shows how the channel gains relate to the panning position. The total perceived loudness will be equal at all panning positions.

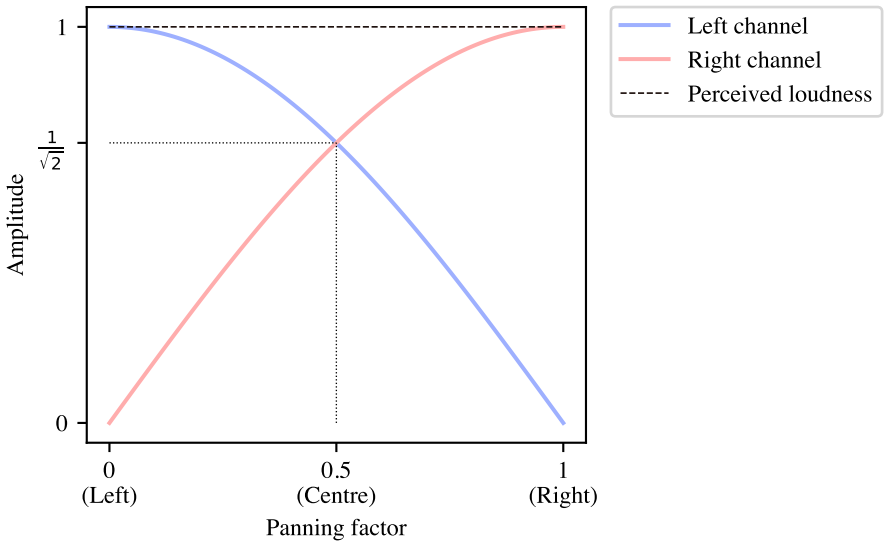


Figure 3.7. Channel gains for equal-power panning. When both channel gains are $\frac{1}{\sqrt{2}}$, the sound will be centre-panned. The perceived loudness is constant.

Panning techniques are applied to single stems only. A typical recording session will contain multiple stems, each with its own panning applied. The combination of all panning and other spatial effects is a focal point of the music production process, and is generally seen as one of the differentiating factors between producers [28].

3.2.2 Stereo image

The term ‘stereo image’ is used to describe the overall distribution of sound across the horizontal plane after panning and any other production techniques have been applied on all stems. One may also use terms like ‘stereo field’, ‘soundstage’, or ‘panorama’, which all refer to the same thing. They are usually described in qualitative terms such as ‘big’, ‘wide’, and ‘ambient’ rather than quantitative measures [29]. When mixing a musical recording, a producer will pan individual stems to their desired position within the stereo image. Those stems which are recorded in stereo have their own stereo image, which can also be controlled by the producer. The combination of all stems into a final stereo mix constitutes the overall stereo image.

Figure 3.8 shows a simple example stereo image, as perceived by the producer. It contains two mono stems – vocals and violin, and one stereo stem – piano. There are some common tropes seen throughout many productions, some of which are shown in the diagram. Firstly, vocals are recorded in mono and panned to the centre. Secondly, since the piano is panned slightly left, the violin is panned roughly the same amount to the right in order to achieve a balanced mix. Thirdly, all stems are placed in the stereo image in a way that emulates a natural performance on a stage, and to avoid the instruments masking each other. Note that while these practices are commonplace, they are purely an artistic choice on behalf of the producer [29]; Recordings breaking the trend are not difficult to find.

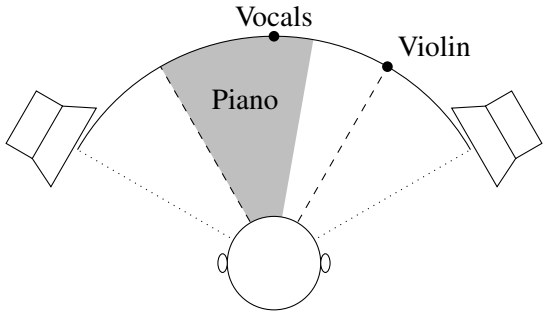


Figure 3.8. Example stereo image created by a music producer. Mono stems are shown as black dots and a stereo stem as a grey fill. Dashed lines represent the extent of the stereo image. Dotted lines represent the maximum boundaries of the possible stereo image.

3.2.3 Production equipment and software

To create a stereo image, a music producer will use the equipment available in a typical recording studio. Before the advent of digital recording, said equipment usually comprised a mixing console where each stem was assigned to one ‘track’. Each track had individual controls for (amongst others) level and panning that were directly linked to the internal electronics. The producers might also use the term ‘fader’ for the level control and ‘pan-pot’ for the panning control[2]. All tracks would be mixed by the console and the combined output fed to stereo speakers for playback. In a digital recording studio, the equipment is largely the same except that the mixing console will be connected to a computer that controls the session and stores the recorded audio. Figure 3.9 shows photographs of a small recording studio where the faders and pan-pots for each track are visible. The music producer will manipulate the physical controls for each track to create a stereo mix.

In modern recording scenarios a music producer typically uses a Digital Audio Workstation (DAW) - graphical software that allows them to record and edit audio. There are many DAWs available, but one such example (Reaper) is shown in fig. 3.10a. In particular, it shows a single recorded audio track which has its ‘Pan’ control automated. In a more complete recording session, the DAW would contain many more tracks, each with their own pan control.

[2]The term ‘pan-pot’ is a shortening of ‘panoramic potentiometer’.

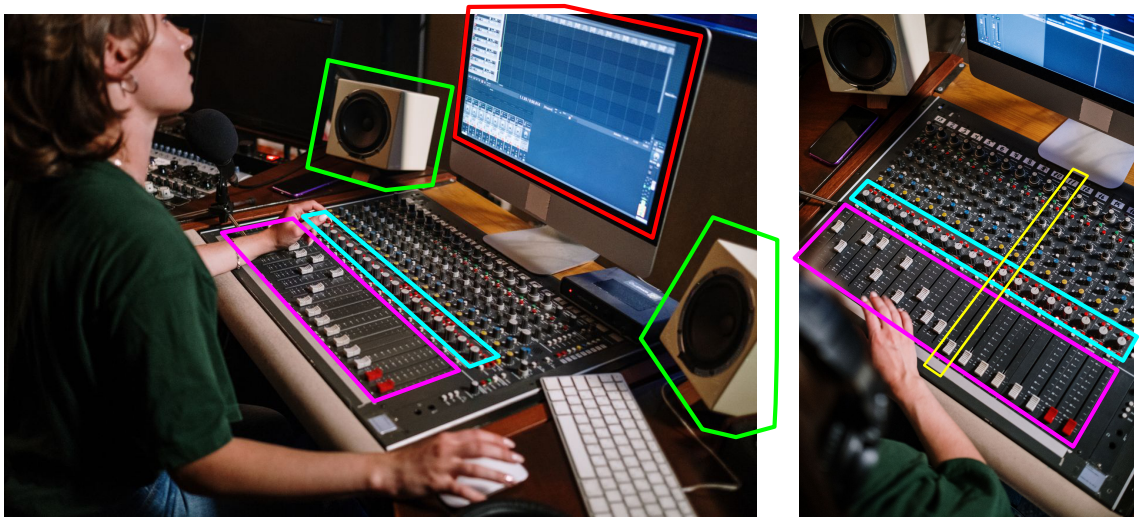
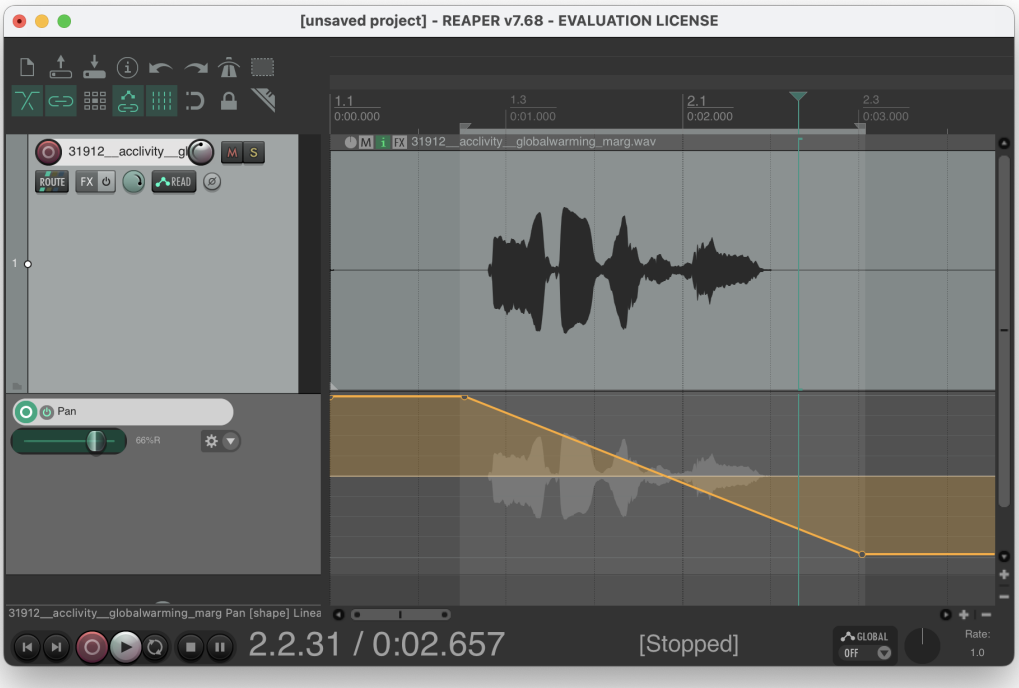


Figure 3.9. Photographs of a small recording studio*. The left photo shows all studio equipment, while the right photo shows a close-up of the mixing console. Faders are highlighted pink, pan-pots are highlighted blue, stereo speakers are highlighted green, and the computer running a Digital Audio Workstation (DAW) is highlighted red. One track is highlighted yellow.

*Original images by cottonbro studio are licensed under Pexels. Annotations have been added.

While there are some typical practices in the industry, it must again be emphasized that panning and creation of stereo field more generally is an artistic process [24]. The producer pans each track individually by either physically turning the pan-pots or adjusting a digital panning control, and judges the stereo image purely by listening. Producers are typically known for their individual style and taste, and the ways in which they use panning in a large DAW session containing many tracks forms part of that style.

In some cases the producer consults monitoring equipment such as a level meter or goniometer to fine tune their mix [9, 32]. They also look for potential quality issues that can occur when the stereophonic recordings are mixed down into a monophonic signal for reproduction, which have been documented well before the advent of digital recording [33]. In more modern production, DAW plugins are generally used to visualize and assess the stereo qualities of the final mix [9, 34]. Figure 3.10b shows one example - the ToneBoosters Goniometer plugin [31]. It is designed to replicate an analogue goniometer, which, taking two voltage signals as input, displays them on a 2D axis such that a music producer might be able to discern the correlation between the signals.



(a) Recording project in the Reaper DAW [30]

(b) ToneBoosters Goniometer plugin [31]

Figure 3.10. Screenshots of example music production tools

3.2.4 Audio representations

Audio representations are standard paradigms that an audio recording can follow. If we consider an audio recording as a set of concurrent time signals, the representation defines how each signal in the set should be interpreted. They are usually grouped into 'channel-based', 'scene-based', and 'object-based'. In channel-based audio, each signal represents the exact waveform that should be played back by a single loudspeaker. In scene-based and object-based audio, each signal represents one part of a larger scene, and rendering equipment is responsible for translating that scene to individual loudspeakers [35]. While the focus of this thesis in on stereo audio, it is important to compare stereo to the wider audio landscape to understand why it is so heavily used for recorded music.

Monophonic

Monophonic recordings contain a single channel to be played back on one speaker. They are only able to convey a small amount of spatial information; Reverberation is the only way to infer a sense of space. Archival/historical recordings will be recorded in mono, as well as a majority of raw studio stems, speech, and telephony recordings.

Stereophonic

Stereo recordings contain two channels: left and right. Compared to mono, stereo has the ability to convey the width and location of a sound 2D space. For music this sense of space is especially important since music signals tend to be highly dense and complex. Stereo mixing therefore allows music producers to add clarity and balance to their recordings. This marked increase in quality drove stereo's commercial adoption in the 1950s, and it has been the dominant format for recorded music since then [2].

Binaural

Binaural can be considered a subset of stereo. Each channel conveys the precise sound wave as heard at the left and right ears. It is typically used when rendering spatial audio[3] for playback on headphones, and in Virtual Reality (VR) or Augmented Reality (AR) scenarios.

Multichannel

Multichannel is a continuation of stereo which adds speakers at more positions. While it could contain any number of channels, there are some standards that are referred to by their channel count and layout, e.g. 5.1 (5 listener-plane channels, 1 Low Frequency Effects (LFE) channel) or 22.2 (10 listener-plane, 2 LFE, 9 overhead, 3 low) [35]. Compared to stereo, it has the ability to convey width, depth, and – depending on the arrangement – height, creating a higher sense of immersion. This ability sees it being used mainly for film and TV soundtracks where the emotional impact of single sound events can be critical. That said, the high setup cost and difficulty positioning extra speakers has historically discouraged its wider adoption [37].

Ambisonics

Ambisonics is a scene-based audio representation where each channel is one directional component of a 3D physical sound field. Recordings are referred to by their 'order', which implies both the number of different directions and the number of channels. The number of required channels increases quadratically with the order according to the formula $C = (O+1)^2$ where C is the number of channels and O is the order. Ambisonic recordings typically excel at representing diffuse sounds, but struggle to represent point sources unless the order is very high. Thus, ambisonic recordings are only used for very specific types of music, such as orchestral performances in large concert halls where accurate recreation of the ambience is more important than individual instruments. Even in those cases, an ambisonic recording will generally be rendered to binaural for mass distribution [38].

[3]The exact definition of 'spatial audio' is somewhat contested in the literature but here I assume Cobos et al.'s definition: *'audio that contains spatial information'* [36].

Object-based audio
In object-based audio, the entire audio scene is composed of individual audio objects accompanied by metadata specifying their location and movement. Rendering equipment is then responsible for translating those objects to speaker channels for playback. It is a relatively new format and while its main use is for cinematic sound, the music industry has recently begun to adopt it. In contrast to ambisonic recordings, object-based audio is distributed in its original form, allowing the same set of signals to be rendered in multiple ways.

Overall, stereo recordings are able to deliver high enough spatial fidelity for the majority of music listeners. The newer formats, while gathering interest in pockets of the music community, have yet to receive wide-spread adoption from consumers. The result is that the mainstream distribution of music still happens in stereo.

3.2.5 Lossless and lossy encoding

Regardless of which audio representation is chosen, the resulting recording will most likely be encoded to a compressed format for distribution. There are two types of compression: 'lossless' and 'lossy'. A lossless format, when decoded, will produce a bit-for-bit recreation of the input signal. Lossy formats on the other hand will not produce the exact same signal, but will discard information that they consider irrelevant according to a psychoacoustic model. There are many lossy formats each suitable for different types of audio, but for the purposes of this thesis it is just pertinent that spatial information can be lost during lossy encoding.

3.3 Analysis of digital recordings

The output of a recording session is usually only the stereo mix. Stems and DAW sessions are generally kept private, with precious few being available under licensing terms that are conducive to academic research [39]. Thus, we assume that all analysis can only be performed on the final stereo mixes that are distributed for playback. The most common distribution format is a digital audio file, and there are some fundamental analysis techniques that can be applied to a digital audio signal.

3.3.1 Digital audio signals

When a sound wave interacts with a microphone, it generates an electrical signal in continuous time. To convert that analogue signal into a digital one, it is sampled at regular intervals. The sampling frequency (f_s) is related to the sampling interval (T_s) by the relationship $T_s = \frac{1}{f_s}$.

Shannon's sampling theorem states that a signal can be perfectly recreated when sampled at twice the highest frequency contained in the signal. Conversely, the highest frequency that can be represented by sampling a signal at a given f_s is half of that frequency. This limit is called the 'Nyquist frequency' and is calculated by $f_N = \frac{f_s}{2}$ [40]. This is illustrated by Figure 3.11, which shows the sampling of three waveforms. The upper plot shows a signal at the Nyquist frequency (f_N), which can be perfectly reconstructed from the digital samples. The lower two waveforms have frequencies equidistant but in opposite directions from the Nyquist frequency ($f_N \pm f_\delta$). The digital samples shown in circles could represent either of the two waveforms. In practice, when the analogue waveform is reconstructed, the lower frequency will always take precedence.

A typical f_s for music is 44.1 kHz, with a resulting f_N of 22.05 kHz. The prevalence of this particular number stems from the first widespread distribution mechanism for digitally encoded music - the Compact Disc (CD)[4]. While higher frequencies may be used in the recording and production process, an exceedingly large proportion of recorded music has been distributed at 44.1 kHz, even after CDs were displaced by other distribution methods.

[4] 44.1 kHz was chosen by a consortium of CD manufacturing companies as a trade-off between technical requirements and the limits of human perception [41].

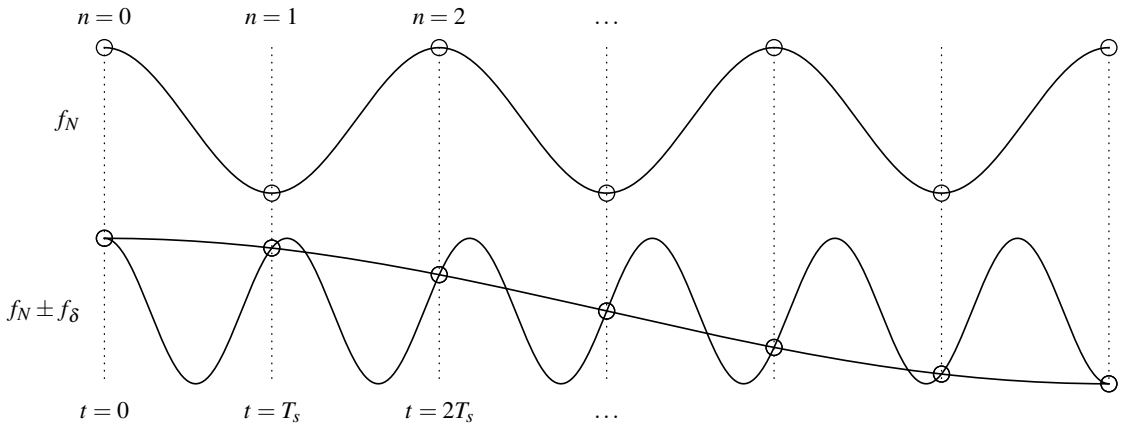


Figure 3.11. Illustration of the Nyquist Frequency in digital audio signals. Dotted vertical lines represent sampling points with a separation of T_s, and monotonically increasing index n. The upper plot shows a cosine wave at exactly f_N. The lower plot shows two cosine waves with frequencies equidistant above and below the Nyquist frequency (f_N). Sampled values are shown as circle marks on the plots. The two lower signals can both be represented by same digital samples.

3.3.2 Time-Frequency representation of audio

In order to perform many types of audio analysis, the audio signal can be converted from the time domain into the time-frequency (TF) domain. In perceptual acoustics research there are several common forms of TF representation, each with its own nuances. This section provides those details, and a side-by-side comparison is shown in appendix B.

Short-time Fourier Transform (STFT) spectrogram

The STFT spectrogram is one of the most popular TF represention across all audio processing work due to its computational efficiency. The fundamental principle is that a time-domain signal is split into short (1 to 100ms) overlapping chunks, and the Fast Fourier Transform (FFT) is performed on each chunk, thereby converting that chunk into the frequency domain. By arranging subsequent frames along a time axis it is possible to represent the original signal as a 2D image known as a spectrogram. When the time-domain signal is entirely real, (which is the case for monophonic audio signals), the result of the FFT will be a complex signal which is symmetrical around the Nyquist frequency and the upper half is therefore dropped. Furthermore, the complex spectrum is usually represented in magnitude/phase form to better match the human auditory response.

A fundamental property of the FFT is that it assumes the signal repeats outside the time frame that is being analysed. Since this is generally not the case for audio signals, it is standard practice to apply a time-domain windowing function to each frame so that large transients do not occur at the boundaries. Several standard windowing functions exist, each with their own nuances, but for most audio applications the 'Hann' window can be used [42].

The nature of the spectrogram is affected by two parameters: the frame size and hop size. 'Frame size' refers to the number of samples in each analysis frame. To obtain maximal computational efficiency this must be a power of 2. 'Hop size' refers to the number of samples between the start of subsequent frames and is sometimes expressed as a percentage of the frame size. The frame size dictates the resolution of the spectrogram's frequency axis since the frequency bins will be spaced evenly between 0 Hz and the Nyquist frequency. Importantly, a longer frame size will result in lower time accuracy within each frame; Sharp transients in the time domain will appear as wideband energy in the frequency domain, but information about exactly where the transient occurred within the frame is lost. To counteract this inaccuracy, a smaller hop size can be used so that transients might be detected as sharp changes between STFT frames. One major drawback of the STFT spectrogram is that the linearly-spaced frequency bins do not align well with the human perception of frequency, which is logarithmic. The STFT therefore contains redundant information in higher frequencies. Several common alternatives exist to account for this.

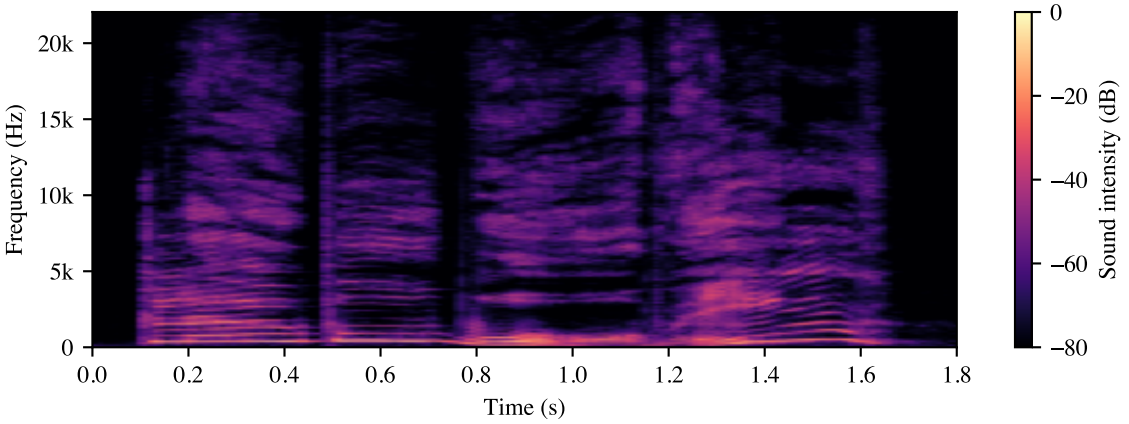


Figure 3.12. STFT spectrogram of a speech recording.

Mel spectrogram

The ‘Mel scale’ is a quasi-logarithmic frequency scale derived from perceptual experiments. Several definitions exist in the literature [43], but popular implementations like that of librosa [44] and Auditory Modelling Toolbox (AMT) [45] tend to underpin most audio experimentation.

A ‘Mel filter bank’ is a set of band-pass filters that are spaced equally on the Mel scale. Practically, they are implemented as sets of weights that are applied to an STFT spectrum with the relationship $M_n = \sum_{k=0}^{N} S_k W_{k,n}$ where M_n is the nth Mel coefficient, S_k is the kth STFT bin, N is the number of STFT bins, and $W_{k,n}$ is the kth weight of the nth Mel filter. An example Mel filter bank with eight filters is shown in fig. 3.13. While 8 filters are shown for demonstration purposes, audio analysis typically requires over 100. When this filter bank is applied to each time step of an STFT spectrogram, the result is a ‘Mel spectrogram’, an example of which can be seen in fig. 3.14. Observe that the higher frequencies have been condensed into fewer frequency bins compared to the STFT spectrogram.

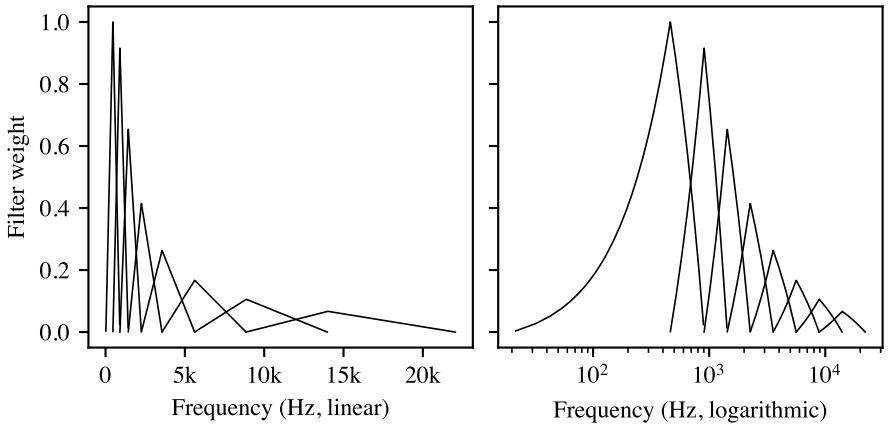


Figure 3.13. Example Mel filter bank containing eight filters, shown with linear (left) and logarithmic (right) frequency scales. The bandwidth of each filter is roughly equal in logarithmic space, and the amplitudes are scaled so that the sum of the weights in each band is equal. Filter weights equal to zero are removed for clarity.

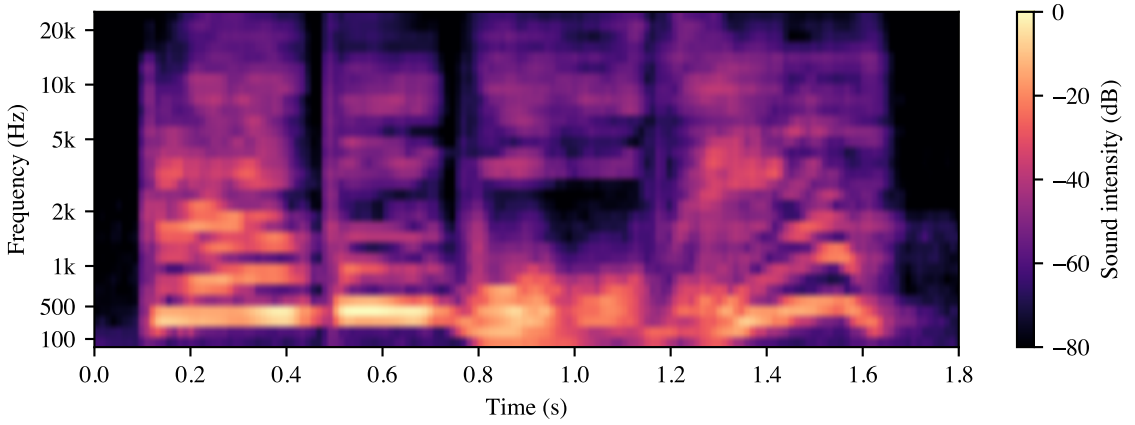


Figure 3.14. Mel spectrogram of a speech recording, using 32 bands.

Cochleagram

As an alternative to the two spectrogram representations, a cochleagram aims to make a TF representation that models the acoustic response of the human cochlea more closely. To that end, all processing is performed in the time domain and is sometimes referred to as a 'peripheral ear model'. Several examples appear in the literature [46–48]. In the classical method [46] the original signal is first processed by a bank of gammatone filters with their centre frequencies placed according to a perceptual scale (such as the Equivalent Rectangular Bandwidth (ERB) scale [49]), as shown in fig. 3.15. As with Mel filters, several definitions of gammatone filters exist in the literature, but the popular implementations (e.g. scipy [50] and AMT [45]) preside. Figure 3.16 shows a cochleagram of the example speech recording.

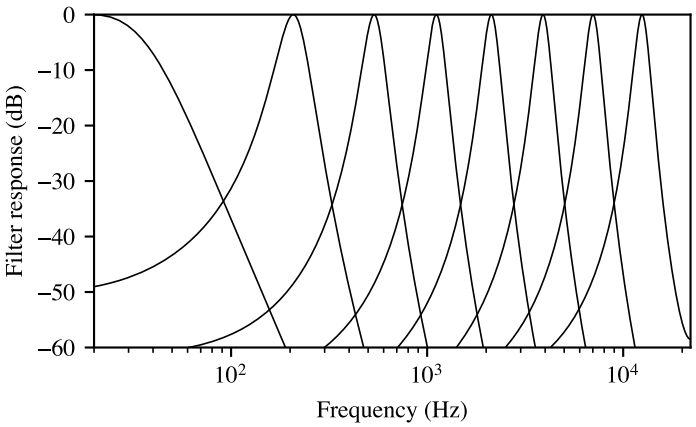


Figure 3.15. Example gammatone filter bank with individual filters spaced equally on the ERB scale. The filters are Infinite Impulse Response (IIR) and are applied to an audio signal in the time domain.

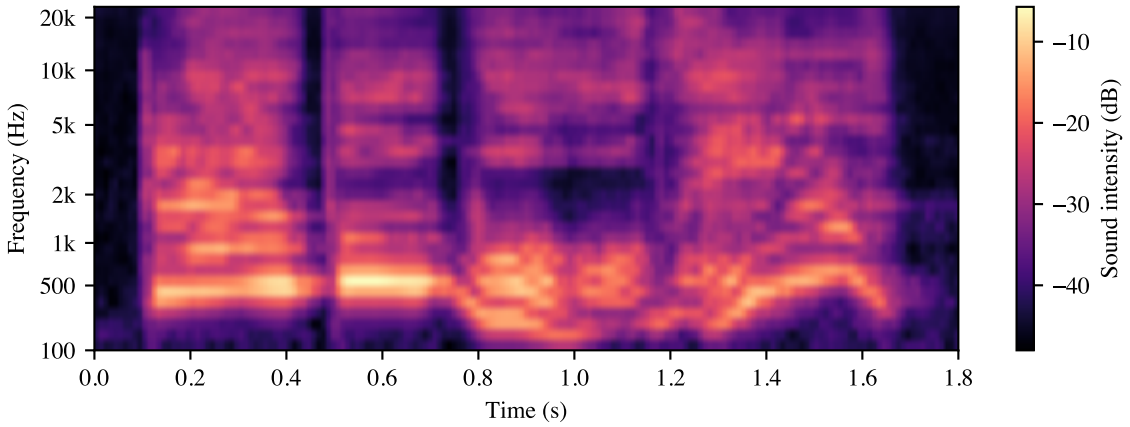


Figure 3.16. Cochleagram of a speech recording, using 32 bands.

Figure 3.17 shows the intermediate steps to generate a cochleagram. Each sub-band signal is processed by a half-wave rectifier, square root, and low-pass filter to mimic the excitation patterns of the cilia and basilar membrane. The result is a signal representing the envelope of each sub-band. Those envelopes are then segmented into overlapping time chunks. The amplitude of each chunk is then calculated as the mean of all its values, forming one TF point in the cochleagram.

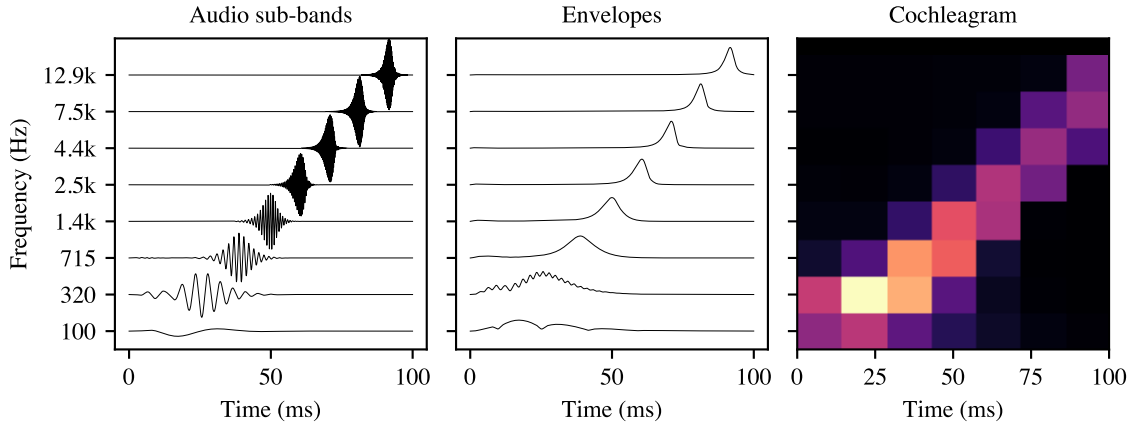


Figure 3.17. Steps of an example decomposition from time-domain audio to a cochleagram. The audio being filtered is a frequency sweep from 100 to 20 000 Hz. First it is processed by a gammatone filter bank, generating eight sub-band signals which are spaced equally on the ERB scale. Each sub-band has its envelope extracted. The envelope is then split into overlapping time chunks, which are averaged to form the cochleagram.

3.3.3 Machine learning

Any data-driven audio research, particularly that using vast datasets, will naturally involve the use of ML. Broadly speaking, ML techniques can be classified as either supervised or unsupervised. When training a supervised ML model, the training dataset needs to include ground-truth labels such that the network can objectively discern via a loss function whether the predictions it is making are correct. Unsupervised ML on the other hand does not require any labels. Instead, it seeks to form a representation of the underlying dataset via techniques such as clustering or dimensionality reduction, thereby learning the structures and patterns of the data while retaining as much key information as possible.

Striking a middle ground between the two is 'self-supervised' learning, which addresses the issue of creating objective ground-truths from unlabelled data. Instead of relying on external labels, self-supervised learning generates pseudo-labels from the dataset itself, and then uses those as the input to a 'pretext task', which is tackled in much the same way as a supervised task.

As an example we can consider the current largest dataset of recorded music: FMA [7]. The dataset contains the stereo audio of approximately 100 000 tracks, along with metadata such as genre tags. An unsupervised task might involve learning a high-order representation of the raw spatial cues present in the stereo signals. A supervised task might involve predicting genre using the audio signal as input. A self-supervised task might involve masking off a short period of audio and filling it back in using the surrounding audio as context.

In relation to the overarching RQ in section 2.1, we assume that a music producer's DAW session is the only source of objective ground-truth data regarding stereo field. Furthermore, we assume that DAW sessions are not publicly available, as was found in [51, 52]; The only public-facing output of a recording session is the mixdown. It follows that either unsupervised or self-supervised learning is necessary to draw any conclusions to the RQs.

4. Related works

In this chapter I break down the details of existing algorithms for stereo field analysis, as well as demonstrating those techniques on example audio recordings. To provide additional context, I also present several high-level initiatives within the audio industry, showing how stereo field analysis has been used up until the present day. Throughout the chapter I use three recordings as examples. They all come from the FMA dataset [7], and each one presents a different use of stereo image. In the text they are referred to simply as track 'A', 'B', and 'C'. Full details for all tracks can be found in appendix A. Note that in order to discuss and demonstrate the algorithms in this chapter, I implemented them from scratch in a monolithic software library. By breaking each algorithm down into its constituent parts it was possible to draw links between them, since they often share underlying DSP techniques.

4.1 A brief history of stereo field in industry

Research into the stereo field of recorded music specifically (as opposed to generic spatial properties of audio signals) has its roots in the early 2000s. At the time, stereo recording was a well-established norm and audio distribution was handled by the early lossy encoding formats such as MPEG layer 3 (MP3) and Advanced Audio Coding (AAC) [22, 53]. Analysis and description of stereo field was used to increase the coding efficiency of these codecs, for example by Parametric Stereo (PS) coding [54–56]. In parallel, new audio encoding schemes such as Binaural Cue Coding (BCC) (later ratified as 'MPEG Surround' [57]) were developed based on the work of Baumgarte and Faller [58].

The standardization of multichannel audio presented an opportunity: new reproduction equipment needed to remain compatible with the existing back-catalogue of stereo recordings, and vendors could compete on sound quality. There was therefore a need for algorithms that upmixed stereo audio to multichannel. Analysis of the stereo field, source separation, and primary/ambient extraction were seen as effective ways to achieve this [59]. In fact, such research is still ongoing today [60].

In the prevailing years, object-based audio has been developed and standardized as e.g. Dolby Atmos [61] and MPEG-H [62], and the same early research into stereo field analysis is still relevant to these new formats for all the same reasons. Nowadays, it is commonplace to use ML or DL to perform the same tasks, although the results only outperform the DSP-based predecessors in specific cases [36].

With access to these efficient distribution formats, large digital music collections emerged and brought new problems to solve: collectors needed ways organize and categorize their music. Stereo field analysis was just one of the mechanisms used to facilitate this, paving the way for research into e.g. Automatic Genre Recognition (AGR), MIR, and music recommendation [7, 63, 64].

4.2 Spatial cues for stereo image

A ‘spatial cue’ is a low-level analysis of spatial information in a short (roughly 1 to 100 ms) audio snippet. While those mentioned in section 3.1.2 aim to describe binaural sound as perceived by humans, there are several examples of spatial cues tailored specifically to the analysis of recorded music.

The ‘panning index’ and ‘ambience index’ specified by Avendano and Jot [65] are loosely analogous to ILD and IACC in the acoustics domain. To estimate the ‘panning index’, a signal model is assumed whereby a musical source is panned within the mix using the ‘energy preserving pan law’, in which each channel is scaled such that the sum of the squared amplitudes equals 1 (see section 3.2.1). While this panning technique does not affect the phase of either channel, existing phase differences between channels are measured using the ‘ambience index’, which is defined such that it is low when the stereo channels are correlated (indicating a direct sound source), and high when channels are uncorrelated (indicating an ambient or reverberant sound source). An example of the panning and ambience indices for a musical recording can be seen in fig. 4.1.

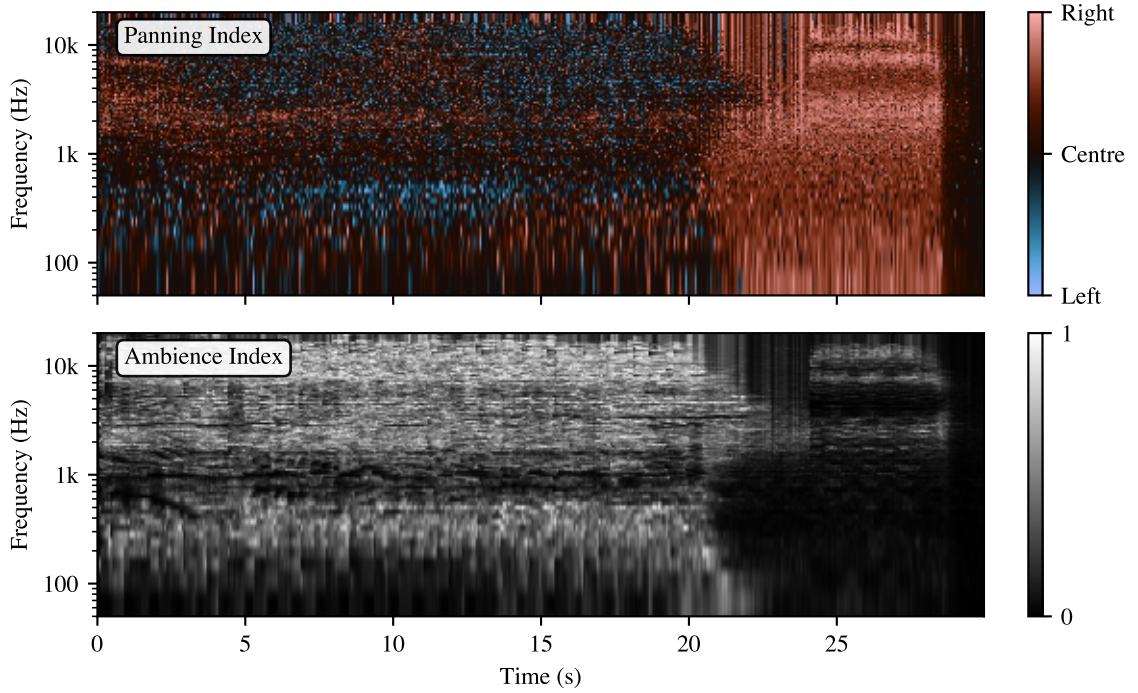
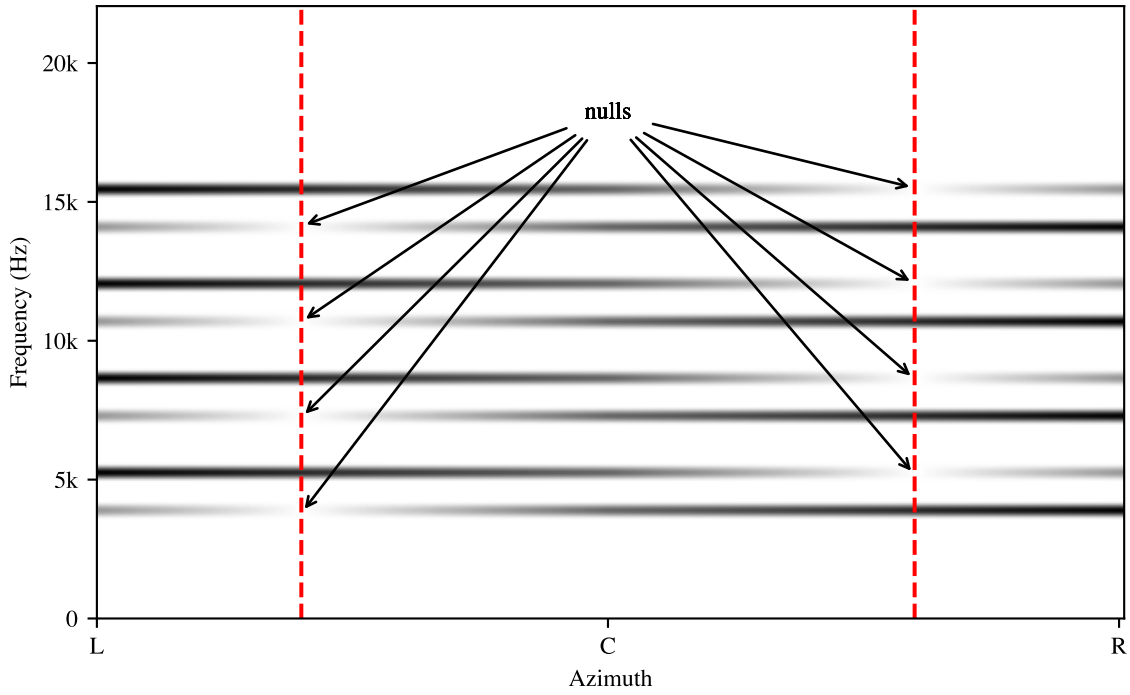


Figure 4.1. Demonstration of ‘panning index’ and ‘ambience index’ on Track C. Several panned sources are visible as red or blue bands in the ‘panning index’, and at roughly 20 s most sources fade out leaving a single one on the right. The ‘ambience index’ plot shows that the track starts off with no single dominant direct sound source. At 20 s when most sources fade out, the ‘ambience index’ drops to 0 across the lower half of the spectrum, indicating that the remaining source is direct. In the upper half of the spectrum, the source remains slightly ambient.

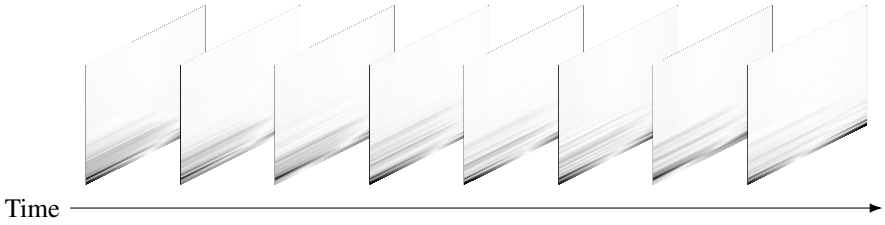
Presti provides a re-contextualization of the panning and ambience indices in his work on Bivariate Mixture Spaces (BMSs) [66, 67]. He demonstrates that it can be used to analyse the panning position of individual TF bins, plus the clarity (or trustworthiness) of that measurement. Thus, it is useful for tasks such as primary/ambient decomposition, stereo field manipulation, and visualization.

Barry and Lawlor present another seminal technique in their paper defining the Azimuth Discrimination and Resynthesis (ADRess) algorithm [68]. The fundamental principle is to estimate the panning position of frequency bins in a stereo spectrogram by scaling one channel, inverting it, and mixing the channels back to mono. The resulting phase cancellation will determine the scaling factor that was applied to the original audio source during amplitude panning, and hence will show where the source was panned in the stereo field. To demonstrate, fig. 4.2a shows an ex-

ample Frequency-Azimuth (FA) plane for a combination of panned sine waves, where nulls in the plane can be observed at the azimuthal positions that the sinusoids are panned to. To demonstrate a realistic use case, fig. 4.2b shows an FA analysis signal for a music recording. While this was technique was originally intended to be used for audio source separation, it has been repurposed many times since.



(a) FA plane for a signal comprising eight sinusoids at differing frequencies. Half are panned left and half are panned right. Red dashed lines show the panning positions detected by looking for nulls in the FA plane. This FA plane is formed from a single frame of an STFT spectrogram.



(b) FA analysis signal constructed from consecutive FA planes of track C.

Figure 4.2. Frequency-Azimuth (FA) planes and analysis signal.

Importantly, all the previously mentioned analyses are performed in the TF domain and result in one value per TF bin. Their use cases typically involve reconstruction or editing of the original audio source as opposed to *description* of it since they inherently entangle semantic content with spatial information. They have, however, become the basis of other algorithms that are designed specifically for description.

4.3 Stereo image description

Stereo image description refers to the generation of a numerical vector that represents the stereo image in an arbitrary feature space. Recordings with similar stereo image should be close within the space, although the distance metric is application-specific. A typical technique is to construct the vector from statistical summaries of low-level spatial cues. In the acoustics domain where stereo image description is typically used for analysis of concert halls and music reproduction systems, the previously mentioned spatial cues (ILD, ITD, and IACC) typically form the basis of the analysis. However, as stated by Sarroff and Bello, the task of describing recorded music is inherently different since said recordings may exhibit spatial aspects that cannot occur in physical acoustic space [69]. Throughout this section, these descriptions are referred to as Stereo Image Profiles (SIPs), and examples of each one are given on the three example tracks.

4.3.1 DSP-based SIPs

SIPs based on amplitude panning

Tzanetakis et al. provide the earliest example of a technique specifically designed to interpret the use of panning in recorded music: Stereo Panning Features (SPF) [3]. It targets amplitude panning only, using Avendano and Jot's 'panning index' as the basis. Additionally, it separates the frequency spectrum into three sub-bands – lows, mids, and highs – following the common practice of music producers. While not specifically stated by Tzanetakis et al., I interpret the vector to represent the amount and variance of 'fast panning' and 'slow panning' in each frequency band. Tzanetakis et al. later expanded the features by adding an estimation of the number of individual sources in the recording [8]. The resulting SIP can be used to compare the recording and production styles of different music tracks, as shown in fig. 4.3.

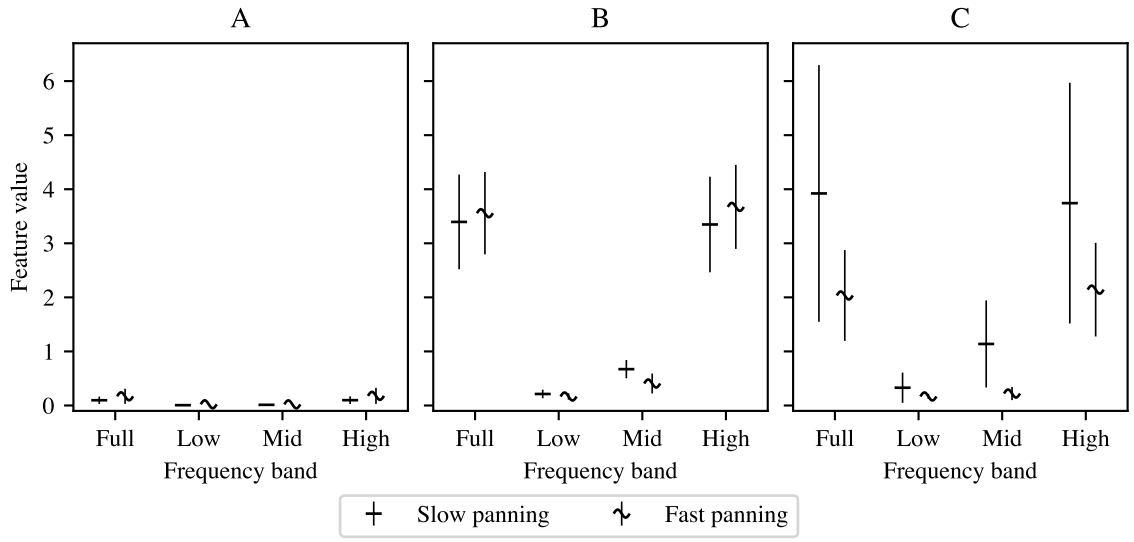


Figure 4.3. Stereo Panning Features for three tracks from the FMA dataset [7]. All tracks clearly show different panning and production styles.

While SPF shows production style at a high level, other descriptors contain more detail. Gomez et al. [10] and Sarroff and Bello [4] both present separate methods to form a Panning Histogram (PH), which displays the likelihood of a sound source being panned to a specific azimuth. Figure 4.4 shows examples of both methods for the same three example tracks. As Sarroff and Bello state, the 'azimuth' value should not be interpreted as a true angular position relative to the listener, but rather as a loose placement of a sound source within the stereo field. Note that compared to SPF, PH visualizes the location of sound energy in the stereo field rather than just the 'amount of panning'. On the other hand, PH looses information about the *dynamics* of the panning, which

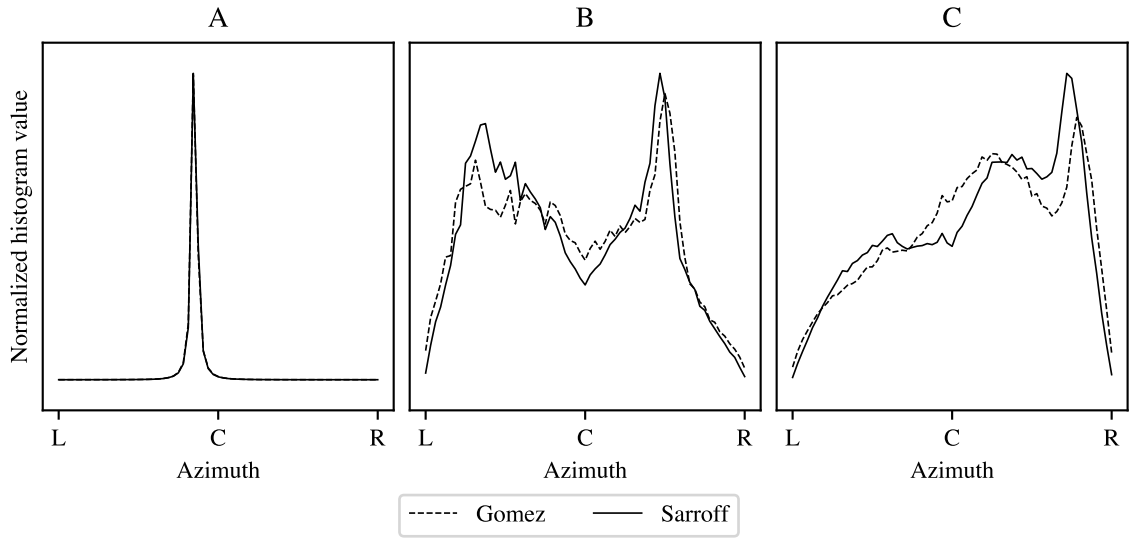


Figure 4.4. Panning Histograms for three tracks from the FMA dataset [7]. All tracks clearly show different shapes of stereo field. The two different algorithms (Gomez et al. [10] and Sarroff and Bello [4]) produce slightly different fine detail, but they agree on the general azimuthal distribution.

SPF does not. Sarroff and Bello subsequently use the PH to calculate an estimate of stereo width [4], which is an important perceptual measure (see section 4.4).

Gomez et al. present a continuation of their PH algorithm. They note that the histogram itself can be quite large, and that standard distance metrics may not correlate with perceptual differences. To avoid this, they propose converting the PH to a vector of 'Panning Coefficients (PC)' in a non-perceptual domain and truncating it. Equation (4.1) shows this conversion where $\boldsymbol{C}$ is the vector of coefficients, $\mathscr{F}^{-1}$ is the inverse FFT, L is the truncation length, and $\boldsymbol{HH}^{\leftarrow}$ is a concatenation of the panning histogram and its reverse.

$$\boldsymbol{C} = \left\{ \mathscr{F}^{-1} \left\{ \log(\boldsymbol{HH}^{\leftarrow}) \right\} \right\}_{1:L} \tag{4.1}$$

The efficacy of this technique remains under-tested, and my work in paper III showed that PC were unable to accurately represent PHs with sharp peaks. It should also be noted that, as revealed by paper II, the term 'Panning Coefficients' is used frequently in the literature to mean other things [70–73].

Tardieu et al. present another assessment of amplitude panning named Amplitude Stereo Features (ASF) [5]. Differing from the previously mentioned techniques, Tardieu et al.'s method focuses on providing measurements of the amount and variation of panning in many frequency bands. The basis of the algorithm is the ILD of a stereo cochleagram, as opposed to an STFT spectrogram. Tardieu et al. use 70 frequency bands equidistant on the ERB scale, each of which produce multiple features. Thus, the resulting SIP vector is an order of magnitude larger than SPF or PH. As fig. 4.5 shows, ASF's large size allows for a very detailed assessment of panning across the frequency dimension, although it runs the risk of including noise and redundant information.

SIPs based on studio equipment

Ziemer et al. argue the case in [9, 11, 32] that analyses based on a recording studio goniometer can aid in describing the stereo properties of music, particularly in differentiating between the artistic intentions of different recording sessions. The underlying logic is that since producers consult goniometers in their creative process, their readings must have inherent descriptive value. To perform the analyses, a virtual goniometer readout consisting of a lissajous figure and channel correlation measurement is constructed for non-overlapping time frames of an audio recording. The lissajous figure is quantized and reduced to a single number by counting the unique X/Y positions. Channel correlation is defined as the Pearson correlation coefficient [74] between the left

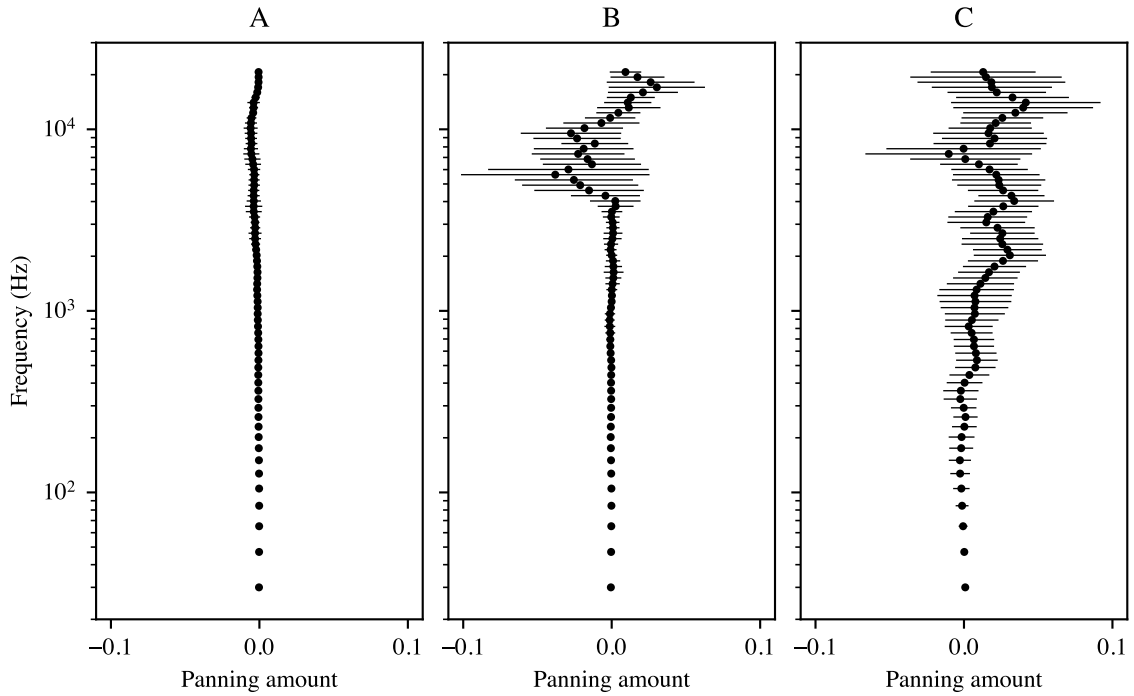


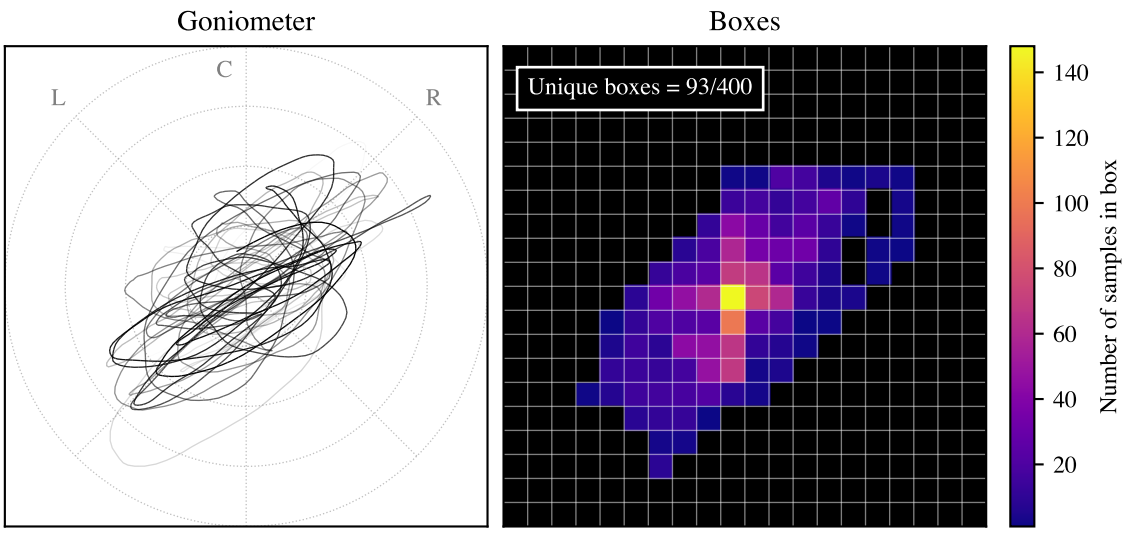
Figure 4.5. Visualization of ASF the three test tracks. For demonstration purposes, only a subset of the calculated features are shown. The amount and variation of panning is visible in fine detail across the frequency spectrum.

and right channel signals. Figure 4.6 shows how an audio signal can be converted to goniometer features.

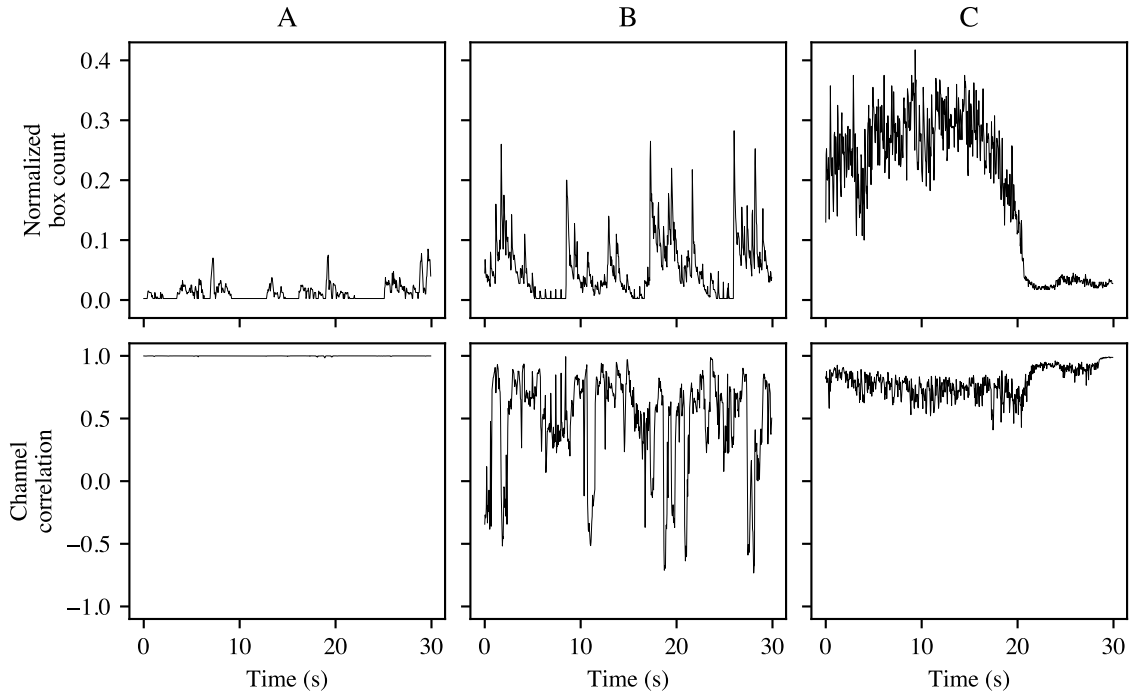
Other techniques

Härmä's work on the classification of stereo audio in [13, 14, 64] presents a slightly different take on the problem. He proposes a signal model where the stereo mix is formed from a combination of a single centre source, a significant amplitude-panned source, plus left and right uncorrelated sources. He subsequently presents a list of eight pre-defined classes representing various combinations of those sources. Furthermore, he shows that stereo recordings can be classified into six out of eight of those classes by estimating the variances of the underlying source signals. The classes are visualized as areas on the surface of a unit sphere, with each dimension representing the variance of the three source types, as shown in fig. 4.7. It should be noted that Härmä does not give a method for the classification of the final class – a mixture of all three source types – deeming it too complex to detect. In the context of stereo image description, Härmä's work is somewhat different from the others since the descriptions (i.e. the eight classes) are pre-defined; The main innovation is the method to assign an audio signal to a class.

While all the aforementioned studies focus on descriptions of amplitude panning in stereo mixes, recording engineers can make use of phase-based techniques to add spatial feeling to their mix. Tardieu et al. also suggest a measurement for these named Phase Stereo Features (PSF) [5], underpinned by an STFT spectrogram derivative named Spectral Stereo Phase Spread (SSPS). While the idea of measuring these common production effects has merit, I unfortunately could not confirm that it worked in practice because the method to calculate SSPS as defined by Tardieu et al. produced spurious results for all recording that I tested.



(a) Example goniometer reading for a 40ms segment of recorded audio. The left plot shows an original goniometer as the producer would see in their DAW. Darker lines represent audio that is more recent. The right plot shows the boxes that the lissajous figure passes through, counting the number of samples in each box.



(b) Goniometer features for the three example tracks. Each track clearly shows a unique profile in how the features vary over time.

Figure 4.6. Example goniometer features as defined by Ziemer [9].

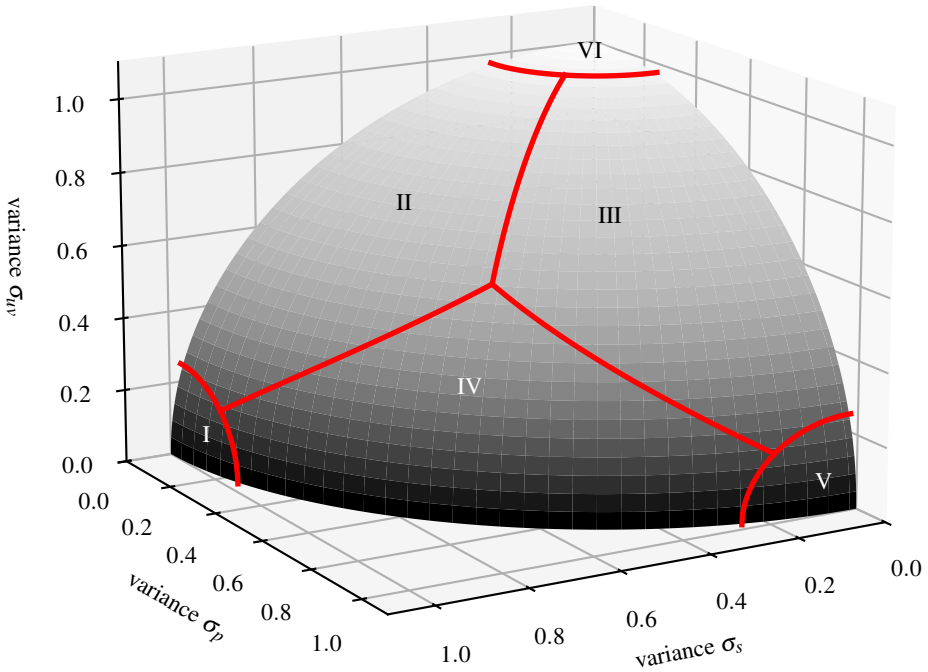


Figure 4.7. Visualization of six stereo mixture classes, recreated from [64]. The axes σ_s and σ_p represent the variances of the centre source and panned source, whereas σ_{uv} represents the covariance of the left and right uncorrelated sources. Classes are indicated as Roman numerals on the sphere's surface and separated by red dividing lines. Class 0 is silence and is hence trivial to detect. Class VII is cannot be detected by Härmä's method.

4.3.2 ML-based SIPs

All the previously mentioned algorithms use classical DSP to create their spatial descriptors, but more recently there have been efforts to use Deep Learning (DL) to disentangle spatial information from audio content. Yang et al. presents one such example in [75]. While the original aim was to upmix stereo music to multichannel, a necessary component of the resulting neural network was a latent space that represents the spatial information of the multichannel signal. A simplified architecture diagram of Yang et al.'s method is shown in fig. 4.8.

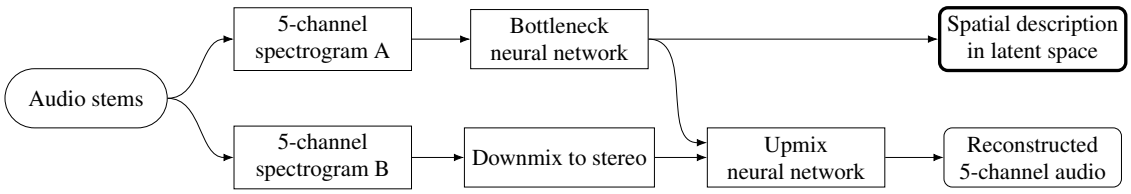


Figure 4.8. Architecture digram of Yang et al.'s neural network that learns a spatial representation of its audio training data. The latent vector (shown in a thick outline) can be used as a description of the spatial image. Spectrograms A and B use the same stems but different panning parameters. Recreated from [75].

DL has also been used outside the music domain to e.g. learn spatial relationships between individual microphones in an array [76]. While such tasks do not directly address the research questions in this thesis, they demonstrate that learning spatial features on large amounts of unlabelled audio data is possible given the appropriate methods and pretext task.

Neural generation of spatial audio also continues to be a hot topic up to the present day, with many attempts to generate high-quality binaural, ambisonic or multichannel audio from mono sources [77–83]. In doing so, the networks must infer spatial information from a source other than audio (typically an accompanying video or text prompt), and in the process will learn the spatial qualities of their audio training data.

4.4 Prediction of perceptual scalars

A perceptual scalar is a measurement that would usually be made via a perceptual listening test. Berg and Rumsey give a long list of typical perceptual measures that can be derived from the low-level cues [84], although later literature has focused on Apparent Source Width (ASW) and Listener Envelopment (LEV) [85]. Listening tests are known to be slow and costly [86], so prediction of the perceptual scalars via objective means is particularly useful to researchers as it allows for faster experimentation.

ASW is one of the more common values that has been addressed in the literature. The early work of Supper [87], Mason [88], and Dewhirst [89] detail the construction of 'binaural analysers' that are able to estimate ASW. Dewhirst compares several methods to estimate ASW given ILD, ITD, and IACC cues from binaural audio signals. The prediction results were promising, but mixed. Dewhirst even notes that the original values of ASW given by test participants were somewhat varied, concluding that the test itself was a difficult task.

Since these early experiments the analysis of ASW has remained a topic of interest, with Käsbach et al. [90], Mieth and Zölzer [72], and Antoniuk and Zieliński [91] all proposing ways to calculate it. It is not restricted to the binaural domain, with Sarroff and Bello [4], Wilson and Fazenda [92], and Mourgela et al. [93] also proposing methods for generic stereo signals.

LEV on the other hand has prompted slightly more varied discussion over the years. An early study by Soulodre [94] compared several objective measures for it. The methodology was to fit a correlational model that predicted the output of perceptual test from objective audio measures. The objective measures were based on either the IACC or Lateral Energy Fraction (LEF) of the audio signal. Soulodre notes that the late-arriving sound (more than 80 ms after the sound's onset) are most relevant to LEV, regardless of which objective measure is used. Sunish et al. [95], Power

et al. [96], Jackson et al. [97], and Nowak and Klockgether [98] later expanded the same concept by fitting models on different objective audio measures.

Dewhirst et al. propose a different measure in [99] which involves calculating the area covered by the plurality of individual sound sources after estimating their directional localization. Although, by the authors' own admission, this is a crude measure and was not validated by perceptual tests.

Notably, the prediction of LEV appears to be used more often to assess spatial audio reproduction (sometimes rendered to binaural), rather than plain stereo. Power et al. [96] even include both mono and stereo reproduction in their experiment, concluding that their subjective ratings for LEV were considerably lower than all spatial formats. Sunish et al. [95] also state that LEV might not be cleanly separable from ASW.

Within the context of this thesis, the absolute accuracy of any prediction is not necessarily of primary importance. Rather, it matters that the value calculated for each recording in a corpus provides a suitable comparison to the other recordings. Mourgela et al.'s work on trends in music production [93] is one example of such a study. The study groups raw stereo width calculations into four buckets, thereby showing that finding recordings with similar stereo width is of more interest than the absolute value.

4.5 Spatial quality comparison

The SIPs described in sections 4.2 to 4.4 are often used to assess the spatial degradation of an audio signal after some sort of distortion has been applied. A typical method is to calculate the same spatial descriptor on both a reference and degraded signal, then calculate the difference in those descriptors. There is sometimes an additional aggregation step to reduce a multidimensional descriptor down to a scalar.

The types of distortion vary, although a major industrial use is in the development of lossy codecs. The work of You and Cheng [100], Wang et al. [101], Delgado and Herre [102], and Delgado and Herre [103] do exactly this. Figure 4.9 shows a typical workflow for the measurements of spatial distortion used in all these studies.

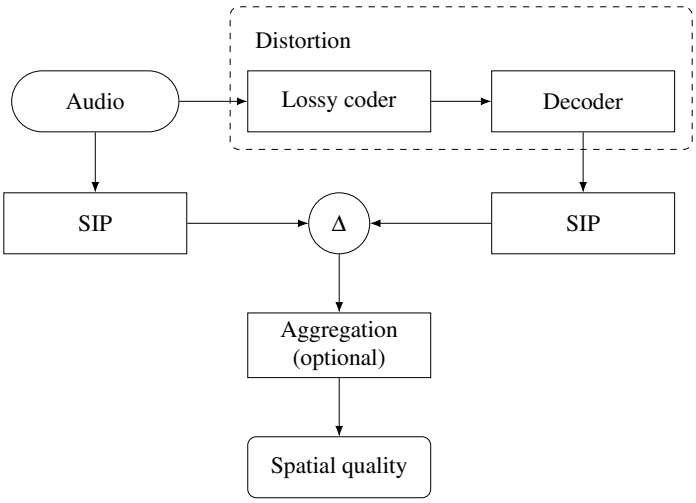


Figure 4.9. Example architecture to measure the spatial quality of a lossy audio codec. The two SIP blocks must use exactly the same method, although the choice of method is application-specific. The 'Distortion' block may be replaced when using the workflow for other tasks.

Spatial distortions other than lossy coding have also received attention in more recent years. For example, Barry et al. present a comparison of binaural rendering and downmix algorithms (as well as lossy codecs), showing that minute differences in binaural cues can be easily detected using their methods [104]. Taking the opposite approach, Watcharasupat and Lerch demonstrate

a new metric for assessing spatial quality and demonstrate it using several common, but artificial distortions [105].

In all the above studies, the detail and implementation of the spatial descriptors vary. Some studies (e.g. [103, 104]) opt to use high-level SIPs such as 'directional loudness maps' or ILD histograms, while others (e.g. [101]) use low-level spatial cues such as ILD and ITD. In short, both approaches are valid and relevant to the RQs presented in this thesis.

Following from this establishment of objective measurements for spatial quality, it is increasingly common practice to use them as additional terms in DL loss functions, thereby allowing the networks to learn spatial features. Erruz [106] demonstrates a loss term based on the ILD of binaural audio, whereas both Lim and Nam [77] and Giampiccolo et al. [107] implement an Inter-channel Level Difference (ICLD) loss term for various combinations of ambisonic channels. In all these studies, the underlying industry motivation was the generation and rendering of new spatial audio formats, similar to the upmixing research mentioned in section 4.1.

4.6 Analysis of large musical datasets

Many of the studies cited in section 4.3 use a large dataset of recorded music to demonstrate the effectiveness of their SIPs. For example, [3, 5, 8] tackle genre recognition, [32] predicts artist or DJ, and [5] determines the decade of the recordings. Similar studies such as [108, 109] show that the geographic origin of a recording can be determined from audio features, although they don't include any spatial features.

Ziemer et al.'s continuation of their work on goniometer features (see section 4.3) uses a dataset comprising songs that were either produced or rapped by specific artists [110]. They train a Self Organising Map (SOM) on the goniometer features of the dataset. The results firstly show that specific artists have specific sonic profiles, and furthermore that when those artists collaborate, the sound profile of the resulting track is dominated by the producer.

Collins et al. collated a dataset of historical electronic music [111]. As part of that work, they extracted two spatial metrics: 'channel loudness difference' and 'stereo spatial ebb'. The example experiment given to demonstrate the dataset showed that 'stereo spatial ebb' changes according to the year of the recording, thus indicating an artistic trend over time. 'Channel loudness difference', on the other hand, remained static.

In Panteli et al.'s work on the classification of singing style [112], the classes – instead of being determined by known labels – are learned by first extracting low-level audio cues, then finding common features within those cues. The features are then used as the input to unsupervised clustering algorithms to determine the relationships of the recordings to each other. Panteli et al. continued the work in [113] where outliers in the global dataset where found using similar techniques and links between different musical cultures were drawn. While neither of these studies addressed spatial information, they nevertheless present use cases where unsupervised learning on features extracted from large audio dataset generates cultural knowledge that extends beyond the musical domain.

On a cautionary note, Chauhan and Pundhir recently showed that ML-based tasks taking audio input from multiple cultures are biased towards Western sources if they use the Mel spectrogram as the audio frontend [114]. The authors recommend using alternative frontends, such as ERB.

What all these studies have in common is an interest in the dataset itself regarding both the relationships between individual recordings, and the relationship of the entire corpus to the wider world. Low-level audio features are of lesser importance, other than to construct the high-level metrics that enable comparison between recordings.

5. Research Methodology

In this chapter I break down the reasoning behind the RQs given in section 2.1, both from philosophical and practical standpoints. I argue that the methods I chose for the research in this thesis allow for a timely and achievable answer to the overarching RQ.

The overarching RQ – 'How do music producers use stereo field' – could be approached in many ways. However, the work in this thesis is steeped in the context of large audio datasets. Indeed, the reason for performing the research at the current point in time is the ease of access to said datasets along with the availability of computer processing power to analyse them. From an ontological standpoint this most closely fits the Parmenidean viewpoint of *being*. According to Gray, Parmenidean ontology posits that reality is composed of clearly formed entities with identifiable properties [115]. If the producers' collective musical output is 'reality', then the recordings in the datasets certainly fit that description.

The two smaller RQs represent different epistemological perspectives on the overarching one. RQ_1 is based in *constructivism*, where the meaning of the stereo field is imposed on the recordings by the listener. Conversely, RQ_2 is rooted in *objectivism*, where the properties of a recording's stereo field are an objective truth that can be measured. Not only that, but generalizations about the use of stereo field across the entire population of recordings are also objective and measurable. While the two epistemologies are somewhat adverse to each other, they both fall under the ontology of *being* [116].

Under *constructivism*, one valid research methodology is 'heuristic inquiry' – the act of totally immersing oneself in the phenomenon being researched and allowing intuition to guide the understanding of the RQ. I used this technique for a sizeable part of the work described in this thesis. For example, paper I is a retrospective account of building a new musical instrument where the primary sonic material was the signal from a remote stereo microphone. The act of building the instrument, listening to its output, and forming ideas about which sonic phenomena were interesting was a direct way to relate the stereo field with musical knowledge, addressing RQ_1. While this provides a basis for other research, critics of heuristic inquiry will point out that it is deeply personal and therefore not particularly generalizable [115].

To complement the *constructive* approach and form more robust conclusions about an entire population of music producers, I concurrently embarked on a more objective research path. The epistemology of *operationalism* is defined by the principle that in order to understand something, it must be measurable [117]. Thus, to understand stereo field, a set of operations to measure it is necessary. In practice, that means a software library capable of analysing music recordings and extracting information about their stereo field. The benefits of creating this software are threefold. Firstly, it enables comparative measurements of disparate music recordings, allowing one to validate and challenge the existing research described in chapter 4. Secondly, the methods used for any experimental research must be controllable and repeatable. A software library is a fantastic way to achieve this. Thirdly, the act of writing the library can be seen as a similar act of immersion to that previously described in the *constructive* research. By breaking down the measurement of stereo field into simple instructions, one is forced to think about quantifiable relationships between audio phenomena. Wiltschko argues that language is a tool for both thought and communication in equal measure [118]. It follows that computer code (which is a written language like any other) acts as both a representation of the creator's thought process, and the communication of those thoughts to any readers. The software written for this thesis was therefore fundamental to the research process, as well as underpinning the objective results of paper III.

This combinatorial approach is not a new idea. The 13th century philosopher Nicolaus Cusanus spoke about two foundations of knowledge - 'ratio' (that which you measure) and 'intellectus' (the relationship to the unknown) [119]. It captures the dichotomy that when measuring

a phenomenon, more measurements will give one a more accurate picture of the truth, but one must have the forethought to imagine that the measurements might not be able to represent the entire truth. Cusanus argued that both are necessary to make sense of the world, and modern academics propose that as our world is increasingly measured and rationalized, intellectus becomes increasingly important [120].

One potential criticism of this combinatorial approach is that an answer to the overarching RQ could instead be gained by gathering interview or observational data (using Yin's taxonomy of 'evidence sources' [121]). As Gray's assessment of the strengths and weaknesses of each evidence type states [115], the former (interviews and observation) result in deep, contextually rich data, but are time-consuming to collect and subject to biases from the act of interviewing or observing itself. More specific to the music community, there is an inherent understanding that music is a form of language [122]. Adding a layer of indirection (interviews conducted in spoken language) only serves to muddy the resulting data. Indeed, Hospers argues that an artist's expression is inherently intertwined with the creation the original art form, noting that the intentions are sometimes so complex and multifaceted as to defy explanation even to the artist themselves [123]. In contrast, the musical datasets that we are presented with would be classified as 'archival records' or 'physical artefacts', which according to Gray are precise and technically insightful, but prone to selectivity and reporting biases [115].

I argue in this thesis that such biases can be partially mitigated by combining research from multiple epistemologies. While *objective* research can determine what music producers do from a technical perspective, *constructive* research can offer insight into the choices and trade-offs they made to end up at their final artistic output. The work I have performed for this thesis is a combination of immersion in musical practice, thorough reviews of existing literature, and detailed implementation of audio DSP analysis techniques. This combination presents a well-rounded viewpoint on the original RQs, as well as giving rise to new questions that could be answered in the continuation to doctoral studies (see section 7.1).

6. Paper summaries

Paper I

'Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications'

Authors
Austin Franklin, Richard Mitic, Daniel Hedin, Rikard Lindell and Henrik Frisk

Status
Published in *Journal of The Audio Engineering Society*, Mar. 2025 [124].

Research methods
This paper presents a retrospective of a project to build a novel musical instrument for a distributed audio performance, sending stereo audio over mobile networks using the WebRTC standard protocol. It follows the practice-based research methodology as defined by Candy [125].

Results and conclusions
The research journey documented in the paper serves as both a map and proverbial 'early warning system' for future researchers or artists who are undertaking similar audio projects. It highlights that while focus might initially be on the sonic material, practical and technical issues have the potential to become disproportionately engaging. One particular conclusion was that network reliability was a limiting factor of the audio performance, sometimes leading to a degraded spatial impression in the transmitted audio. This addressed RQ_1 in proving that the technological context surrounding the audio signal itself can make an equally large impression on the stereo field as the musical content.

My contribution
Member of the team that conceptualized the instrument and made the initial survey and prototyping of existing audio transmission technologies. Additionally responsible for the post-prototyping of alternative network approaches and custom audio software.

Paper II

'A Systematic Literature Review of Methods to Describe the Stereo Image of Recorded Music'

Authors
Richard Mitic and Andreas Rossholm

Status
Accepted to *Journal on Audio, Speech, and Music Processing*, May 2026

Research methods
SLR following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) standard to gather all currently available methods for determining spatial features of stereophonic musical audio.

Results and conclusions
Addressing RQ_2, the review revealed a comprehensive set of techniques that can be used to make spatial descriptions of stereo music, encompassing low-level spatial cues, high-dimensional descriptors, and predictions of important perceptual values. Despite the high number of studies that were assessed, there were relatively few that were 'very relevant' to recorded music specifically. The SLR also revealed that while ML/DL have been used on spatial features in the past, both *unsupervised* ML and very large music corpora remain fairly unexplored. It also details a framework for how to create a SIP algorithm that is useful for any given dataset and use case. It acts as a one-stop-shop to springboard any future research into stereo field.

My contribution
Main author, responsible for all data collection and processing, and writing.

Paper III

'Towards an Objective Comparison of Panning Feature Algorithms for Unsupervised Learning'

Authors
Richard Mitic and Andreas Rossholm

Status
Published in *Proceedings of the 28th International Conference on Digital Audio Effects (DAFx25)*, September 2025

Research methods
Several DSP techniques for determining audio spatial features were performed on a small dataset that had predictable musical characteristics. The different techniques were compared using a new objective metric.

Results and conclusions
The paper revealed that although there are several DSP techniques specifically designed for the extraction of recording and panning features, they are all limited in their own ways. One particular technique (SPF from [3]) was particularly promising as a candidate for use later in the doctoral phase of studies, and the others could be used with some modifications. Overall it highlighted the importance of choosing both a SIP and a distance metric that are appropriate to the overarching task, thereby addressing RQ_2.

My contribution
Main author, responsible for all data collection, experimental design, and software implementation. Presented the paper at the 28th International Conference on Digital Audio Effects in Ancona, Italy.

Paper IV

'Perceptual Evaluation of Binaurally Panned Speech and Music Mixtures'

Authors
Richard Mitic, Majed Elwardy, Veronica Sundstedt, Andreas Rossholm and Yan Hu

Status
In review at *Journal of The Audio Engineering Society*, August 2026.

Research methods
A perceptual test was used to determine listeners' reactions to carefully constructed audio stimuli. Standard statistical analysis was performed on the raw test results.

Remaining Work
Await peer review and address any comments.

Results and conclusions
The paper concluded that when listeners are presented with voice overlaid on top of music, the best panning position for the voice was not always in the default centre position. The data showed that test participants held strong individual preferences regarding which panning position sounded best, although the demographic data collected on them did not explain their preferences. Counterintuitively, the data did not reveal any link between the music track and the preferred panning position. We speculated that performing a much wider study using more music tracks with more controlled vocal elements might reveal more.

My contribution
Joint main author alongside Majed Elwardy. Contributed to experiment design, data analysis, software implementation, and writing.

7. Discussion

As mentioned in chapter 5, the work listed in this thesis happened in the context of easy access to vast music datasets. This access certainly influenced both the methods I used for each paper, and the research questions themselves. One can envision a quasi-mirror-image research process where instead of beginning with a research question and collecting data to find an answer, one must ask *'what knowledge can I pull from the available data?'*.

It was precisely this setting that made the combinatorial philosophical approach detailed in chapter 5 so suitable. By immersing myself in the world of stereo image, the research questions germinated, evolved, and eventually coalesced into their current form. The need to consider what kind of knowledge was hiding in the data spurred a downstream need to understand what musical knowledge the producers and musicians could have imparted via their creative process. RQ_1 was the outcome of that thinking. By contrast, RQ_2 was a relatively straightforward concoction once RQ_1 started to materialize. One could even consider RQ_2 to be part of the answer to RQ_1.

The fact that I focused on stereo image was somewhat of a practical choice. Firstly, it had a handful existing industrial use cases but potential for more. Secondly it was relatively understudied in previous literature - certainly not at the scale that is possible today. It became apparent from reading said literature (see chapter 4) that while the original authors were able to use their analysis methods for specific tasks, there was little discussion of what audio features the algorithms were able to pick out. In effect, each study had its own view of an objective reality, making it impossible to compare their results. Likewise, there was precious little discussion of the global gamut of stereo image features in real recorded music. The work in this thesis firstly highlights these limitations to the MIR community, and secondly forms a computational basis that can be used for further, consolidated research into stereo field.

With that said, we must now address the proverbial elephant in the room. Despite what one might imagine from the title of this thesis, the included papers do not deal with 'vast repositories of recorded music' directly. Rather, they all focus on understanding what is possible to find in those datasets, and subsequently how to find it. I do not consider this a contradiction in terms, but rather an optimistic indication that answers to the two minor RQs contribute to answering the overarching RQ as well as exposing new knowledge gaps that require more research.

Previous research into stereo field generally followed a simple pattern: A novel algorithm is proposed, calculated on a mid-size music collection, and then used to predict labels that were known a priori. The gap I identified was a lack of understanding into what musical features the existing algorithms have the ability to recognize. Once that is established, it would be possible to examine the available datasets with the intention to understand how a huge population of creatives choose to express themselves though their music. That research will reveal the scale and variation of stereo techniques that musicians are able to employ in their work. Given the basic knowledge gained from the work in this thesis, it might then be possible to reason about how musicians and producers sculpt their stereo fields, relating it both to the musical content and the wider context in which it was created.

7.1 Future work

The work documented in this licentiate thesis provides the theoretical and practical basis for analysis of stereo image in music recordings. The focus was on amassing the signal processing techniques that enable the analysis of stereo field, and testing their applicability to various types of recordings where were already known to interesting examples of stereo field. The next stage of research will involve applying the knowledge from this thesis on much larger musical corpora. The signal processing techniques will be honed and run on vast musical datasets, secure in the knowledge that they will generate data appropriate to the experiment at hand.

During the licentiate, several points arose that provoked further investigation. Firstly, the available public datasets of recorded music are generally encoded using lossy compression. While lossy compression algorithms undergo perceptual tests in their development, it is unclear how they affect spatial cues at the scale of said datasets (which encompass much musical variation). The spatial effects of lossy compression therefore warrant further investigation. Secondly, the public datasets are generally formed from online services where musicians have uploaded their own recordings for free. In contrast, professional recordings (which are much more numerous) are not included in any public datasets. Since professional recordings make up the vast majority of recorded music, it needs to be assessed whether public datasets are representative of professional recordings.

In short, the doctoral research will tackle a larger problem of assessing human musical creativity though the recorded output. The techniques surfaced by the licentiate will provide the foundation, and will be applied to real-world data.

A. Demonstration Audio Files

Demo tracks

These tracks from the FMA dataset [7] were used to generate diagrams of spatial cues and SIPs in chapter 4.

Track ID	Artist	Title	Genres	Letter
123899	Rachel Mason	Rebel Angels	Singer-Songwriter	A
117614	Blue Dot Sessions	Red City Theme	Soundtrack, Instrumental	B
122809	The Carnys	foolsgame	Experimental Pop	C

Freesound files

These sound clips from Freesound [129] were used to generate diagrams in chapter 3.

Sound ID	Name	URL
581451	kick	`https://freesound.org/people/johnnydekk/sounds/581451`
581473	hihat	`https://freesound.org/people/johnnydekk/sounds/581473`
31912	speech	`https://freesound.org/people/acclivity/sounds/31912`

B. Spectrogram comparison

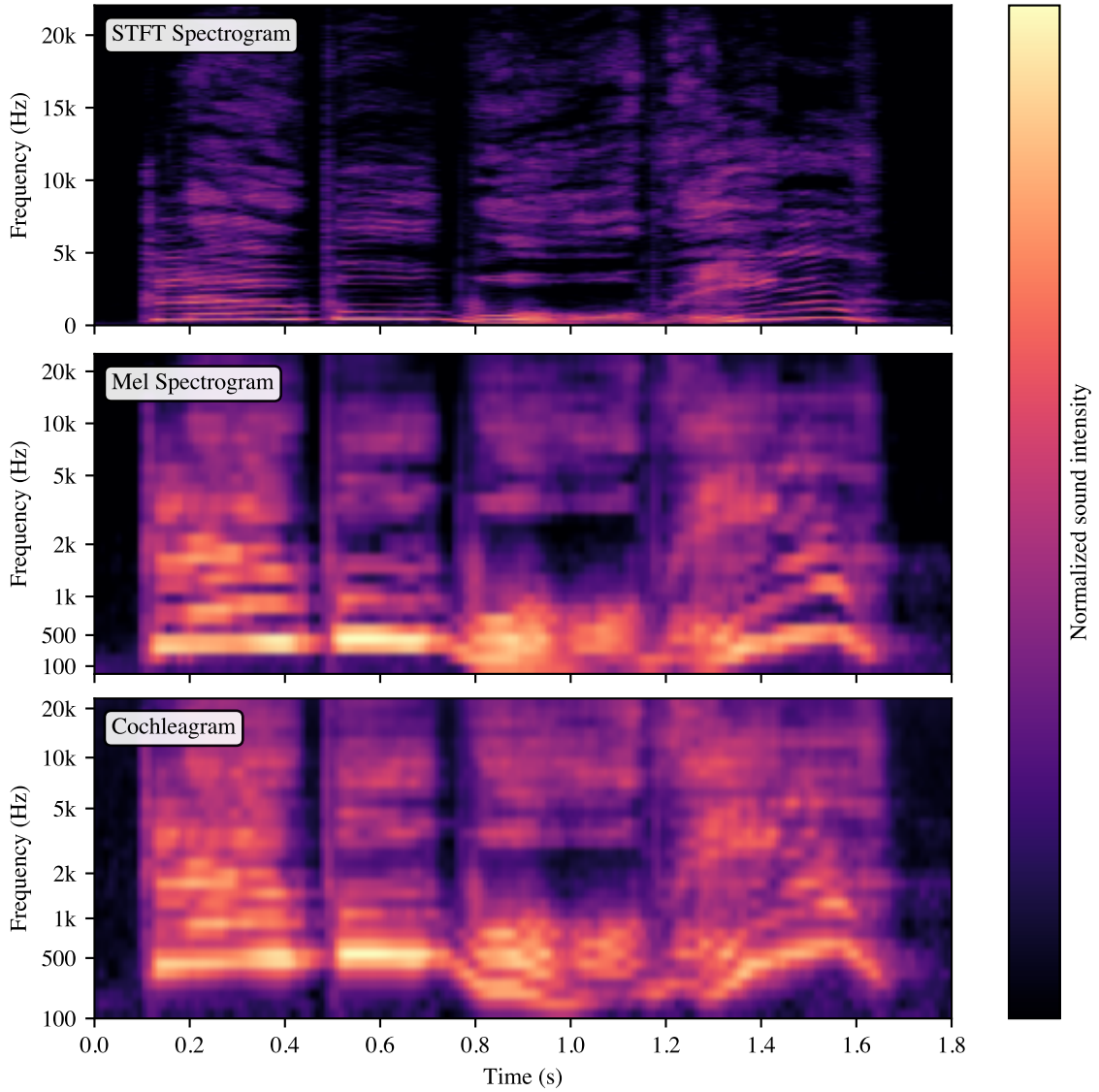


Figure B.1. An example audio signal displayed as STFT spectrogram (above), Mel spectrogram (mid), and cochleagram (below). The spectrogram frequency bands are spaced linearly between 0Hz and the Nyquist frequency, which is not coherent with human perception. The Mel spectrogram is created from the original spectrogram based on linear combinations of its frequency bands and retains more detail in lower frequencies than higher frequencies. The Cochleagram is created directly from time domain audio by mimicking the physical response of the human ear.

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Part II

Included Papers

A. Franklin, R. Mitic, D. Hedin, R. Lindell, and H. Frisk,
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Development of a Web-Based Real-Time Distributed Reverberation Chamber for Live Concert Applications

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This paper documents the challenges encountered during the development of a mobile web-based, real-time distributed reverberation chamber for a live concert setting. Initial efforts focused on leveraging existing open-source and commercial solutions, but these proved inadequate due to limitations in network compatibility, audio routing, and mobile device support. WebRTC emerged as the most viable option, offering peer-to-peer communication and firewall traversal capabilities. However, significant obstacles persisted, including network reliability, audio hardware/software ecosystem limitations, and debugging complexity. The project highlights the fragmented nature of real-time audio streaming technologies and the need for more transparent, reliable, and well-documented tools. Despite these challenges, the authors developed a functional web application prototype using WebRTC, Web Audio API, and custom Session Description Protocol modifications, offering insights into the complexities of deploying real-time audio systems over mobile networks. The paper concludes with reflections on speculative approaches, such as decentralized audio routing and native app development, and calls for improved documentation and standardized debugging tools to support future innovations.

0 INTRODUCTION

The authors began a project in February 2024 to create a simple proof-of-concept: distributing a live reverberation chamber as a stereo audio signal over a mobile network from a concert hall to up to three remote locations and back to the original site, where an audience would experience the distributed performance alongside a live music event. The end goal was to create a merged reality for the audience and to explore their experience of the site-specific aural qualities of the remote locations merged with those of the concert hall. The prototype, named Auxtrument, was officially used in concert in November 2024 following a successful demonstration at the 2024 International Conference on Quality of Multimedia Experience [1].

It was anticipated that this project would be a relatively straightforward endeavor, given the established use of network-based tools for collaborative performance. Available open-source and commercial solutions were initially explored, but each proved inadequate for various reasons. The authors' efforts extended to experimenting with various protocols and network stacks, leading them to settle on Web Real-Time Communications (WebRTC) as the most viable option. WebRTC is marketed as a reliable, standardized solution that appeared to address the core challenges faced: seamless real-time communication across devices, browsers, and networks [2]. However, even after adopting WebRTC, significant obstacles persisted, particularly in three key areas: (1) network compatibility and stability, (2) limitations within the audio hardware and software ecosystem, and (3) debugging complexity and software fragmentation. Many of these challenges remained unresolved throughout the project, contradicting WebRTC's reputation as a mature and reliable technology.

Boem et al. state that platforms based on WebRTC were not originally designed for networked music experiences but rather as "environments where a single data stream...

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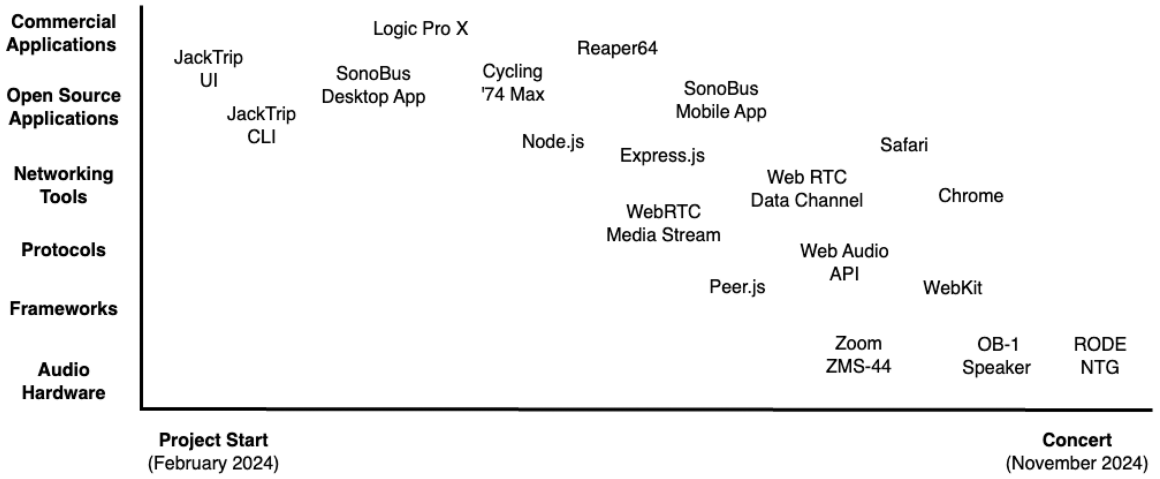

Fig. 1. The timeline of development for the Auxtrument.

is broadcast to all users simultaneously" [3]. Additionally, they suggest that overcoming these limitations using custom servers and protocols, while possible, remains challenging. These issues raise critical questions about the reliability, scalability, and practical limitations of these tools—where things like frequent version updates and deprecations can be difficult to track across multiple web stacks and may render software components, or even entire applications, unusable. While the Auxtrument application provided a workable solution, the development and testing process exposed significant gaps in documentation, unpredictable "black box" network behaviors, and difficulties in achieving consistent performance across different devices and environments.

Fig. 1 shows the development. Each item represents a tool, protocol, or software/hardware component laid out in the order in which it was experimented with. Given the aims and timeline of the project, a proper state-of-the-art investigation into the full gamut of mobile web audio streaming technologies was not conducted. Given the reputation of WebRTC as tried-and-tested software, and the saturation of open-source and commercially available audio streaming solutions, it was not reasonable to assume that many of the problems encountered were not already solved. Instead, the journey is documented, highlighting motivations and processes, aiming to shed light on the broader complexities of deploying real-time audio systems over mobile networks using exploratory and practice-based methods.

1 METHODOLOGY

Exploratory research, as described by Stebbins, emphasizes open-ended investigation into uncharted territories, allowing for documenting and reflecting on unexpected outcomes during the process [4]. Practice-based research methods, in which creative practice serves as both the process and outcome of research, are also applied, generating new insights through artistic exploration and critical reflection [5, 6]. To distinguish this work as practice-based, the methodology was structured to align with the criteria proposed by Candy [5]:

1. Define the objectives and research question(s).
2. Document the tools, setup, and actions used.
3. Analyze the results, noting the challenges.

This research began with an artistic question: "How can the real-time distribution of site-specific acoustic reverberation between a concert hall and multiple remote locations create a merged reality audience experience?" However, as development progressed, a series of challenges was encountered—including network reliability, limitations in existing audio ecosystems, and debugging complexities—and the focus necessarily shifted. In response, the research question evolved: "What are the practical and technical constraints of implementing a mobile, real-time, distributed reverberation system for live performance using existing audio streaming technologies?"

To investigate this question, three objectives were established for achieving a high-fidelity user experience: (1) the audio signals must be high-resolution to preserve the inherent quality of the transmitted sounds and accurately relay spatial properties of the acoustic environments, (2) the system should be easy to set up and use and work over different types of networks, and (3) to support use in any location, it must be compatible with mobile devices. This framing allowed for critically assessing available tools and infrastructures as active conditions that shape the artistic possibilities of the original research question. In writing this paper, the authors set out to retroactively analyze and better understand the causes behind several of these challenges. The findings are documented here in the belief that they contribute to a broader understanding of how current technologies shape the possibilities—and limits—of mobile networked music performance, while also illuminating the assumptions embedded in their design and use.

2 SURVEY OF TECHNOLOGIES

In evaluating potential software solutions, hands-on tests were conducted with a variety of audio applications, focusing primarily on their networking capabilities and suit-

ability for real-time audio transmission of multiple channels and audio routing capabilities across different platforms and clients. Given the goal of integrating mobile devices with the Auxtrument, applications with mobile support were also explored. However, mixer apps on mobile phones were not tested, as these typically require custom plugins for advanced routing and signal processing—something that mobile operating systems restrict. As the authors are primarily composers, sound artists, and researchers, they explored solutions well-known in their respective field(s), and the list of software is by no means exhaustive. These findings guided the decision to further explore alternative networking approaches for real-time audio streaming.

2.1 Existing Audio Applications

Although the aim was to have Auxtrument work on mobile networks, initial tests involved using JackTrip [7] and SonoBus [8] on MacBook laptops. Tests started with laptops and Wi-Fi connections, anticipating this would be a quick and practical first step in prototyping while allowing for evaluating the software. Although JackTrip's user interface (UI) was straightforward, it lacked the flexibility needed for audio routing between clients, sending the same audio stream to all participants without the ability to control volume or mute channels on the host side. It also did not have an implementation for mobile phones. These limitations made it unsuitable for the authors' purposes. SonoBus, on the other hand, addressed this issue by offering a mobile app and multichannel input/output capabilities, allowing users to specify channel destinations.

The authors attempted to connect using SonoBus over a mobile hotspot using the Swedish Tele2 service provider, with one laptop connected via eduroam. Eduroam is a global network roaming service for students and employees in academia [9]. Its authentication requirements are standardized across all networks, but individual providers can and often do add additional security features. The setup using SonoBus with Tele2 and eduroam proved unsuccessful. In contrast, connections between two laptops granted eduroam certificates worked, and even certified mobile devices connected to eduroam were able to stream audio via the SonoBus iOS app without issues.

These initial findings led the authors to suspect that the security features implemented at Mälardalen University and The Royal College of Music in Stockholm, where eduroam networks are accessible, were blocking peer-to-peer (P2P) connections from external networks. A major obstacle was SonoBus's lack of options to specify ports or other network settings, making it ineffective when dealing with firewall restrictions. This posed a significant obstacle, as eduroam was the primary testing network, and there was no reliable way to determine in advance whether an external network would permit a connection.

Digital audio workstations are powerful tools that feature a wide range of audio plugins. Reaper [10] and Logic [11] can host Apple's AUNetSend and AUNetReceive plugins, which are used to send audio between applications, hardware instruments, and more on a Mac [12]. However, these plugins use zero-configuration networking through Bonjour [13] and multicast Domain Name System service records to send audio over a local area network only. Other well-established live audio tools such as Cycling '74 Max have networking capabilities, limited primarily to control data via User Datagram Protocol and Open Sound Control protocols [14]. One workaround for audio streaming is to convert audio vectors to jitter matrices and send the video stream over the network. However, Max is not compatible with iOS.

Following the completion of the prototype, the search of existing audio streaming applications was broadened. The Virtual Rehearsal Room and a list of alternatives [15] were found. Virtual Rehearsal Room is based on Audio over Open Sound Control, the underlying protocol behind SonoBus, using Pure Data as a plug-in for the Mozilla browser [16]. These alternatives—including LoLa, NinJam, and Soundjack—reflect a range of design priorities and technical trade-offs. LoLa, for example, requires dedicated hardware and a stable 1-Gbps connection, typically only available on academic networks. While Internet service providers (ISPs) advertise such speeds, real-world performance varies. NinJam, bundled with Reaper, addresses latency by aligning performers to musical measures, sometimes increasing latency by one or more bars. Soundjack uses P2P networking but requires User Datagram Protocol port 50050 to be forwarded—an issue on networks like eduroam, where port forwarding is often blocked for security reasons.

What became clear through this review was that no single tool fit these needs perfectly. SonoBus remains the closest to meeting the criteria due to its cross-platform support and intuitive interface. Yet the inability to manually configure network ports or tunnel through more restrictive firewalls ultimately limited its utility in real-world settings. As a final thought, the process of evaluating these alternatives confirmed that while the existing ecosystem of remote audio tools is mature in some respects, it remains fragmented, with little attention paid to mobile-first, multi-endpoint collaboration under restrictive network conditions.

2.2 Existing Web Technologies

To create a realistic reverberation chamber effect, minimizing network latency was essential. Although a strict target was not set, the goal was to reduce the delay below the threshold of the precedence effect of approximately 40 ms for complex sounds such as speech and music [17]. Beyond this threshold, the delayed sound is perceived as a distinct echo rather than a single auditory event. Given this requirement, it was determined that P2P connections would be the most suitable approach.

Existing technologies and protocols were explored. For instance, an implementation was made that relies on WebSockets through Node.js by setting up a remote server and having clients connect to a central computer. The as-

sumption was that WebSockets, being a well-established technology, would provide a reliable foundation for audio transmission. However, a significant challenge was quickly encountered—WebSockets by themselves were unable to reliably connect two devices on disparate networks. Given these limitations, it was eventually decided that building a mobile web app using WebRTC would be the most feasible approach due to its native P2P and firewall traversal capabilities and broad browser support.

WebRTC relies on the clients negotiating a suitable way to connect to each other. The process of negotiation, known as signaling, takes place outside the WebRTC framework (e.g., using a custom-built signaling server based on WebSockets). The first step exchanges session descriptions in the form of a Session Description Protocol (SDP) offer that describes the type of connection (if it is video or audio), the codecs and codec parameters to be used, and more. The second step of signaling uses Interactive Connectivity Establishment (ICE) to establish a way for the peers to stream to each other.

ICE looks for the lowest-latency connection by trying to (1) establish a P2P connection using a Session Traversal Utilities for NAT (STUN) or Traversal Using Relays around NAT (TURN) server, (2) establish a direct Transmission Control Protocol (TCP) connection via the HTTP and HTTPS ports, and (3) fall back on an indirect connection via a TURN server. This means that even in cases where network address translation (NAT) prohibits a P2P connection using STUN, the TURN relay should act as a failsafe that permits the connection, albeit with increased overall latency, provided that it is explicitly configured to do so.

3 THE AUXTRUMENT AND THE NETWORK

From an implementation perspective, the final version of the Auxtrument is a web application based on WebRTC that is deployed using Express.js [18] running on top of Node.js [19]. The Peer.js library is used for establishing the connection [20]. Peer.js is a free library and service intended to provide easy-to-use WebRTC connections. On the client side, it provides a library that wraps WebRTC, allowing clients to establish connections using only an ID. This requires the use of a Peer.js signaling server that uses the IDs to automate the connection process. Peer.js provides its own signaling servers and the source code for the server to enable projects to host their own applications.

This stack helps streamline and simplify the number of actions a user must perform to establish a connection. Users simply navigate to the Uniform Resource Locator and select whether they are the central hub or location peers. The signaling and P2P initiation is automatically handled in the background, and the audio input and output devices are specified by each client in the UI prior to establishing a connection. The concert-ready version of the Auxtrument is limited to three remote clients and one central hub, but nothing prevents further expansion into more complex configurations.

3.1 Network Challenges

Due to the limited number of IPv4 addresses, most ISPs rely on some form of NAT or carrier-grade network address translators (CGNAT). This allows multiple clients to share the same external Internet Protocol (IP) address, thus conserving the address space. NAT and CGNAT typically prevent connections to clients to be initiated from the outside, which causes problems if two peers situated behind NAT or CGNAT want to establish P2P connections. Similar situations may occur when firewalls are used to protect local networks. The first priority of the ICE protocol is to find ways to establish P2P connections in the presence of NAT, CGNAT, or firewalls (e.g., using “hole punching”). Somewhat simplified, hole punching is based on indirectly manipulating the state of the firewall or NAT by initiating communication from the inside of a network in such a way that the initiator is exposed on the external network [21].

Experiments with the Auxtrument show the complexity of the network setup and identifying and mediating issues. Tests with Tele2 and Tre for example, some of Sweden’s largest mobile service providers, did not work with this WebRTC implementation whatsoever using STUN. Because of the fact that most of the testing occurred between eduroam networks, where the authors are based, and other ISPs, it was not immediately apparent that the mobile network itself, or combination of ISP and eduroam networks together, was the problem over firewall restrictions when used with laptops.

This led to tests with other mobile network providers in Sweden, primarily by visiting service provider stores, setting up some basic equipment, and running the Auxtrument using borrowed Subscriber Identity Modules. Based on these tests, it was discovered that Telia worked perfectly in all the situations where other providers previously had not. The frustration with mobile network providers is that there is no way to know exactly why Telia works and the others do not, in terms of how specifically each CGNAT is configured, and what penetration methods might be successful prior to experimentation. To complicate the matter further, the authors were able to get a connection working behind CGNAT, but via a fiber-based ISP.

The authors experimented with exploring the networks using traceroute, but the results were inconclusive. While there are other tools for mapping the architecture of a network, they were not pursued due to the complexity of the setup, in particular, the interaction between the IP layer and mobile network that, among other things, allows for roaming between carriers. As Kanaris and Pouwelse note, there are many different network configurations and NAT architectures that make penetration techniques difficult. However, they document an issue where both peers behind the same CGNAT cannot establish a connection using STUN since the server is outside of the intranet and only sees the “middle” network [22]. This explains, at least in part, some of the issues when testing with Tele2 mobile networks.

3.2 Webrtc Media Stream

WebRTC presented challenges with minimizing latency using its adaptive jitter buffer. The buffer automatically resizes when latency exceeds its capacity and does not reset unless the web page is refreshed. Setting the jitter buffer is "best effort," meaning that WebRTC may not honor the request and prioritize its own settings in cases where there are network interruptions. There are other not-so-well-known methods for minimizing latency, such as implementing a playout delay—a Real-time Transport Protocol header extension, which provides the sender's intent to the receiver on how quickly a frame needs to be rendered [23]. Community forums and bug reports show mixed reviews with playout delay implementation, with some suggesting the feature does not work with Chrome's Javascript API layer at all. This mirrors the authors' experience using this feature in Chrome.

3.3 WebRTC Data Channel

Using the WebRTC Data Channel was explored to overcome latency issues and have explicit control over the jitter buffer. By sending segments of audio data as bytes via this channel, the authors hoped to piggyback on its low-latency performance and then create an audio channel for playback on the other end. Initially, the latency was acceptable for this use case, between 10 and 40 ms; however, it would unexpectedly increase to around 20–30 s for brief moments of time. When this occurred, the audio was significantly affected, with clear indications of data loss. The authors were initially astonished: where do those packets live for such a long time? While the issue has been unable to be reproduced in a controlled manner, it seems to be easier to trigger on unreliable networks (i.e., mobile networks).

The authors hypothesize that it is a combination of network quality of service and WebRTCs use of Stream Control Transmission Protocol for reliable data channel messaging [24]. Specifically, when a transmitted message exceeds the maximum transmission unit of the network, it is segmented into smaller portions. Variable latency in each segment can be due to scheduling messages in the network stack, congestion, or differing processing times between network nodes, and this latency can easily compound. When packets arrive out of order, additional buffering, reordering mechanisms, or even retransmission may be needed.

Even though the channel can be configured to be "unordered" to reduce latency, lost packages still need to be retransmitted to preserve audio quality. It could be the case that high network congestion causes frequent retransmission, further exacerbating the congestion and data loss. It is assumed that quality of service plays a part in this process since WebRTCs media stream does not suffer from the same issue. Furthermore, there are unknowns about the ways in which mobile network topology contributes to latency, given that some tests with 5G encountered greater variance, while 4G seemed to provide more reliable results.

Moreover, when implementing the authors' own jitter buffer using the data channel, an unexpected issue was discovered where playback sounded like white noise rather than the intended audio stream, even when testing using a local area network. The issue occurred when the buffer size grew too large, with the exact threshold depending on the mobile device's browser running the node client. Strangely, setting the buffer past this threshold caused the client to misinterpret the type of data, specified as frames of 128 bytes, from an `ArrayBuffer` to an `UInt8Array`, causing the audio data to be in the range of 0–255 instead of the intended normalized range of –1 to 1. The WebRTC data channel effectively supports dynamic typing through Javascript, but it is not explicitly defined by the API itself. Instead, the data type is dynamically determined at runtime based on what is passed to `send()`, and the receiver must handle different types accordingly. The authors have no explanation as to why this process fails in their own implementation using the data channel.

4 THE AUXTRUMENT AND AUDIO

The Auxtrument uses the open-source and royalty free Opus codec that is preferred by WebRTC [25]. It operates with a sampling rate of 48 kHz to match that of most web browsers to prevent resampling at any stage in the transmission process, along with variable bitrate encoding. The recording for both the locations and central hub is handled using the web browsers' `getUserMedia()` javascript function of the MediaDevices API, which returns an audio stream from the selected audio input device.

At the same time, audio playback requires the Web Audio API. Once the audio has been received from all locations, the hub parses the channels from the streams of each location and merges them into a single multichannel stream. For instance, three locations with three interleaved stereo streams are merged into a single stream with six channels that is sent to a multichannel audio device and controlled via a mixer (Fig. 2). This process is performed using the splitter and merger web audio graph nodes [26]. The splitter transforms the interleaved stereo signals into separate mono signals, and then the merger node mixes all six mono signals for the left and right channels, respectively.

4.1 WebRTC Session Description Protocol

When setting up the aforementioned WebRTC connection, the protocol requires stream configuration information. SDP munging is required to configure a media stream to contain more than the default single channel [27]. The SDP is the standard that describes media communication sessions, and it contains the codec parameters, source address, and timing information of audio and video streams. For the Auxtrument, streaming stereo audio—to be diffused in the concert hall—was chosen as the minimum requirement for achieving an immersive experience, while remaining lightweight enough to minimize hardware demands and avoid potential bandwidth issues. Using WebRTC, streaming a stereo signal requires forcing the boolean `stereo = 1` flag to the description. However, the WebRTC documentation makes no mention of editing the SDP or specifying stereo transmission, and in hindsight, this makes

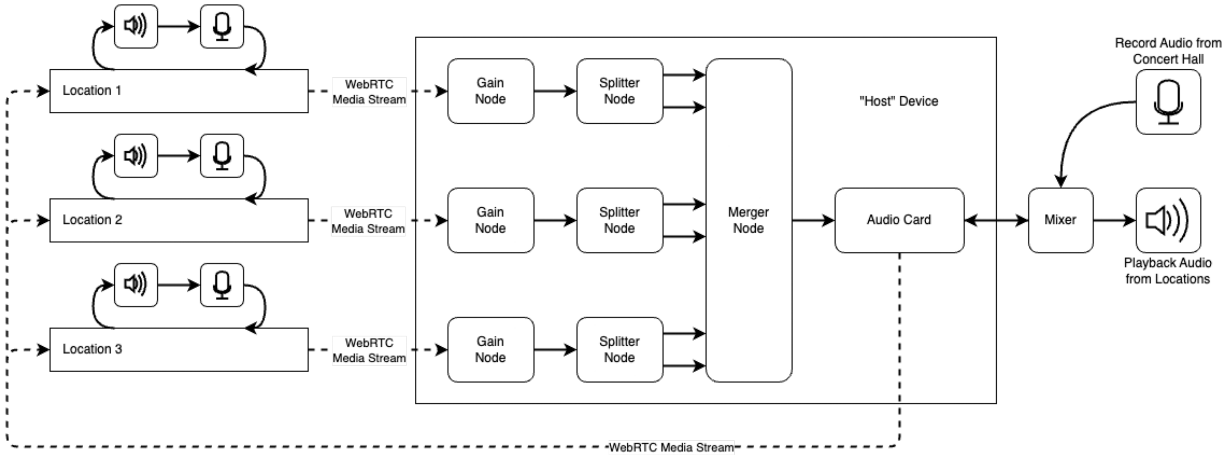

Fig. 2. WebRTC and Web Audio API between Auxtrument peers.

```
function modifySDPForStereo(sdp) {
  const sdpLines = sdp.split('\r\n');
    for (let i = 0; i < sdpLines.length; i++) {
      if (sdpLines[i].startsWith('a=fmtp:111')) {
        sdpLines[i] += 'channel_mapping=0,1,2,3,4,5';
        num_streams=4;coupled_streams=2;
        maxaveragebitrate=510000';
      }
      if (sdpLines[i].startsWith('a=rtpmap:111')) {
        sdpLines[i] = 'a=rtpmap:111 multiopus/48000/6';
      }
    }
  const modifiedSDP = sdpLines.join('\n');
  return modifiedSDP;
}
```

Listing 1. SDP munging function.

sense given that most WebRTC applications use mono audio streaming for video conferencing.

In addition to this, WebRTCs `getUserMedia()` function includes a `channelCount` constraint that allows users to specify the number of channels in the local stream. This does not, however, determine the number of channels one is expecting to send and be received by another peer over the network. That property must be munged in the SDP file negotiated between two peers during signaling. Without this, a stereo signal is taken from one peer and a mono signal is received by the other. For multichannel audio specifically, the required codec name is `multiopus` as opposed to `opus`, which also must be munged in the SDP offer along with the channel configurations. However, WebRTC does not mention the codec, and in both cases, the necessary solutions were found in community forums. Listing 1 shows a function that edits the SDP file for multichannel streaming.

4.2 Audio Ecosystem Limitations

To ensure ease of use and portability, mobile device support was prioritized—enabling connections from remote locations with minimal hardware while maintaining reliable connections. iOS development was focused on primarily because most of the team uses Apple devices. Apple's restrictive approach to external microphone compatibility is part of its broader sandboxing and security philosophy, which prioritizes system integrity over user flexibility. While this makes iOS more secure and stable, it also creates significant challenges for real-time audio applications—especially those relying on WebRTC and web-based tools rather than native apps [28]. Unlike macOS, where WebRTC freely accesses external audio devices via standard APIs, iOS restricts audio input at the system level. Even when a device is selected using WebRTCs input selection function, iOS does not always honor the selection. This was the case with a Zoom ZMS-44 two-channel audio interface, which promised iOS compatibility.

After testing other devices, the RØDE NTG mobile boom mic proved compatible with Google Chrome and WebRTC on iOS. However, iOS treated the external mic as an additional input rather than a full replacement. The internal mic did not turn off completely and continued picking up audio, albeit at a lower volume. The streams for both inputs were merged into a single channel and sent over the network, leading to an undesirable mix of sources. In the end, the investigation ended here, mainly due to time constraints. However, the RØDE NTG proved otherwise successful given the project goals. The RØDE NTG also features a 3.5 mm line output that was used with an OB-1 speaker by Teenage Engineering for playback.

5 DEBUGGING

While developing the Auxtrument in line with their objectives, the authors encountered numerous challenges and gained valuable insights. However, debugging proved particularly difficult due to the complex and fragmented nature of the technology stack, which involved multiple layers and components. Furthermore, the lack of comprehensive and accurate documentation often hindered progress, forcing reliance on community forums and other unofficial sources.

5.1 Debugging Complexity

Debugging presented a huge challenge throughout the project, particularly because of the "black box" nature of these technologies and the fact that hearing no sound from another location during testing could represent one of dozens of potential issues, many of which are extremely

time consuming, if not impossible to find. Problems could lie with audio hardware, operating system, browser type and version, WebRTC, Web Audio API, the authors' own implementation, a library, or mobile network(s). In most browsers, for example, WebRTC and Web Audio API work together seamlessly, whereas in the latest version of Safari at the time (17.5.1), multichannel through the Web Audio API became nonfunctional after updating.

Furthermore, a known regression in Safari 17.5.1 causes the `AudioContext()` to stop after the browser loses and then regains focus (e.g., minimizing the window) [29]. Even when interacting with UI elements designed to produce sound, the audio remains unresponsive for the rest of the session. This is indicative of broader challenges with WebKit, the engine underlying Safari, which has historically lagged in implementing certain web standards. For instance, the `AudioWorklet` interface, crucial for low-level audio processing, was not implemented in WebKit until 2021, hindering developers relying on this feature for enhanced audio capabilities [30]. These inconsistencies across the Apple ecosystem add complexity to debugging, as behaviors can vary significantly between WebKit and other browser engines and devices, sometimes necessitating browser-specific workarounds.

Another major challenge is logging and monitoring in a real-time WebRTC environment. Traditional in-browser debugging tools often fall short because WebRTC operates asynchronously and across multiple layers, meaning errors can occur at any stage—signaling, negotiation, network transport, or playback—without clear visibility into where the failure originates. While tools like `webrtc-internals` (Chrome) and `about:webrtc` (Firefox) can provide useful insights, they require detailed expertise to interpret, and they do not always reveal the full picture when dealing with custom audio routing or multichannel implementations. They also do not communicate any information about how, if at all, WebRTC and Web Audio API are collaborating.

The abstraction process used by Peer.js makes establishing connections across devices and networks easier, but more difficult to control and debug. For instance, there are bug reports that express difficulty connecting certain mobile providers. There are even reports with Peer.js and iOS version compatibility, suggesting it is not guaranteed to work across all variations reliably. Furthermore, Peer.js includes its own list of STUN and TURN servers to use as defaults. In many reports, mirrored by the authors' own experience, the Peer.js TURN relay tries repeatedly to negotiate between candidates before eventually failing to establish a connection. As a post concert test, the authors set up their own TURN relay through Metered, a WebRTC TURN relay service, and noticed a bug in the Auxtrument code that was not properly overwriting the default Peer.js servers with their own server specifications. After fixing the issue, a connection was able to be established using the TURN relay on both eduroam and Tele2 networks. This points to an issue with the TURN relays provided by Peer.js.

One reason Peer.js was not suspected sooner was because it does not log anything, and there was no reason to expect that the servers were not even being considered. When development began, using a TURN relay was explicitly decided against due to its added latency (an average roundtrip time of approximately 115 ms based on some of the tests). It is possible that considering this option sooner, rather than for trying to debug network issues after the fact, might have helped discover the problem with Peer.js earlier in the development process. In any case, this says nothing about the use of STUN with different network architectures and NATs, which still presents significant challenges.

5.2 Documentation

There were many moments throughout development where a property, method, or constraint was not documented through the WebRTCs documentation repository. Instead, these were found elsewhere on community forums such as Stack Overflow, CodePen examples, and Reddit threads. A recent Editor's Draft from World Wide Web Consortium (W3C) that was published on February 13, 2025, includes the most complete overview of the more esoteric capabilities of WebRTC that the authors have been able to find [31].

Although this is an incomplete draft as of the day of writing, many unknowns about WebRTC features are revealed, although the information is largely inaccurate. For example, the `channelCount` constraint is initially listed as a boolean, but another place on the same web page correctly lists it as unsigned long. This is the case with all constraints WebRTC describes in this draft. The latest published version from October 8, 2024, makes one mention of the `channelCount` constraint, but does not define it or reference how it should be used and with what data type.

6 OTHER APPROACHES

There are several speculative approaches and improvements that could have been considered to answer the research question and address the challenges presented in this paper. While WebRTC provided a workable solution, the unpredictability of its implementation and distribution, particularly for a live concert setting, highlights the need for more user-friendly and reliable solutions.

6.1 Decentralized Audio Routing

Instead of relying on a single WebRTC-based application to handle all connections, separate devices could have been used for each pair of connections. This approach would have simplified the network topology and reduced the complexity of debugging, as each connection would be independent. However, this would have required more devices. Although the authors chose to develop a web-based application for its cross-platform compatibility, they could have considered developing a native app for iOS and Android. A native app would have provided better access to device hardware and operating system features, custom handling of jitter buffers, and compensation for packet losses, thus potentially overcoming some of the limitations presented using web-based solutions.

6.2 Virtual Private Networks

Routing all connections over a virtual private network (VPN) could have simplified peer discovery since each device can be assigned a hostname or static virtual IP address. This option was decided against in practice simply because the WebRTC application was already developed. It was also posited that the underlying network routes of a VPN would effectively be the same as those negotiated by WebRTCs ICE protocol and, therefore, the VPN would serve no additional benefit. However, the VPN's data encryption may slightly increase overall transmission latency.

As a retrospective test, the authors attempted to recreate the Auxtrument between two SonoBus clients connected to a VPN—a laptop connected to a corporate WiFi network and an iPhone connected to the Tre mobile network. Using SonoBus' *connect to group* functionality, the laptop connected successfully. When the phone attempted to connect, an error message appeared, and no audio was transferred. Using SonoBus' experimental *direct connection* functionality using the IPs provided by the VPN, audio was able to be sent in only one direction—from phone to laptop. There was no further debugging information available from SonoBus, so it is unclear exactly where the error occurred.

6.3 Alternative Protocols

As a potential alternative to WebRTC, the authors experimented with creating a transmission protocol similar to Reliable (Unreliable) Streaming Protocol (RUSH) [32], which is based on QUIC [33] and allows fine-grain control over delivery guarantees and avoidance of congestion. However, it was determined that implementing an entire streaming protocol from scratch was not feasible in the time available. One of the marketed benefits of WebRTC was that the implementation already exists in browsers, and for a networked music performance, the authors were inclined to favor tried-and-tested software over an experimental option.

7 THE AUTHORS' VISION

The grand vision of the research project (Information Retrieval in Embedded Systems for Audiovisual Artistic Processes) is the ambition to have a musical performance in remote locations where the performers and audience are interconnected through a mobile interface. This interface will allow access not only to the sound of the performance but to the data involved in producing the performance, thereby blurring the performer-audience boundary. In addition, participants are also envisioned coming and going from the performance ecosystem in real time. Currently, Internet of Things technologies are the only viable option for such a system, placing great importance on established technologies such as WebRTC. This vision could, for example, be a merged reality concert, where the functionality of the Auxtrument allows any user to filter sound through the physical environment of another user at any time. For this—and other experimental network music performance applications—to succeed, the technologies discussed in this paper need to be reliable, robust, and scalable.

8 CONCLUSION

One of the primary challenges faced was the lack of comprehensive documentation for WebRTC, especially multichannel audio and SDP munging. In the future, the authors would expect more detailed and accessible documentation for developers. In addition, they hope for the development of standardized debugging tools that provide better visibility into the interaction between WebRTC and Web Audio API in the browser. WebRTC has been available since 2011 and stable since 2018. Nevertheless, it is suggested that web browser developers increase their efforts to make their implementations compliant with W3C standards. Finally, addressing the issues with mobile networks regarding transparency and compatibility are paramount, particularly with STUN server signaling, but also wishful thinking. Perhaps until the transition to IPv6 address space is complete, or until NAT vendors become more aware of the requirements of significant P2P applications, many of these mobile network challenges will remain unsolved.

The journey creating a web-based, real-time distributed reverberation chamber reveals both the potential and limitations of current technologies for real-time audio streaming over mobile networks. While WebRTC provided a foundational solution, its implementation exposed significant challenges, including network compatibility issues, and hardware/software ecosystem limitations, and debugging complexity. These challenges highlight the need for more robust, transparent, and well-documented tools to support real-time audio applications. Looking ahead, the development of standardized debugging tools and more comprehensive documentation for WebRTC and related technologies will be critical for advancing real-time audio streaming applications. This project serves as a case study in the complexities of web audio technologies and offers valuable insights for future research and development.

9 ACKNOWLEDGMENT

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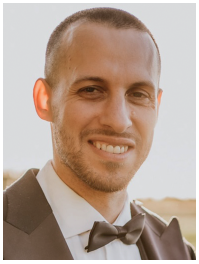
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REVIEW

Open Access

A systematic literature review of methods to describe the stereo image of recorded music

Richard Mitic[1*] and Andreas Rossholm[2]

Abstract

Analysing spatial information within recorded music is increasingly important for audio research given the explosion of available datasets, but relevant techniques are spread over diverse fields. To address this, a systematic literature review (SLR) was conducted to ascertain all existing methods for describing the spatial features of stereophonic music. The Scopus, Web Of Science (WOS), and Semantic Scholar (SS) databases were searched on 26th February 2026, and articles were deemed eligible if the method they described included any extraction of spatial information from a music audio signal. Links and similarities were drawn between studies to identify the core methods of describing stereo image. One hundred seventy-eight articles were included, with 29 of those deemed "very relevant". The fundamental methods of extracting spatial information are detailed and a framework for creating a custom stereo image profile (SIP) is drawn up, followed by suggestions for future research. The review was limited by the substantial volume of literature to be assessed by few researchers, although this wide breadth served to mitigate bias by gathering at least one example of all existing SIP methods. Inconsistent application of terminology within the existing literature presented a significant challenge. The work was funded in part by The Knowledge Foundation (KKS).

Keywords Music information retrieval, Stereo image, Systematic literature review

1 Introduction

The extraction of spatial information from musical audio signals has long been a goal for digital music practitioners, with downstream uses such as classification, recommendation, format conversion, and even musicology. However, until the recent development of object-based audio, it was not common practice to include any such information in music metadata, and it must therefore be derived after-the-fact from the final musical mix. With the new availability of very large musical datasets there exists an opportunity to gain insights into musicians' use of spaciousness in their work.

It is imperative to recognize that the gamut of spatial processing in audio research is enormous, and the terminology, keywords, and article structure somewhat non-standardized. Many studies include the extraction of spatial information in their methods, and while said information may be enough to describe stereo field, the articles are written from the perspective of the downstream task rather than the spatial features themselves. Consequently, *very relevant* literature is difficult to find using simple searches, yet increasing the search radius results in excessive amounts of *semi-relevant* literature. This represents a missed opportunity for "cross-pollination" between adjacent (but usually independent) audio research fields, and to that end, this SLR aims to determine all currently available methods to describe the stereo field of recorded music. It follows the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines [1].

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1.1 Previous reviews

Reviews of techniques to extract spatial information from audio are generally rare, with the few examples being tailored to specific use cases. For example, [2] gives an overview of machine learning (ML)-based techniques for the capture, processing, and rendering of spatial audio. The targeted use cases are gaming, augmented reality (AR)/virtual reality (VR), teleconferencing, and entertainment applications. While binaural audio is mentioned as a spatial format, other stereophonic recordings are not, nor is the focus on musical audio. The authors also state that their review is "representative... [but not] exhaustive". Most interestingly, they conclude that tasks such as "analysing the spatial properties of sound [had] not been... disrupted by ML methods". In a similar vein, [3] is an SLR of the technologies and applications of spatial audio in AR. The authors state that it is the first systematic review in the field, but the audio sections are focused on synthesis rather than analysis, and (like the former) musical audio is not targeted. A more recent review [4] focused specifically on spatial audio, mentioning music as a use case. It too only references binaural audio (and not stereophonic more generally) as a spatial format, and is focused on research performed from 2022 onwards. It does, however, give an excellent overview of input and output audio representations, spatial audio tasks, available datasets, and evaluation metrics.

1.2 New aims and scope of this review

To fill gaps in previous reviews, this SLR focuses specifically on techniques to describe the spatial features of stereophonic musical audio, extracting them from previous literature and gathering them in a single place for reference by the audio research community. The aforementioned lack of standardization will be normalized such that future researchers can share a common footing and language. See Sect. 2.5 for a detailed explanation of specific outcomes.

A list of sonic qualities that represent audio "spaciousness" is defined in [5], from which it is possible to infer that any audio signal with more than one channel is eligible to contain spatial information. Rumsey [6] states that even though it is possible to perceive distance (which can be considered a spatial attribute) in a monophonic recording by assessing the relative loudness of sources or extent of reverberation, *width* is a far more prominent spatial attribute that is impossible to reproduce monophonically. To keep the scope of this SLR reasonable, monophonic audio is therefore deemed out of scope despite the fact the assessed techniques would be inapplicable to older and historical recording collections. Conversely, while many studies act on audio with more than two channels, they are still deemed relevant if the underlying techniques could be applied to stereophonic audio.

2 Methods

This section defines all processes for performing the review, and discusses some known limitations.

2.1 Search process

The review follows the Population, Context, Outcome (PCO) protocol [7]. The population is defined as "music", the context as "stereo", and the outcome as "spatial information". Since these terms and their synonyms are often not standardized, a list of negative search terms were also included to exclude some obvious false positives. Several very specific terms that were already known to the author were also included, and any article using those terms was included in the initial search result. The full lists of synonyms, specific terms, and negative terms used to form search queries, plus the search queries themselves can be found in Appendix 1. The search was performed on 26th February 2026.

Table 1 shows the databases that were searched, and their precedence. Both Scopus and WOS have a browser-based search functionality accepting advanced queries, from which the results can be exported as Comma separated variable (CSV)/Tab separated variable (TSV) files. SS provides an application programming interface (API) endpoint that can be used to perform similar advanced searches [11]. It was found during initial searches that although the SS API accepts negative terms, it often wrongly excluded relevant articles. Negative terms were therefore omitted from the SS query. The data exported from the searches was universal resource locator (URL), title, abstract, authors, publication year, publication source, Digital Object Identifier (DOI), and citation count of each study. A full dataset of the collected search

Table 1 Databases used for article search

	Database	Description
1	Scopus [8]	The largest database of scientific articles, indexing a majority of computer-science publishers.
2	Web Of Science [9]	Another well-respected general scientific database that acts to fill gaps in Scopus.
3	Semantic Scholar [10]	A general-purpose scientific database created via web-scraping that tends to surface more "grey" publications such as theses, commercial white papers, and pre-prints, but can be prone to errors and false records.

results for all databases is included with this manuscript as an additional file.

2.2 Initial screening

The initial list of articles was filtered to remove duplicates, firstly identifying them by DOI, and then by title and publication year when DOI was not available. Since there are can be differences in formatting between the databases, the titles were first "normalized" by converting to lower case and removing all non-alphanumeric characters. Articles were marked as duplicates if the normalized title and publication year exactly matched one previously seen in the list. See Appendix 4 for pseudocode that describes the automatic filtering process.

A first pass was then performed to make obvious exclusions. The title and abstract of each paper was read, and the article was rejected if any of the following were not true:

- Database record was valid.
- Title and abstract were written in English.
- Document was available for reading.
- Document described some research[1].
- Subject of the study was audio.
- Described method included the extraction of spatial features from an audio signal at any point.

A record was considered valid if it linked to a published document. In practice, invalid records were only found in the SS database, which (as previously mentioned) is created via web scraping and is hence prone to errors. For example, SS record 196093012 was included in the search results, but is considered invalid. Some duplicates were also identified manually and rejected at this stage. Note that the conditions of the audio comprising at least two channels and being musical were not assessed during the screening as it was very often not obvious from the title and abstract. If the researcher was unsure about whether an article should be rejected or not, it was marked as "unclear" and revisited at a later stage.

2.3 Review process

2.3.1 Primary review

After the initial pass, all non-rejected articles and all articles marked as "unclear" with a publication year of 2020 or later were obtained and read in full. They were assessed (and possibly rejected) based on the same criteria mentioned in Sect. 2.2, plus two additional conditions:

- The audio signal comprised at least two channels
- The audio signal was musical

To assess whether an audio signal is musical or not, it must meet the condition that it contains a recording of at least one musical instrument (including singing voice), or that the study specifically stated that it was applicable to music. Studies using both musical and non-musical signals were also included.

While it is common to exclude reports based on the impact factor or citation count [12], an initial scan of the search results revealed that some very obviously relevant studies had very few, or even zero citations (e.g., [13]) or were published as pre-prints only (e.g., [14]). Therefore, citation count and publication venue were not taken into account.

2.3.2 Post-hoc screening test

As a post-hoc test, a second researcher was given a random 5% sample of the search results (with automated rejections already removed) and independently screened them based on the conditions specified in Sects. 2.2 and 2.3.1. Discrepancies between the researchers' opinions were flagged for discussion only if the second researcher included a study that the first had previously rejected. Table 2 shows the results, and ultimately, only one such study [15] was found. The first researcher marked it as "unclear" and later rejected it because the publishing year was missing from the database search result. After reading the full paper both researchers agreed to reject it because it did not describe the extraction of any spatial features. Overall, the post-hoc test did not raise any issues with the primary review.

2.3.3 "Very relevant" studies

Since the number of relevant studies from the initial assessment was high, a further selection process labelled those deemed "very relevant" to the aims described in Sect. 1.2. This process filtered studies based on their "downstream task" (see Table 3), selecting tasks that are relevant to the extraction or interpretation of spatial information in large audio corpora. As per the PRISMA

Table 2 Results of post-hoc screening test

Outcome	Count	Treatment
Initially rejected, post-hoc included	1	Researchers discuss
Both included	3	OK
Initially included, post-hoc rejected	13	OK
Rejected for same reason	68	OK
Rejected for different reasons	68	OK
Total	153	–

[1] Other document types were e.g. book review, patents, contents of conference proceedings, etc.

Table 3 Data extracted from reviewed articles

Data	Description	Examples
Audio model	The channel count or standard paradigm that the audio signal follows	stereo, binaural, ambisonic, mic array
Contents of signal	A high-level description of what the audio contained	music, speech, noise
Specific audio configurations	More detail about the signal contents or how it was recorded/created	reverberant, synthetic binaural, orchestral
Datasets	Any audio datasets that were used in the creation of the audio signal	FMA, MTGJ, MedleyDB
Downstream tasks	The specific tasks that the spatial information was used (or was intended to be used) for	Source separation, MIR, classification, compression
Novel mechanism category	A high-level category of the articles main contribution	DSP, ML, psychoacoustic test
Spatial attributes	The specific spatial information that was extracted from the audio signal	DOA, panning histogram, cochleagram
Novelty	A record of whether the article describes a new spatial feature or referenced one from the literature	novel, referenced, modified
Evaluation technique	The technique by which the scientific claims were verified	Objective metrics, subjective test, simulations

guidelines, those studies are listed and summarized in Sect. 3.2 — Readers may use them as a main reference point for existing techniques that directly relate to the aims stated in Sect. 1.2. All studies are still deemed "relevant", forming the basis for the syntheses described in Sect. 2.5, and providing the building blocks for readers to construct spatial descriptors specific to their task.

2.4 Data extraction

For each study included in the review, data was collected according to Table 3. Note that the options in each category are not mutually exclusive, e.g., [16] describes Binaural Cue Coding (BCC) which, while originally intended for the compression of multichannel signals, is also used for stereo. An additional example is [17] which states that its downstream task is music information retrieval (MIR), and then provides proof of the method's effectiveness on a classification task.

2.5 Syntheses

The primary synthesis is an exhaustive list of methods to generate a SIP for musical audio, plus a thorough analysis of their underlying extraction techniques, and what musical features they are able to detect. Complementary to that is an explanation and normalization of their terminology, drawing links where studies are using similar techniques and differentiating between homonymous terms.

Following that, a framework of how to choose or form a SIP that is suitable for any particular task. The data extraction process differentiates between traditional digital signal processing (DSP) and newer ML/deep learning (DL) spatial features, and it will be noted which tasks are currently benefiting from modern techniques, thereby giving inspiration for how to incorporate them in other tasks.

In addition, a list of useful audio datasets and experiment validation techniques will be formed so that any future researchers can find tools suitable for their tasks. Propositions for future research will also be given based on what is missing from the search results.

2.6 Limitations

Several limitations were uncovered during the review process. Firstly, due to the wide research questions and non-standardized formats of audio-related article abstracts, the number of articles requiring review was very high. Even though a second researcher performed a post-hoc test to judge article screening, only a single researcher was tasked with extracting data from all articles. This introduces a potential source of bias. However, the review's wide breadth was also a point of strength since the aim was to gather at least one example of all existing SIP algorithms. It is difficult to assess the impact of audio processing-related articles since they typically do not receive many citations and the publication venue and other related information is often missing from the databases. The solution was to assess each article equally, which created more manual work.

In addition, the non-standardized terminology meant that the database search queries (see Appendix 2) contain many synonyms for "spatial information". The list was built iteratively over several initial searches, but there is still a chance that some missing keywords would have been beneficial to include. However, since this review is not concerned with the absolute number of search results for each term, this limitation is largely mitigated as long as at least one example of any given spatial descriptor was

present in the results. Similarly, while the search process included specific terms for commercial spatial audio formats, information about them (and their underlying algorithms) is typically not public and hence was absent from the search results.

There is likely a bias regarding the publication venue of the articles sought for retrieval since some specific journals were not available through library subscriptions[2]. Articles appearing potentially relevant were sought out via other methods (internet searches or direct contact with the authors) as a mitigation. Several doctoral theses were surfaced in the search results, and while they were relevant they were difficult to categorize in the review's framework since they typically contain multiple studies and hence fall into multiple categories. The accuracy of the data extracted from them could therefore be unreliable.

The assessment of DL-related papers presented a few challenges. Firstly it is quite common to simply cite the audio dataset used, rather than describing the format and contents of the audio. Additionally, due to the black-box nature of DL, the trained network often *does* learn spatial features, but it is debatable whether that constitutes a spatial description (see Sect. 3.1 for examples).

3 Results

The search initially identified 4678 studies, of which 1548 were removed before screening. Two thousand three hundred sixty-seven were excluded after reading the title and abstract, leaving 763 that were sought for retrieval. Out of those, 48 could not be retrieved and 537 were rejected, leaving 178 that were included in the review. Twenty-nine of those were deemed "very relevant" due to their downstream task. Using those 178 studies as a seed, 15 studies were added by following references, 4 studies were found using the SS recommendation service, and 5 studies were recommended by peers. A full flow diagram of study inclusion and rejection is shown in Fig. 1. In total, 14 studies were pre-prints and had not seen peer review at the time of the search or recommendation.

As mentioned in Sect. 2.3, some studies were marked as "unclear" during the initial screening. As a metric for the judgement of the researcher, the number of studies assessed and eventually included are shown in Table 4. It shows that a far lower proportion of "unclear" studies were eventually included, implying that the researcher's judgement was largely correct.

[2] The most prominent unavailable journals were "Proceedings of Forum Acusticum", "Proceedings of the International Congress on Acoustics", and "Journal of the Acoustical Society of America".

3.1 Borderline studies

Some studies presented edge cases during assessment. The following non-exhaustive list gives some examples of such studies, explaining why they were either included or excluded.

3.1.1 Inclusions

- Pollow et al. [18] describes a method that acts on audio from a large microphone array rather than recorded music, but it is included because it mentions the nuances of using phase and intensity differences to estimate sound directivity.
- Bourbon et al. [19] uses multichannel renderings of object-based audio as the source, but was included because the method acts on pairs of channels as opposed to all channels simultaneously.
- Fernandez et al. [20] is primarily focused on the rendering of binaural microphone array signals, but was included because it references several useful algorithms for calculating spatial attributes of audio.

3.1.2 Exclusions

- Lindau et al. [21] contains a useful list of spatial audio terms, but does not describe any audio processing.
- Colone and Reiss [22] describes a method for predicting panning positions of stems in a stereo mix, but required access to those stems in order to function.
- Felipe et al. [23] uses stereo features to aid in the classification of binaural scenes, but does so by extracting monophonic features from each channel.
- Sierra et al. [24] is focused on capturing spatial audio for MIR purposes, but simply outlines the capture process and choice of spatial format rather than deriving any spatial information.
- Pulkki et al. [25] is a software implementation of the underlying spatial processing technique, which is described academically elsewhere in the literature.

3.2 Very relevant studies

This section will highlight the studies that are very relevant to the review's aims given in Sect. 1.2, summarize each study, and draw links between them. Generally speaking, very relevant tasks are those that relate to creating a high-level spatial description of a recording, or describing the spatial features of large musical corpora. The tasks deemed not very relevant are more concerned

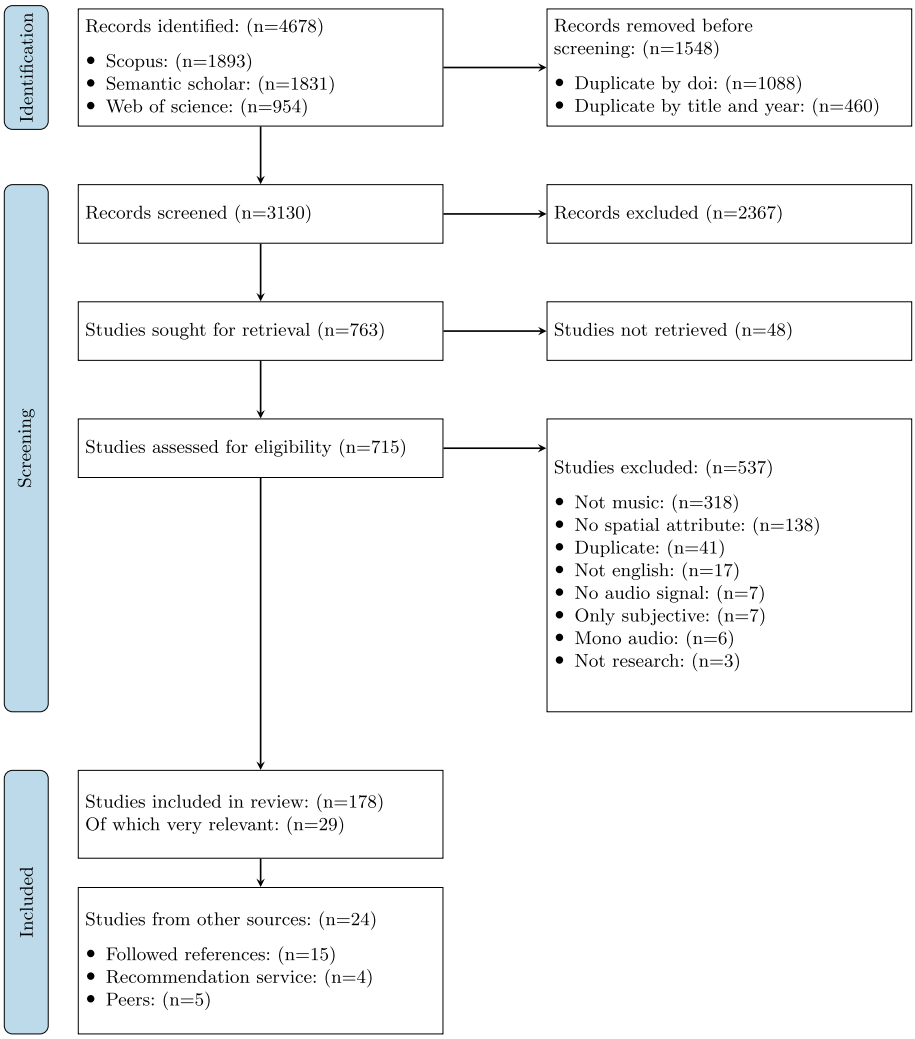

Fig. 1 PRISMA flow diagram

Table 4 Proportions of studies that were included in the review, split by clarity

	Assessed	Included	Proportion
Obvious	631	175	28%
Unclear	84	3	4%
Total	715	178	25%

with low-level spatial features and often require the audio to be reconstructed or altered after analysis. A full list of downstream tasks can be found in Appendix 5. In this SLR the 29 studies deemed to be very relevant are those whose downstream task was at least one of "MIR", "spatial analysis", "musicology", "visualization", or "classification". These studies tend to fall into tightly-related cohorts, often referencing each other.

The earliest is [26]—a doctoral thesis defining methods for spatial analysis of binaural signals for use in the broadcasting industry. Dewhirst [27] (another doctoral thesis) expands on [26], combining it with a binaural hearing model and an interaural cross correlation (IACC)-based stereo width estimator [28] to create an artificial listener with the intention of assessing perceived sound at arbitrary (i.e., non-sweet-spot) listening positions.

A method to estimate amplitude panning position in the time-frequency domain (named panning index (PI)) was defined in [29, 30] and demonstrated for upmix and source separation tasks. Tzanetakis et al. [31] and its extension [17] define the stereo panning features (SPF)—a descriptor of amplitude panning based on the PI (although renamed to stereo panning spectrum (SPS)). In turn, [32] uses SPF as a measure of similarity between two mixes of the same musical stems, and [33] used the

SPS to visualize artistic audio recordings and SPF as a measure of similarity between those recordings.

Sarroff and Bello [34] defines a spatial descriptor named panning histogram (PH), which is based on the Frequency-Azimuth (FA) plane defined in [35]. The PH is also collapsed into a single measure of "stereo width", which the same author uses in [36, 37] to create a computational model of perceived spaciousness. Notably, these studies also use monophonic descriptors, including a prediction of the extent of reverberation in the recording.

The concept of a PH is used by several other studies, although under different names and with different mathematical definitions. Takanen and Lorho [38] generates likelihood histograms of the direction of arrival (DOA) of sound sources from measurements of the interaural level difference (ILD) and interaural time difference (ITD) in binaural correlograms, from which a measurement of stereo width is also derived. A major difference from the previously mentioned works is that the method requires calibration using a recording of background noise, and reference ILD/ITD values from which the DOA is calculated. Gomez et al. [39] defines another PH which is based on the PI from [29]. Furthermore, it defines a method of PH quantization with the intention of allowing simple similarity measurement of music recordings in a lower-dimensional, non-perceptual domain. Differing slightly from other studies, [40] focuses mostly on the visualization of stereo audio. It transforms time-frequency (TF) domain interaural intensity difference (IID) and interaural phase difference (IPD) into two cues — "position" and "clarity" — which are then used to create several visualizations including a PH. The same author expanded on this idea in [41], defining a transformation from time-domain stereophonic signal into a generic bivariate mixture space (BMS), which in turn is used for visualization and panning estimation (among others).

Tardieu et al. [42] provides two alternative descriptors of panning features—amplitude stereo features (ASF) for amplitude panning and phase stereo features (PSF) for phase panning. They are created from time-domain cochleagrams and frequency-domain spectrograms respectively, and are demonstrated on a decade classification task.

Härmä [43, 44] collectively demonstrate a new paradigm of SIP classification. As a precursor, a novel signal model is introduced that defines seven classes of stereo field. Subsequently, it is shown audio clips falling into six of the seven classes can be identified, first using a feed-forward neural network and then purely analytically. The inputs to the classifiers are cross correlations between the two channels and their variances (in TF domain). The downstream tasks are the aforementioned classification itself, as well as audio enhancement, audio manipulation, and upmixing.

You and Cheng [45] presents a method of comparing a degraded signal with its original in the spatial domain. It uses differences in IID and ITD within audio sub-bands, the primary use case being objective assessment of spatial compression algorithms. With a similar use case of predicting the perceptual quality of binaural renderings of microphone arrays, [46] uses both textural features defined by the Objective Measurements of Perceived Audio Quality (PEAQ) standard [47] and spatial features defined by its multichannel extension PEAQ-MC [48]. The spatial measures of PEAQ-MC are differences in ILD, ITD, and IACC between an original and degraded signal.

Chen and Hasegawa-Johnson [49] exploits the common mixing practice of contrasting highly centred sources with panned sources to tackle the task of speech-to-music-ratio prediction. After converting a stereo signal to mid/side configuration, a metric representing the ratio of centred to panned audio energy is defined. Somewhat similarly, [50] uses mid/side encoding of Mel-spectrograms to aid a DL classification of specific classes of binaural music recordings. Importantly, both studies perform the conversion to mid/side format in the time domain and use the resulting signal to construct features for input into an ML model.

Focusing on the enabling of new musical production techniques, [51] used analysis of goniometers and a derived measurement of stereo width to demonstrate the effectiveness of "dynamic polar patterns" in stereo microphone recordings. In a similar vein, [14] tackles a genre classification task based on measurements derived from virtual goniometers and channel correlation. Both studies use the goniometer output as an analogue to stereo width, noting that since mixing engineers refer to such tools during their work, there must be value in their readings. Regrettably, neither study considers that a goniometer's output inherently entangles phase and amplitude such that a gain alteration might induce a stereo width change, and vice versa.

Another estimator of stereo ensemble width is given in [52], based on "spatiograms" (originally defined in [53]) derived from a binaural signal, and the same author provides a comparison of different ensemble width estimators in [54]. Interestingly, spatiograms were originally intended to estimate DOA, are derived from the ITD, and are hence "phase-only".

Acting on stereo stems that have been separated from a mix using DL, [55] uses ILD to form panning maps of individual musical components, subsequently comparing songs from different decades and finding trends in music production. ILD is also used in [56] to determine the

amplitude panning azimuth of individual notes in recordings of polyphonic music, with the caveat that their fundamental frequencies have been determined beforehand.

In somewhat of a departure from the other studies surfaced in the review, [13] (while initially studying a DL upmixing task) demonstrated that the latent space learned by the neural network successfully disentangled spatial information from musical information, thereby enabling a vector within that space to represent the spatial qualities of the input audio. In a similarly unique task space, [57] uses the classification of stereo features to detect audio forgeries, although the features used are not learned, but rather are derived from classical DSP spatial cues such as ILD and IPD.

3.3 Fundamental concepts and terms

Many of the studies gathered in this review make use of similar fundamental audio concepts and signal processing techniques. The naming conventions are somewhat fluid, with a single term often used for variations of an underlying concept, or a single concept being referred to by multiple terms. This section aims to clarify these ambiguities.

3.3.1 Binaural and spatial cues

The terms "binaural cues" and "spatial cues" are umbrellas referring to any low-level spatial information extracted from two or more audio channels. They are used very broadly and do not have specific definitions, although some subcategories are common. Among the most used are interaural level difference (ILD) and interaural time difference (ITD), which are usually (although not always) extracted from frequency subbands of a stereo audio signal. They have previously been used in the context of psychoacoustic experiments as ways to explain and quantify the perceived effect of lateral reflections in acoustic spaces [58, 59]. Even though they contain the word "interaural" (Latin for "between the ears") they are freely used throughout the literature to mean amplitude or time difference between any two channels of an audio recording, e.g., binaural [60], stereo [45], and multichannel [61] audio. Barry and Lawlor [35] uses ILD as a ratio instead of its absolute form and [62] coins the term interaural level ratio (ILR) to formalize this, noting that since ratios are bounded in the range -1 to 1 they are usable as inputs for ML tasks. (In the classical form, ILD is usually given in decibels and is theoretically unbounded.)

The terms often appear with slight naming variations—synonyms for ILD include interaural intensity difference (IID) [26, 27, 63, 64], inter-channel level difference (ICLD) [65–67], and channel level difference (CLD) [68]. Likewise, a common synonym for ITD is inter-channel time difference (ICTD) [69–71]. Another formulation of ITD is IPD [20, 72, 73] (or inter-channel phase difference (ICPD) [74]) which is a frequency-dependent time difference. Notably, ITD can be calculated without knowledge of a center-frequency, as in [45] which uses transients instead.

Another common family of cues is IACC [46, 60, 75, 76], inter-channel correlation (ICC) [77, 78], inter-channel coherence (IC) [73], or interaural coherence (IAC) [79]. These cues indicate the correlation of 2 audio channels and are closely related to a perceptual audio property–listener envelopment (LEV). Faller and Merimaa [79] notes that IAC can be used to determine whether ITD and ILD represent a strong directional or more reflectional audio source. De Sena et al. [80] provides a similar perceptual measure of "location uncertainty". On a cautionary note, [76] mentions that IACC values measured from the same acoustical space are affected by the underlying musical material.

In the classical perceptual model of directional hearing it is accepted that IPD is only relevant for frequencies under 1500 Hz [81, 82]. However, there are several examples in the literature of IPD for all frequency bands being treated equally [40, 42, 83]. One reason for this is that while the directionality of a single narrowband source cannot be determined by a human using phase differences above 1500 Hz, reverberant or decorrelated sources often result in a broadband mixture of phase differences [84].

Importantly, there are various ways to calculate all these parameters, and [85] even notes that different definitions result in binaural/spatial cues with different underlying properties, hence subject to variation from different influences. For example [86, 87] both use the term ILD but calculate it in the time and time-frequency domains respectively. For an excellent overview of several classic methods for calculating ILD, ITD, and IACC the reader is directed to [26], and block diagrams showing some generic methods are shown in Fig. 2.

As a rule of thumb, it is advisable that when using any binaural or spatial cue for research, the author should preferably state an exact mathematical definition directly in the text, otherwise referring to a specific definition from another publication. In particular, it is advised that authors should use "interaural" terms for binaural audio only, opting for "inter-channel" in other cases.

3.3.2 Panning parameters

Another cohort of algorithms is dedicated to estimating the panning that has been applied to monophonic stems in a stereo mix. Those related to amplitude panning differ from the binaural ILD in that they account for a standard signal model in which the amplitude of the two channels follow the "energy-preserving panning law" whereby the

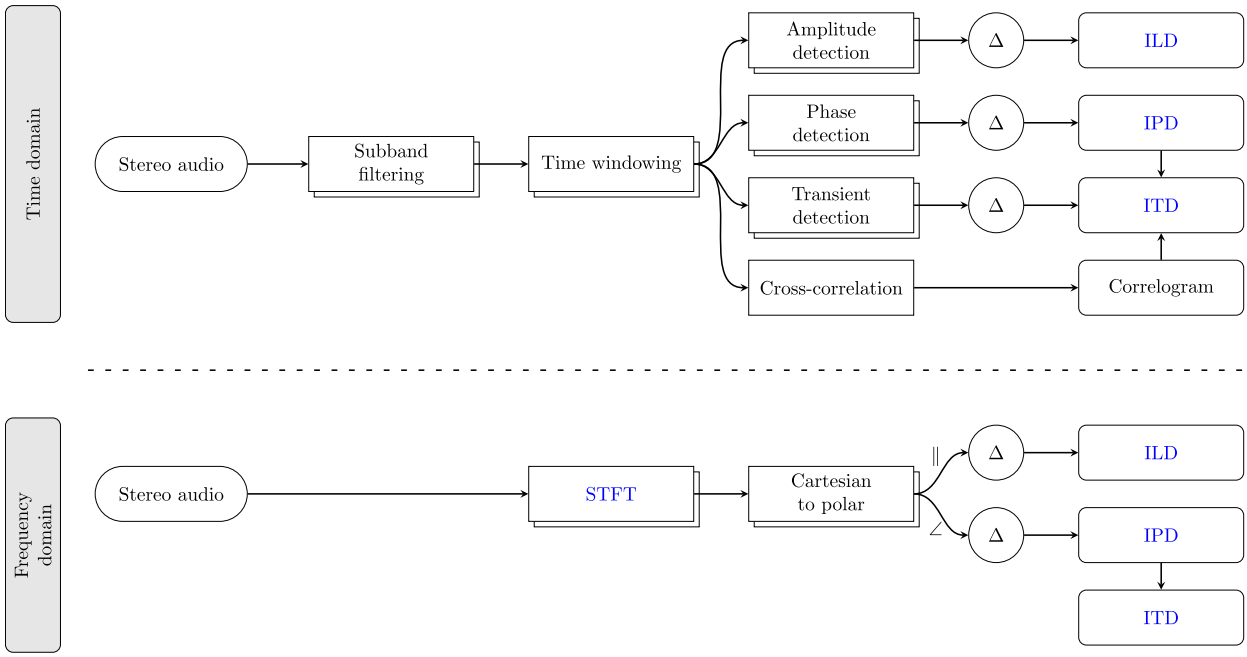

Fig. 2 Block diagrams of common spatial cues. Δ represents the inter-channel difference. ‖ and ∠ represent the magnitude and phase components of the short-time Fourier transform (STFT) respectively

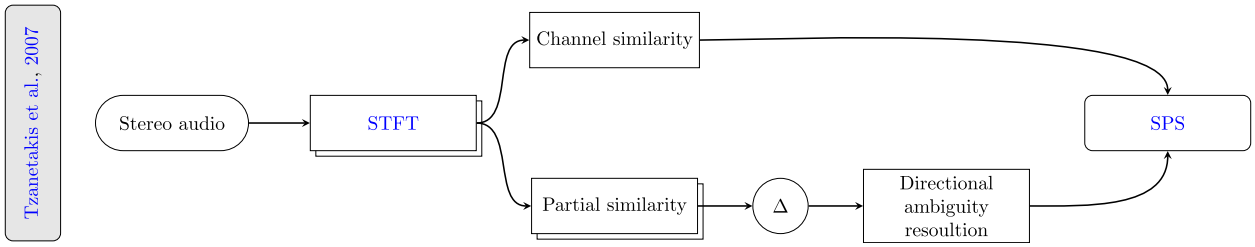

Fig. 3 Block diagram of SPS [31]. Δ represents the inter-channel difference

sum of the squared channel gains is 1. A formal mathematical formulation is given in [29, 30, 88] under the name panning index (PI), and [89–91] use the same technique unaltered. Similar formulations are found under multiple names: "panning parameters" [92–94], "stereo panning spectrum" [31], "pangram" [95], "azimugram" [96, 97], "amplitude panning factors" [98], "pan map" [64], "pan time-frequency filter (TFF)" [84], "mixing parameters" [99, 100], and "panning coefficients"[3] [101–105].

While all the aforementioned parameters have the intention of estimating the panning that was applied to a mono stem, the exact mathematical methods vary tremendously; interested readers are directed to the original studies, although as an example, a block diagram for SPS [31] is shown in Fig. 3. Note that unlike the frequency-domain spatial cues shown in Fig. 2, the stereo spectrogram is not converted to magnitude and phase components, but instead is processed by the complex functions "channel similarity" and "partial similarity".

A general theme is that estimations of amplitude panning can be determined through ILD-related methods and measurements of "ambience" can be obtained using techniques based on correlation between the channels [29]. An interesting, but rarely-used technique is [102] in which the stereo subband signal has principal component analysis (PCA) applied to find a dominant signal and a panning coefficient is subsequently estimated using a Lagrange multiplier. Woodruff et al. and Woodruff

[3] Note that [39, 101] also use the term "panning coefficients" to describe a higher-level panning descriptor derived from a PH (see Sect. 3.2). This use of the term does not describe the same concept as the ones listed above.

and Pardo [99, 100] demonstrate the estimation of both amplitude and time panning, utilizing TF-domain ILD and IPD directly.

In all cases there is an implicit assumption that each frequency subband within the mix contains only a single direct sound source. For some use cases (e.g., source separation) this results in an underspecified problem, but for spatial *description* this is not necessarily a limitation. In fact, one can often expect recorded music to be spatially dense, and any description created from panning parameters should reflect that.

3.3.3 Time-frequency decomposition

In all the cases mentioned in Sect. 3.3.2, the panning parameters are calculated on individual TF bins. Naturally, there are several ways to perform the TF decomposition, each with its own nuances. The majority of algorithms described use the STFT as a starting point due its computational efficiency, although it is noted in several studies that the Fourier transform's lack of perceptually-relevant frequency bands is a problem and should be corrected for. Typically, fast Fourier transform (FFT) frequency bands are combined into groups according to a psychoacoustic model, such as equivalent rectangular bandwidth (ERB) or Bark critical bands [106, 107]. Tzanetakis et al. [31] takes an even more drastic approach, forming just 3 subbands (lows, mids, and highs) according to the typical practice of mixing engineers.

Li and Driessen [102] takes a similar approach but instead of using STFT, the frequency decomposition is achieved via a pseudo quadrature mirror filter (PQMF) filterbank, as defined in the MPEG audio standard [82]. Tardieu et al., De Sena et al., and Zhu et al. [42, 80, 108] choose instead to calculate stereo "cochleagrams" in the time domain; a bank of gammatone filters is applied to the original signal and the resulting subbands segmented in time. This has the benefit of being a much closer approximation to human hearing, but comes at the expense of higher computational complexity. Subsequently, it is common to apply an auditory model, i.e., a computation that models the cochlea and its neural connection to the brain. Several versions can be found through the literature, although a classical version is presented in [109], and [26, 75] include comparisons of several models.

3.3.4 Frequency-azimuth analysis

After converting audio into bin-level spatial cues, it is common to visualize them in a 2D spectrogram-like plot. A theme across most studies is that it is generally useful to view the change in spatial parameters change over time. We therefore see most often that time is displayed on the X-axis of the plot, with frequency and azimuth being assigned to either the Y-axis or colour/brightness. Further analysis of these plots can generate musical insights, as in [96] which uses changes over time to achieve song-structure segmentation of recorded music, and [62] which uses the visualizations directly to compare the rendering quality of different spatial audio formats and lossy compression algorithms. Focused purely on visualization, [40] adds the concept of bin-level "clarity", effectively determining the trustworthiness of a panning parameter estimation. As with the previously discussed spatial cues, the naming of these visualizations varies freely (see Sect. 3.3.2), and interested readers are directed to the original papers for examples.

3.3.5 High-level descriptors and perceptual scalars

Subsequent to bin-level analysis, there are several algorithms that condense the derived parameters down into a compact representation for the purpose of describing a certain spatial aspect of the original signal. In this context, "high-level descriptors" are those mentioned in Sect. 3.2, e.g., PH, ASF, PSF, and SPF.

"Perceptual scalars" are the spatial qualities often evaluated in perceptual testing, which can be predicted both from low-level spatial cues or from the high-level descriptors. Two common ones are apparent source width (ASW) and LEV. Again, there are many synonyms or near-synonyms for these terms: "stereo width", "source width", "auditory source width", "ensemble width", and "stereo image width" for ASW "envelopment", and "immersion", and for LEV.

ASW is usually calculated as a summary of either panning or DOA calculations. For example, [34] defines ASW as the full width half maximum (FWHM) of a Gaussian curve fitted to a PH, [26] uses the interquartile range (IQR) of an IID histogram, and [27] uses the standard deviation of a combination of IID and ITD histograms. Mieth and Zölzer [104] on the other hand uses IACC and lateral energy fraction (LEF) as proxies for ASW and LEV respectively. Continuing from the diagrams given in Sect. 3.3.2, Fig. 4 shows two exemplary block diagrams of algorithms to calculate ASW. Note that each algorithm contains an intermediate PH which can be used as a high-level descriptor directly. Importantly, the algorithms are completely disparate and the resulting ASWs/PHs cannot be directly compared, despite having the same name.

It has also been shown that perceptual scalars such as "immersion" can be predicted via ML from combinations of low-level features, including those which do not contain any spatial information [37]. Figure 5 shows a block diagram for this. Similarly, [110] develops a computer model of "envelopment" based on IACC, PCA,

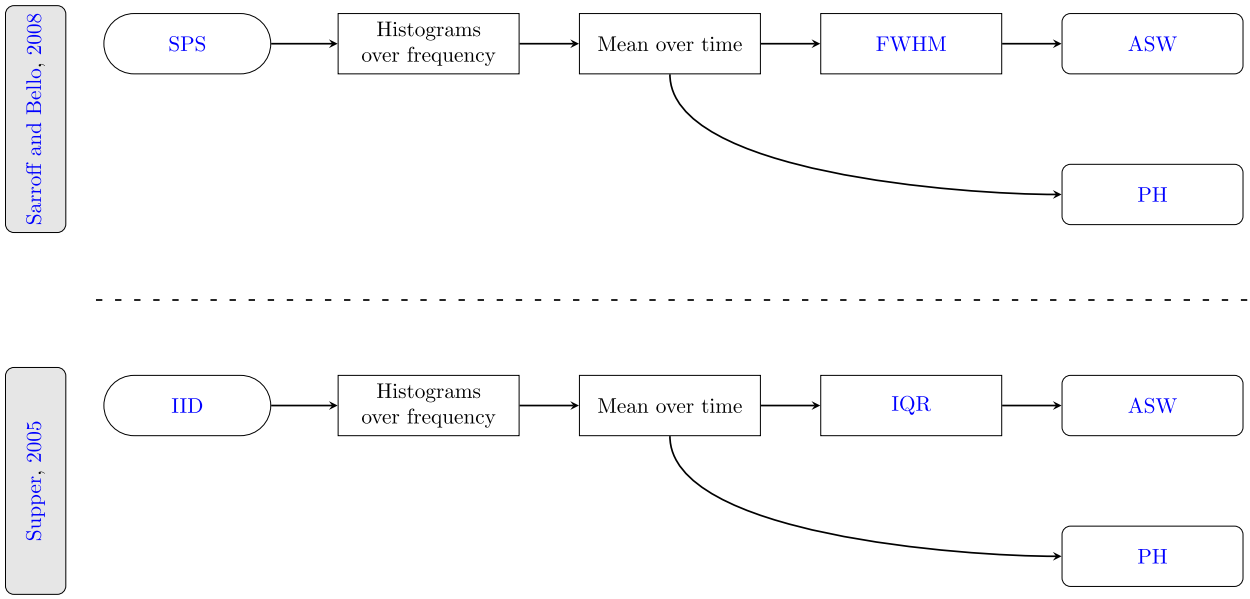

Fig. 4 Block diagrams of two algorithms to estimate ASW. Disparate algorithms are separated by a dashed line

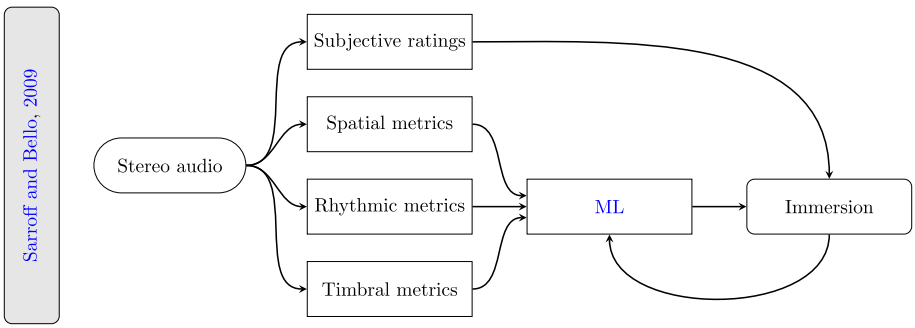

Fig. 5 Block diagram of prediction of immersion via ML on objective metrics [36]

and channel energy distribution (albeit for multichannel audio rather than stereo).

The lack of mathematical standardization in these studies emphasizes the point from Sect. 3.3.1—future research should preferably state formulas explicitly in the text, falling back to a specific definition in a cited publication.

3.3.6 Machine-learned spatial descriptors

Departing from the traditional DSP spatial descriptors, a new family of descriptors generated directly though ML and DL have emerged. As mentioned in Sect. 3.2, [13] encodes the spatial information present in multichannel audio to a vector in latent space by first downmixing the input audio to stereo and then forcing the neural network to upmix back to the original. Manocha et al. [111] takes a very similar approach, but acting on binaural input and its accompanying monophonic mixdown. In both cases, content invariance of the network is one of the optimization goals, and is essential when using the latent vectors as SIPs. The former achieves it by creating two random 5-channel mixes and using the stereo downmix of one to resynthesize the other, while the latter utilizes an additional training task which optimizes the overall network's spatial embeddings to match that of a pre-trained neural binaural decoder which takes only directional coordinates as input. Simplified block diagrams for both methods are shown in Fig. 6; the reader is directed to the original papers for details. A slightly different method is taken by [112], in which stereo audio captured with accompanying video is modified by either swapping the left and right channel, or using audio from a different video clip. The network then learns to spot said modifications, hence learning the spatial relationship between sound events and visual events.

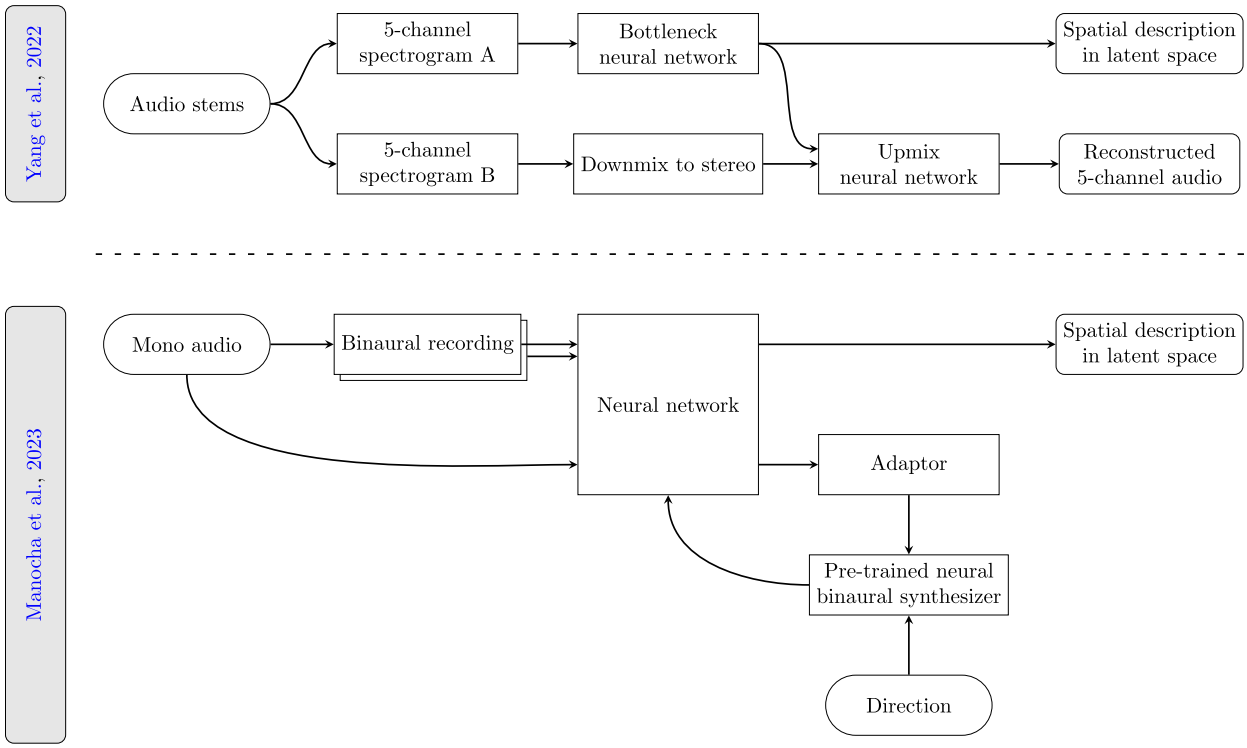

Fig. 6 Block diagrams of spatial description via DL [13, 111]

Where latent features are used a spatial descriptor, it is pertinent to remember that the latent space is a product of the training data and loss function, as opposed to the physical or psychoacoustic principals that DSP-based descriptors are typically based on. While the inherent dimensionality reduction of latent spaces is useful for comparing spatial features between recordings, individual dimensions of the space do not usually have a physical analogue. Analysis and interpretation of latent spaces is an ongoing research topic that has seen promising results in other research fields [113–115], and creating links between latent and physical space is viewed as critical to the applicability of the learned model [116].

Huang et al. and Zieliński et al. [117, 118] both attempt to learn spatial features of multichannel spectrograms using convolutional neural networks (CNNs). The former exploits an over-fitted network on short audio frames to generate inter-channel transfer functions (ITFs) in an effort to generate new coding/compression possibilities. The latter (which compares traditional DSP-based spatial features with those learned by a CNN for a binaural front-back classification task) offers a cautionary note—it found that while a CNN could produce features that resulted in higher accuracy for one specific subtask, the traditional features were more generalizable. Somewhat similarly, [119] creates estimations of fine-grained directional audio by creating neural embeddings of multichannel spectrograms generated from distributed microphone arrays.

On the side of generative ML/DL, [73] shows that a network can be trained to generate plausible Parametric Stereo (PS) (i.e., IID, ITD, and IC) values from a monophonic input (with upmixing as the intended use). More recently, Large Language Models (LLMs) have been used to generate textual descriptions of the spatial qualities of audio recordings; a good summary of the various different implementations is given in [120].

3.3.7 Standardized spatial cues

This review surfaced two spatial measurement systems that have attempted to become industry standards. The Quality Evaluation of Spatial Transmission and Reproduction using an Artificial Listener (QESTRAL) test methodology, detailed across 4 parts [121–124] is designed to measure spatial distortions present in audio processing, transmission, and reproduction. Additionally, Objective Measurements of Perceived Audio Quality-Multi Channel (PEAQ-MC) [48] is a multichannel extension to the PEAQ international standard [47], in which subband ILD, ITD, and IACC differences are calculated between a reference and distorted signal. In both cases the signals under test are expected to be in binaural

format. Other formats should be rendered to binaural by, e.g., convolution with a head-related impulse response (HRIR). It is worth noting that while the development of the measures is well documented and rooted in good theory, the review revealed only a handful of uses for each standard—[125, 126] for QESTRAL and [46] for PEAQ-MC—although this could be because simple use of a standard is typically not written about in academic literature.

There also exist several audio coding standards such as BCC, PS, MPEG-surround, spatial audio object coding (SAOC), and most recently immersive voice and audio services (IVAS) that include the calculation of spatial cues which can be directly integrated into future research (assuming one has access to encoder software). Excellent overviews of these standards and their underlying techniques are given in [127, 128].

Another avenue of standardization is the use of shared open-source software to extract spatial cues. The most established is the Auditory Modelling Toolbox (AMT) [129]—a set of MATLAB routines containing a vast number of algorithms for all aspects of human hearing, include spatial ones. Essentia [130] is a C++ framework for MIR, again containing several spatial algorithms. Released more recently, Binaspect [62] is a Python library designed specifically for extraction and visualization of spatial queues from binaural audio. The Generalized Cross Correlation with Phase Transform (GCC-PHAT) algorithm (which is a measure of DOA usually used in speech-processing, but also with examples for musical audio [52, 131, 132]) has implementations in several major programming languages [133–136]. For research involving transfer functions or impulse responses, the spatially oriented format for acoustics (SOFA) file format has emerged as the de facto standard for distributing spatially oriented acoustical data [137].

3.4 Framework for SIPs

Thus, we can construct a general model of a SIP algorithm according to Fig. 7. Recognizing that SIPs are inherently context-specific, we can consider any method that effectively summarizes bin-level spatial cues as a valid outcome. Section 3.3 details all techniques that were surfaced in this review, and for mathematical definitions the reader is referred to the original studies cited throughout. As seen in many of the reviewed studies, this framework could be applied equally to audio signals of higher order than stereo.

The diagram shows two separate paths for DSP and ML -based features respectively. It is noteworthy that these paths are essentially completely disparate, only converging at the final output stages, where there are examples of both methods being used to generate high-level spatial descriptors. In general, only one path needs to be followed to create a SIP. The lists of techniques given in each block are exemplary (not exhaustive), with their items somewhat interchangeable depending on the intended use of the SIP.

Although not shown on the diagram, it is important to choose spatial features that are relevant for the type of input audio and the intended output features. For example, if the input is standard stereophonic music and the output is intended to convey amplitude panning, then all phase-based spatial cues can be eliminated. However, if the output is intended to show LEV (even given the same stereophonic input), then phase cues cannot be eliminated. If the input audio is binaural, both phase and amplitude differences are relevant, but the frequency bands in which they are relevant differ depending on the intended output. For example, DOA only requires phase below approximately 1500 Hz [81], but other tasks should follow a binaural hearing model with more complex frequency weighting [46, 138].

3.5 Validation techniques

The papers surfaced in this review revealed several high-level categories of techniques to validate the success of the research. They tend to be closely linked to the downstream task, although some are universal. The most prevalent were perceptual listening tests, used in approximately half the studies. Some standardized perceptual tests are ITU-R-BS.1534 [139], which defines the MUlti Stimulus Hidden Reference and Anchor (MUSHRA) test used for measuring "intermediate audio quality", and ITU-R-BS.1116 [140] which is used for measuring "small impairments" in audio quality. In general though, it is more common to set up a custom subjective test asking the participants directly about the audio feature of interest. Several studies [53, 141–143] described their subjective tests as "MUSHRA-like", showing that the standardized frameworks are useful as a basis even if they do not perform the exact task required.

The second popular validation technique is objective metrics. The studies reviewed used a large variety of different metrics, although there is some standardization within specific scientific communities. For example, the blind source separation evaluation (BSS_EVAL) software library is frequently used to measure the similarity of audio waveforms in source separation tasks [144]. The same library can also be used for any task where waveform similarity is pertinent, e.g., [145] which is concerned with audio coding and compression. Another option for comparing the similarity of two waveforms is the "spectrogram difference" [146], which can be considered more in line with human perception but naturally contains a trade-off in time and frequency resolution.

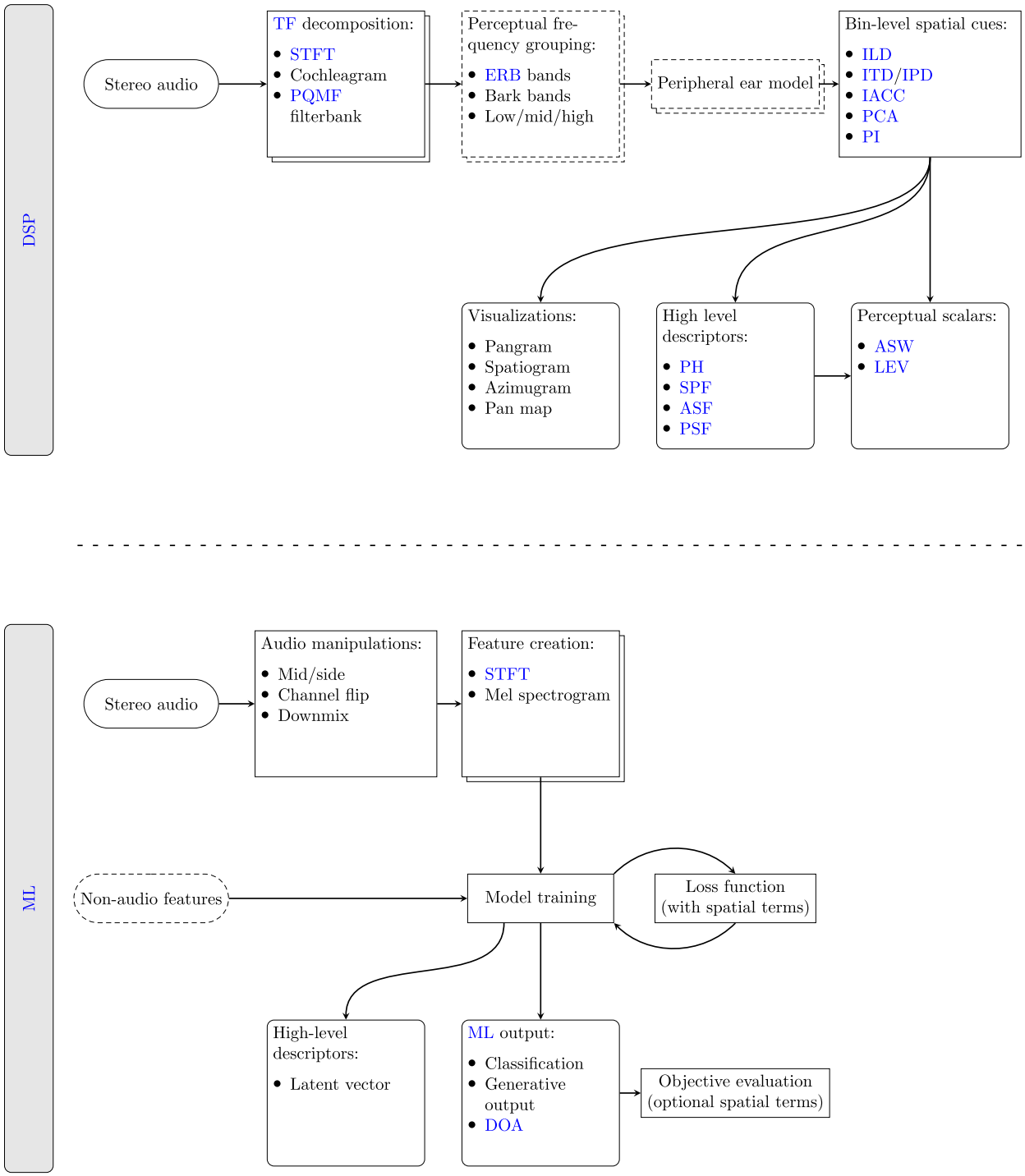

Fig. 7 Flow diagram of a generic SIP algorithm. Doubled blocks denote steps where the same processing is applied to both stereo channels. Dashed blocks are optional. The diagram is split between DSP and ML techniques, only one of which need be applied

Of note here is that the aforementioned techniques are tailored towards monophonic audio (although they can trivially be extended to multiple channels). More specific comparison metrics such as perceptual spectral difference (PSS) [147] or binaural-auditory-model-based quality (BAM-Q) [148] can be used to describe the difference between two stereo (specifically, binaural) signals. For perceptual differences in higher audio models (e.g.,

multichannel or ambisonics) it is typical to record or render the audio to a binaural signal and evaluate that [60, 149, 150].

If the research in question is related to creating or testing an objective metric, it can be useful to reverse the problem; One can perform simulations of real audio data by designing a set of signals with predictable differences, and analyze the results of the metric. Watcharasupat and Lerch [61] is one such example in which a novel metric is defined, and then its real results are tested against theoretical ones. In addition, diagrams are of the simulations are shown for visual inspection. This is another common validation technique in studies where a known, correct, low-dimensional value can be compared against a higher-dimensional metric, e.g., [38] which compares the panning position of an instrument to the peak in a PH, or [151] which compares a known azimuth and elevation with various activity maps generated from ambisonic audio. Note that comparing the similarity of PHs is error-prone as the standard distance metrics (e.g., Euclidean distance) do not necessarily correlate with perceptual difference [152]. Gomez et al. [39] attempts to side-step this issue by converting PHs into a different domain, although the efficacy of this approach remains largely unexplored.

In the ML/DL sphere it is normal to calculate the success of a model by its percentage of correct predictions, both for classical ML [31, 42, 49], and more modern DL approaches [50, 118, 153]. However, this obviously only works with labelled data. In the case that labels do not exist, one option is to create a self-supervised "pretext task" in which labelled data can be generated from unlabelled [112, 154–156]. In cases where there are no known objective metrics for the task at hand (e.g., the plausibility of novel stereo renderings [73]) inspiration can be taken from non-audio-domain tasks, utilizing, e.g., the minimum error of a large sample or the Fréchet distance between real and generated parameter spaces [157].

An under-studied but nonetheless important metric is the computational complexity of the algorithm described. Typical metrics to measure this are the time taken to process a given duration of audio on specific hardware [158], number of floating point operations (FLOPs) [159], or (for ML/DL) number of network parameters [159].

3.6 Audio datasets

3.6.1 Existing datasets

Table 5 shows a comprehensive list of audio datasets that were referenced by papers in this review. The only exception is "mobygratis" [168], which was released after the review was performed but nonetheless presents similar opportunities for research. Datasets that were not available publicly such as those used in Moving Picture Experts Group (MPEG) standardization have been omitted, and the availability of each dataset is noted. The "Size" column represents the number of musical tracks or clips present in the dataset, but does not take their lengths into account. "Duration" is given in days, hours, or minutes as appropriate, and is calculated as the sum of all track or clip durations. Non-musical recordings are not counted. Since it is assumed that larger and more varied datasets are preferable, the table is ordered by "Size". The columns "M", "S", and "B" indicated whether the datasets contain multitrack[4], stereo, or binaural recordings respectively. IEEE Signal Processing Society Online Resources, Li et al. and Bogdanov et al. [161, 182, 192] also contain lists of audio datasets (with some overlap to those listed here) that might be useful for future research.

3.6.2 Synthesized datasets

It is often the case than a dataset matching the exact task at hand does not exist, in which case one is synthesized from other datasets. In fact, some datasets listed in Table 5 are amalgamations of others—see the "Notes" column for details.

When creating a custom dataset, the most prevalent technique surfaced in this review was to convolve monophonic audio with some form of Impulse Response (IR). Said IR can be binaural (e.g., [76]), ambisonic (e.g., [193]), or a microphone array (e.g., [194]). It is also common that these the IRs add reverberation to a recording, as in e.g., [78, 195–197]. For research focused on binaural audio, [118] contains a comprehensive list of publicly available HRIR datasets.

Synthetic datasets of stereo music are slightly different in that they must be manually mixed from monophonic stems, which can be amplitude-panned in any way that the experiment requires. For example, in [44] the stems are placed according to the signal model described in the study, and [13, 198] pan the stems randomly.

Taking a slightly different route, [164] introduces a dataset of song stems synthesized entirely from Musical Instrument Digital Interface (MIDI) files using professional-grade virtual instruments (VIs), motivated by a lack of commercial music datasets offering stems.

3.7 Uses of stereo image profiles

3.7.1 Current uses

Broadly speaking, there are two categories of tasks that spatial information in audio has historically been used for. Firstly there are those focused on a single recording from which the information was extracted. Compression, source separation and upmixing are the most

[4] A Multitrack recording indicates that individual stems of a recording session are available, but not necessarily a stereo mix.

Table 5 Audio datasets, ordered by size (number of tracks or clips)

Name	Audio format	Size	Duration	M	S	B	Availability	Notes
FMA [160]	mp3 various bitrates	106574	343d		✓		Download	Real music. Huge variety. Contains common audio features and tags.
MTGJ [161]	mp3 320kbps	55609	157d		✓		Download	Real music. Wide variety of genres, instruments, and moods.
ccMixter[162]	Various mp3/wav	≈ 48000[§]	?[§]	✓	✓		Download[†]	Real music in many genres. Some stereo, some stems. Vast, but some download links are broken.
ASASCBRM [163]	wav 48kHz 24bit	5928	11h		✓		Download	Synthesized binaural from other multitrack datasets. 30 s clips.
Slakh [164]	wav 44.1kHz	2100	145h	✓	✓		Download	Mono stems. Synthesized from midi.
FAIR-Play [165]	wav 48kHz 16bit	1871	5h		✓		Download	Binaural. 10 s clips. Only 1 recording room.
MAESTRO [166]	wav 44.1/48kHz 16bit	1276	199h		✓		Download	Only classical piano.
MSSS [167]	wav 44.1kHz 16/24bit	625	40h	✓	✓		Download[⋆]	Stems uncompressed. Example stereo mixes in mp3 only.
mobygratis [168]	wav 44.1kHz 24bit	499	35h	✓	✓		Download[⋆]	Stereo stems and mixdowns. Multiple genres but only 1 artist.
RWC [169–172]	CD, 44.1kHz 16bit	465	22h		✓		CD[‡]	Multiple datasets each of a different genre.
MedleyDB 2.0 [173]	wav 44.1kHz 16bit	330	7h	✓			Download[‡]	Real music. Mix of genres. Some instrumental, some vocal. More tracks have been added since publication.
MUSDB18 [174]	AAC 256kbps	150	10h	✓			Download	Only stems. Derived from other datasets such as [175, 176].
MUSDB18-HQ [177]	wav 44.1kHz	150	10h	✓			Download	Only stems. Derived from other datasets such as [175, 176].
MedleyDB [176]	wav 44.1kHz 16bit	132	7h	✓			Download[‡]	Stereo mix + dry and processed stems.
DSD100 [175]	wav 44.1khz	100	7h	✓	✓		Download	Stems and downmixes, derived from [167].
BTPAB [178]	wav 48kHz 16bit	49	49m		✓		Download	60 s clips. Only 1 recording session.
FiloSax [179, 180]	wav 44.1kHz 24bit	48	5h	✓			Download[‡]	Only jazz.
DAOC [181]	CD, 44.1kHz 16bit	44	25m		✓		CD	Anechoic orchestra.
URMP [182]	wav 48kHz 24bit	44	1h	✓			Download	Anechoic classical music. Only stems.
BOMA [183, 184]	CD, 44.1kHz 16bit	31	1h		✓	✓	CD	Anechoic and semi-reverberant orchestra.
TSS [185]	wav 44.1kHz 16bit	23	1.5h	✓			Download[⋆]	Both live and studio sessions.
QUASI [186]	wav 44.1kHz	11	≈ 30m	✓			Download[‡]	Mix of genres.
SQAM [187]	FLAC 44.1kHz 16bit	10	1h		✓		Download	Not all music, some stereo some mono.
RIOOSSR [188]	wav 48kHz 24bit	4	5m				Download	Ambisonic and channel-based versions.
ARSSO [189]	mp3 192kbps 48kHz	4	11m	✓			mp3: Download, wav: Contact author	Anechoic classical music. Only stems.
UIowaMIS [190]	wav 44.1kHz 16bit or 96kHz 24bit	?[‖]	23h	✓			Download[†]	Isolated notes, not musical compositions. ≈ 85% stereo.
Bach10 [191]	wav 44.1kHz	?[§]	?[§]				Contact author	Only Bach chorales.

M Multitrack, *S* Stereo, *B* Binaural

[⋆]Individual tracks. [†]Individual files. [‡]Must request access. [§]Exact size unclear. [‖]Instrument recordings

prevalent examples which arose in this review. In recent years, these have become somewhat intertwined with the advent of new commercial spatial audio formats such as Dolby Atmos and MPEG-H 3D Audio; older content which was released in stereophonic or multichannel format can benefit from combinations of source separation, upmix, and coding/compression techniques that increase the perceptual quality when rendered on modern equipment. The spatial attributes extracted in these cases tend to be the low-level spatial cues detailed in Sect. 3.3.1 since the tasks naturally require that the original audio is re-synthesized in some way.

Quality prediction (and in particular, *spatial* quality prediction) is another area of considerable interest, the driving force being that traditional perceptual listening tests are slow and costly [75, 199]. This differs slightly from the previously mentioned tasks in that it implies a comparison between a reference and degraded signal, and uses summaries of low-level spatial cues combined with non-spatial measurements. The type of spatial attributes used in these cases are more in line with the high-level descriptors detailed in Sect. 3.3.5. In particular, [45, 60, 61, 200] use distortion/difference of ILD/ITD/ITD in their perceptual models, and [26, 27] use "sound field maps" or panning histograms to describe the spatial information present in the analyzed audio. Combining the two approaches, [201] defines the directional loudness distortion (DSD) metric to show the difference between a reference and degraded "directional loudness map".

The studies surfaced in this review reveal that there was a spike of interest in binaural quality prediction and analysis since 2023. This appears to be driven by a renewed interest in spatial audio[5] [75, 149, 150, 202, 203], room acoustics [196, 204] and VR rendering [72, 205, 206], along with the increased ease of gathering or creating binaural audio datasets.

Secondly, there are those tasks which focus on the relationship of a single recording to a larger corpus. Classification is a typical use case, with the list of predicted labels including genre/recording style [14, 17, 31], artist/composer/DJ [17, 207], and decade [42]. Important to note in all of these cases is that the ground truth is known a-priori. However, with the vast size of modern audio datasets, acquiring reliable annotations becomes increasingly difficult. For example, [32, 55] both specifically state that panning parameters, digital audio workstation (DAW) sessions, and musical scores are typically not available for most recorded music, and therefore any such information must be derived after-the-fact from a mixdown.

An interesting middle-ground between the two categories can be found in the aforementioned [96], which uses temporal changes in PHs to segment a song into musical structures (verse, chorus, etc.). Likewise, [43, 44] attempt both a generic classification of stereo field, and a further application to segment longer audio recordings using that classifier.

In addition, a frequently occurring pattern among studies in this review is that spatial cues are used as objective measurements for other tasks. In particular, [178, 208–210] demonstrate that the incorporation of spatial terms into DL loss functions, forcing the networks to learn or generate spatial features.

3.7.2 Potential uses

Compelling areas for further research can be inferred from what is missing from the survey results. In total, 48 studies were deemed to be making use of ML or DL, all of which were performing some form of *supervised* learning. Just a single study [14] mentioned performing *unsupervised* learning on spatial cues—a self-organising map was used to discriminate between two clusters of tracks when simple linear regression failed. Wang et al. [211] presents a similar study, although the audio was non-musical. Mitic and Rossholm [152] gives an overview of the suitability of various panning features with regard to unsupervised learning, and [62] describes some spatial cues that can be used as input to ML routines.

Likewise, relatively few studies perform any spatial analysis of extremely large and varied musical datasets. For example, [146] used music clips from FMA [160] as background noise in a task focused on the binaural rendering of speech, and while the standard spatial cues were extracted, the music was not of primary interest. Tzanetakis et al. and Tardieu et al. [17, 31, 42] use mid-size (thousands of tracks) datasets to demonstrate various classification tasks, although they unfortunately predate the existence of the aforementioned extremely large datasets. Nonetheless, these older studies present an interest in the dataset itself as opposed to low-level features of the audio. A much more recent study [155] expresses a similar interest, measuring production tends in a huge (albeit private) dataset containing hundreds of thousands of tracks. Analysis of the spatial features in vast musical corpora therefore presents an opportunity for further research. In particular, there are no studies on spatial degradation caused by lossy compression of said datasets, even though the largest [160, 161] are both encoded using the MPEG layer 3 (MP3) codec, and [62, 210] clearly show spatial degradation of their test signals

[5] As mentioned in Sect. 3.5, the typical way to measure perceptual effects of higher-order audio models is to render them to a binaural signal.

when distorted with lossy compression and source separation respectively.

Ziemer et al. and Wilson and Fazenda [212, 213] present a further basis for investigation, concluding that two objective measures of spaciousness (inter-channel cross correlation (ICCC) and SPF) did not match the results of a perceptual music similarity test. This issue warrants further investigation, since research into the SIPs of tracks in a large music corpus will naturally involve a measure of similarity.

3.7.3 Example use of SIP framework

To demonstrate the effectiveness of the framework outlined in Sect. 3.4, we will consider a hypothetical experiment on a dataset that has not yet received any investigation into stereo image. Suppose the intention was to recreate the analysis of trends in production techniques from [155], but using a large open dataset such as FMA. Said study employed a simple stereo field classification based on ILD. Each track is classified as either "mono", "narrow", "balanced", or "wide". The exact equation to calculate the stereo width is not given in the study. In order to recreate such tags for FMA we can follow the DSP framework to create a SIP as suggested in Table 6.

4 Discussion

Overall, this review paints a picture of a research landscape that is somewhat fractured. The core psychoacoustic principals and terminology is fairly well standardized, although many researchers offer their own variations on the precise implementations and inferences of both high-level spatial features and low-level spatial cues. The net result is that searching for relevant literature by *downstream task* rather than *spatial attribute* will usually be most efficient. However, this method presents diminishing returns for tasks with few previous examples or non-standard terminology (such as finding SIPs), and the opportunity of "cross-pollination" from other fields of audio research is lost. This review intended to fix the cross-pollination problem, although it turned out that the resulting scope was so large that the researcher themselves became a bottleneck. Experiments in performing audio-related literature reviews using LLMs have shown potential, although it was concluded that they are no substitute for human researchers [214].

Open-source tooling such as those mentioned in Sect. 3.3.7 go some way to towards giving researchers a shared language around spatial aspects of stereo audio. Indeed, many of the studies in this review contained links to both source code and sound samples, but precious few were still active. The move towards shared audio datasets is a step forward though, since until recently large datasets of real music were either completely unavailable or confined to private collections. With the introduction of the datasets listed in Sect. 3.6 it might now be possible for music-specific audio research to share a common footing, just as happened in the image processing community [160].

While this review aimed to provide an exhaustive list of spatial descriptors, the fact that some studies were only added after following references shows that the original search method was not foolproof. On the other hand, while the search result might not have been exhaustive, the goal of gathering at least one example of all spatial descriptors appears to have been met. In the DL sphere this claim can be questioned since each study tends to focus on an individual task using a dataset specific to that task, thereby generating spatial descriptors that are also specific to that task. While classical DSP methods comprise the majority of the included studies, ML and DL are rising sharply, indicating that the latest computing techniques and datasets are enabling research previously out of reach via DSP.

5 Conclusion

A systematic literature review (SLR) following the PRISMA guidelines was performed to find methods of describing the stereo image profile (SIP) of stereophonic music recordings. The terminology, methods, and signal processing techniques are listed in detail, and the conglomeration of data extracted from the included studies provides a framework from which custom SIP algorithms appropriate for any given task can be designed and implemented. Datasets containing suitable audio material are also listed alongside descriptions of the contained audio.

Table 6 Hypothetical use of SIP framework to recreate stereo width analysis from [155] on the FMA dataset

Component	Method	Reasoning
TF decomposition	STFT	Common and computationally efficient.
Frequency grouping	None	[155] does not give a perceptual frequency grouping for stereo width.
Peripheral ear model	None	STFT does not require a peripheral ear model.
Spatial cue	ICLD	[155] states that the stereo width classification is "based on the ICLD".
Perceptual scalar	Stereo width	Calculated by, e.g., mean of all bin-level ICLD values, quantized to 4 values.

While the focus was on stereophonic recordings, many studies are equally relevant to other audio formats such as multichannel and ambisonics. Readers can benefit from inspiration and "cross pollination" of diverse areas of audio research, and the review as a whole enables researchers and practitioners to jump-start new studies into the spatial aspects of music.

Appendix 1 Search terms

Population: "music", "musical", "audio", "sound"

Context: "stereo", "stereophonic", "binaural", "spatial audio", "ambisonic", "ambisonics", "b-format", "multi-channel"

Outcome: "panning", "information retrieval", "spaciousness", "descriptors", "spatial image", "stereo image", "spatial cues", "disentangle", "disentangling", "disentanglement", "spatial decomposition", "spatial information", "spatial descriptors", "spatial quality", "spatial analysis", "binaural model", "goniometer", "lissajous plot", "systematic review"

Negative terms: "stereo vision", "computer vision", "image processing", "stereo camera", "stereo cameras"

Specific terms: "spatial audio coding", "spatial audio object coding", "directional audio coding", "binaural cue coding", "panning spectrogram", "stereo panning spectrum", "object-based audio", "mpeg-h audio", "mpeg-h 3D audio", "dolby atmos", "sony 360 audio", "sony 360 RA", "360 reality audio", "QESTRAL"

Appendix 2 Database queries

Listing 2: Web Of Science query

```
((((TS=("music") OR TS=("musical") OR TS=("audio") OR TS=("sound")) AND (TS=("stereo") OR
TS=("stereophonic") OR TS=("binaural") OR TS=("spatial audio") OR TS=("ambisonic") OR
TS=("ambisonics") OR TS=("b-format") OR TS=("multi-channel")) AND (TS=("panning") OR
TS=("information retrieval") OR TS=("spaciousness") OR TS=("descriptors") OR TS=("spatial image")
OR TS=("stereo image") OR TS=("spatial cues") OR TS=("disentangle") OR TS=("disentangling") OR
TS=("disentanglement") OR TS=("spatial decomposition") OR TS=("spatial information") OR
TS=("spatial descriptors") OR TS=("spatial quality") OR TS=("spatial analysis") OR TS=("binaural
model") OR TS=("goniometer") OR TS=("lissajous plot") OR TS=("systematic review"))) NOT
(TS=("stereo vision") OR TS=("computer vision") OR TS=("image processing") OR TS=("stereo camera")
OR TS=("stereo cameras"))) OR (TS=("spatial audio coding") OR TS=("spatial audio object coding") OR
TS=("directional audio coding") OR TS=("binaural cue coding") OR TS=("panning spectrogram") OR
TS=("stereo panning spectrum") OR TS=("object-based audio") OR TS=("mpeg-h audio") OR TS=("mpeg-h
3D audio") OR TS=("dolby atmos") OR TS=("sony 360 audio") OR TS=("sony 360 RA") OR TS=("360 reality
audio") OR TS=("QESTRAL")))
```

Listing 3: Semantic Scholar query

```
(("music" | "musical" | "audio" | "sound") + ("stereo" | "stereophonic" | "binaural" | "spatial
audio" | "ambisonic" | "ambisonics" | "b-format" | "multi-channel") + ("panning" | "information
retrieval" | "spaciousness" | "descriptors" | "spatial image" | "stereo image" | "spatial cues" |
"disentangle" | "disentangling" | "disentanglement" | "spatial decomposition" | "spatial
information" | "spatial descriptors" | "spatial quality" | "spatial analysis" | "binaural model" |
"goniometer" | "lissajous plot" | "systematic review")) | ("spatial audio coding" | "spatial audio
object coding" | "directional audio coding" | "binaural cue coding" | "panning spectrogram" |
"stereo panning spectrum" | "object-based audio" | "mpeg-h audio" | "mpeg-h 3D audio" | "dolby
atmos" | "sony 360 audio" | "sony 360 RA" | "360 reality audio" | "QESTRAL")
```

Appendix 3 Semantic Scholar API payload

Listing 4: SS search payload. See appendix B for the search query.

```
{
  "query": <see appendix B>,
  "limit": 20000,
  "fields": "influentialCitationCount,citationCount,title,venue,abstract,referenceCount,
citationStyles,year,openAccessPdf,authors,url,fieldsOfStudy,s2FieldsOfStudy,externalIds",
  "fieldsOfStudy": [
    "Computer Science",
    "Mathematics",
    "Engineering"
  ],
  "token": null
}
```

Appendix 4 Search result filtering algorithm

Listing 5: Python pseudocode to filter duplicate database records from search results

```
def normalize_title(title):
    return "".join(char for char in title if char.isalnum()).lower()

doi_seen = set()
title_and_year_seen = set()

for record in search_results:
    norm_title = normalize_title(record.title)
    if doi != "":
        if doi in doi_seen:
            record.reject = "duplicate_by_doi"
        else:
            doi_seen.add(doi)
    elif (this_title, this_year) in title_and_year_seen:
        record.reject = "duplicate_by_title_and_year"
    title_and_year_seen.add((norm_title, record.year))
```

Appendix 5 All downstream tasks

Table 7 List of all downstream tasks and their number of occurrences

Task	Count	Very relevant
MIR	15	yes
classification	10	yes
spatial analysis	8	yes
visualization	3	yes
musicology	2	yes
compression	35	no
source separation	30	no
rendering	23	no
upmix	22	no
quality prediction	20	no
coding	13	no

Task	Count	Very relevant
perceptual model	13	no
localization	10	no
psychoacoustics	8	no
acoustics	6	no
enhancement	5	no
recording	3	no
directional segmentation	2	no
synthesis	2	no
audio format conversion	1	no
mastering	1	no
restoration	1	no
inpainting	1	no
telepresence	1	no
audio object detection	1	no
beamforming	1	no
cross-modal ML	1	no
downmix	1	no
action recognition	1	no
audio forensics	1	no

Abbreviations

AMT	Auditory modelling Toolbox
API	Application programming interface
AR	Augmented reality
ARSSO	Anechoic recording system for symphony orchestra
ASASCBRM	Database for Automatic Spatial Audio Scene Classification in Binaural Recordings of Music
ASF	Amplitude stereo features
ASW	Apparent source width
BAM-Q	Binaural-auditory-model-based quality
BCC	Binaural cue coding
BMS	Bivariate mixture space
BOMA	Bang & Olufsen Music for Archimedes
BSS_EVAL	Blind source separation evaluation
BTPAB	ByTeDance paired ambisonic and binaural
CLD	Channel level difference
CNN	Convolutional neural network
CSV	Comma separated variable
DAOC	Denon anechoic orchestral collection
DAW	Digital audio workstation
DJ	Disc kockey
DL	Deep learning
DOA	Direction of arrival
DOI	Digital object identifier
DSD	Directional loudness distortion
DSD100	Demixing secret database
DSP	Digital signal processing
ERB	Equivalent rectangular bandwidth
FA	Frequency-azimuth
FFT	Fast Fourier transform
FLOP	Floating point operation
FMA	Free music archive
FWHM	Full width half maximum
GCC-PHAT	Generalized cross correlation with phase transform
HRIR	Head-related impulse response
IAC	Interaural coherence
IACC	Interaural cross correlation
IC	Inter-channel coherence
ICC	Inter-channel correlation
ICCC	Inter-channel cross correlation
ICLD	Inter-channel level difference
ICPD	Inter-channel phase difference
ICTD	Inter-channel time difference
IID	Interaural intensity difference
ILD	Interaural level difference
ILR	Interaural level ratio
IPD	Interaural phase difference
IQR	Interquartile range
IR	Impulse response
ITD	Interaural time difference
ITF	Inter-channel transfer function
IVAS	Immersive voice and audio services
KKS	The Knowledge Foundation
LEF	Lateral energy fraction
LEV	Listener envelopment
LLM	Large language model
MAESTRO	MIDI and Audio Edited for Synchronous TRacks and Organization
MIDI	Musical instrument digital interface
MIR	Music information retrieval
ML	Machine learning
MP3	MPEG layer 3
MPEG	Moving picture experts group
MSSS	Mixing secrets for the small studio
MTGJ	Music Technology Group Jamendo
MUSHRA	MUlti Stimulus Hidden Reference and Anchor
PCA	Principal component analysis
PCO	Population, context, outcome
PEAQ	Objective measurements of perceived audio quality
PEAQ-MC	Objective measurements of perceived audio quality-multi-channel
PH	Panning histogram
PI	Panning index
PQMF	Pseudo quadrature mirror filter
PRISMA	Preferred Reporting Items for Systematic reviews and Meta-Analyses
PS	Parametric stereo
PSF	Phase stereo features
PSS	Perceptual spectral difference
QESTRAL	Quality Evaluation of Spatial Transmission and Reproduction using an Artificial Listener
QUASI	QUaero Audio SIgnals
RIOOSSR	Recordings of Italian opera orchestra and soloists in a silent room
RWC	Real World Computing
SAOC	Spatial audio object coding
SIP	Stereo image profile
SLR	Systematic literature review
SOFA	Spatially oriented format for acoustics
SPF	Stereo panning features
SPS	Stereo panning spectrum
SQAM	EBU Sound Quality Assessment Material
SS	Semantic Scholar
STFT	Short-time Fourier transform
TF	Time-frequency
TFF	Time-frequency filter
TSS	Telefunken Studio Sessions
TSV	Tab separated variable
UIowaMIS	University of Iowa Musical Instrument Samples
URL	Universal resource locator
URMP	University of Rochester Multi-Modal Music Performance
VI	Virtual instrument
VR	Virtual reality
WOS	Web Of Science

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Authors' contributions
RM conceptualized the SLR, gathered and interpreted data, formed the syntheses, and wrote the manuscript. AR performed the post-hoc data analysis and edited the manuscript. The authors read and approved the final manuscript.

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Data availability
The dataset supporting the conclusions of this article is included as an additional file. It comprises a concatenation of search results from the three databases searched, in CSV format. The columns are:

Table 8 Explanation of fields in accompanying file of database search results

Header	Explanation
`url`	The URL of the record within the database
`title`	The title of the article
`abstract`	The abstract of the article
`year`	The year of publication
`authors`	The authors of the article, in the raw format given by the search results
`source`	The publication venue
DOI	The DOI of the article
`db`	The name of the database from which the record was extracted
`db_id`	The unique identifier within the database

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The author declares no competing interests.

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TOWARDS AN OBJECTIVE COMPARISON OF PANNING FEATURE ALGORITHMS FOR UNSUPERVISED LEARNING

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ABSTRACT

Estimations of panning attributes are an important feature to extract from a piece of recorded music, with downstream uses such as classification, quality assessment, and listening enhancement. While several algorithms exist in the literature, there is currently no comparison between them and no studies to suggest which one is most suitable for any particular task. This paper compares four algorithms for extracting amplitude panning features with respect to their suitability for unsupervised learning. It finds synchronicities between them and analyses their results on a small set of commercial music excerpts chosen for their distinct panning features. The ability of each algorithm to differentiate between the tracks is analysed. The results can be used in future work to either select the most appropriate panning feature algorithm or create a version customized for a particular task.

1. INTRODUCTION

The vast majority of recorded music available today was recorded and produced in stereophonic format. The musicians and producers involved will often pay close attention to the placement of sound objects in the stereo field, and various pieces of equipment such as goniometers, phase scopes, and stereo balance indicators have been standard equipment in recording studios for decades. There is therefore a wealth of spatial information encoded in the final recording. Extracting and disentangling that information has been the subject of research for many years, in various different guises.

A typical use case is classification, where stereo features have been used as a predictor of genre [1, 2, 3], artist/composer/DJ [2, 4], and decade [5]. Additionally, panning features have been used on their own merit (as opposed to predictors of another feature) for tasks such as source separation [6, 7], upmixing [8], and music similarity/recommendation [9, 4]. The classification of panning features themselves has also been studied as a precursor to upmixing, source separation, and audio enhancement [10, 11], giving the ability to select appropriate algorithms based on audio content.

In the aforementioned cases, the audio analysed has either comprised single excerpts or lists of 1000-2000 tracks with labels assigned. However, in recent years much bigger musical datasets have become available, e.g. [12, 13], which contain 55000 and 100000 tracks respectively. These datasets do not contain any metadata regarding panning features, nor any similar information such as recording methods/style. While they do offer genre and date tags, panning features on their own have previously shown low correlation with genre and decade [2, 5], from which we can infer that genre and date cannot be reliably used a proxy for panning features. If one wishes to use these datasets for the aforementioned use cases, there is hence a need to perform unsupervised learning on the extracted panning features, and for this to be successful we must understand the abilities and limits of the extraction method being used.

Several examples of panning feature extractors for stereo audio signals exist in the literature [1, 14, 9, 5], but as yet there has not been a comparison of them and no discussion of what panning features they are able to detect. The aim of this paper is to provide such discussion. It will begin by summarizing the relevant literature in Section 2, describing and drawing links between existing panning feature extraction algorithms. Section 3 presents examples of commercial music chosen for their different uses of amplitude panning, plus the methods used to compare the panning feature algorithms. Section 4 shows the results of the analysis and Section 5 will discuss some overall trends that emerged across all algorithms. Concluding remarks are offered in Section 6.

2. PREVIOUS WORK

[8, 15] are seminal works describing the 'panning index' – a description of amplitude panning for each time-frequency bin in a stereo spectrogram. [8] also presents the 'ambience index' – a measure of inter-channel correlation in time-frequency (TF) bins. While they target re-synthesis and upmix as use cases, the underlying techniques have become the basis for many applications since. [6] is another seminal work which describes a frequency-azimuth plane constructed from phase cancellation of delayed stereo TF planes. Similarly, it targets re-panning and re-synthesis but has since been repurposed. [1, 2] reuse the panning index from [8] (renamed to 'Stereo Panning Spectrum (SPS)') and derive a set of Stereo Panning Features (SPF) that are used for genre classification and Music Information Retrieval (MIR). [14] details a method to estimate the perceptual stereo width of a music recording by forming a Panning Histogram (PH) based on the frequency-azimuth plane from [6]. Another PH method (this time based on the SPS from [1]) is described in [9, 16], specifying the measurement of music similarity as the primary use case. It contributes an extra step of converting the PH into Panning Coefficients (PC) through cepstral analysis, stating that this allows the resulting vector to be used with any generic classification algorithm. In a similar vein (and with classification as a use case) [5] combines cochleagram differences with an analysis of phase differences between

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channels to form two feature sets - the 'Amplitude Stereo Features (ASF)' and 'Phase Stereo Features (PSF)'.

[17, 11, 10] collectively define a new paradigm of classifying regions of audio whereby stereo mixtures are modelled as combinations of centre, panned, and uncorrelated sources. Even though genre classification is used as an example use case, they specifically state that the class representing a combination of all source types (i.e. most music recordings) is very difficult to identify and therefore omitted from the model.

Similar to the phase-based features defined in [5], there are a group of algorithms based on the emulation of a studio goniometer; [4, 3] use both a goniometer box-counting technique and inter-channel correlation analysis to derive a time-varying measurement of 'spaciousness'. Interestingly though, [18] found that subjective impressions of spaciousness in various musical styles were largely contradictory to the results of goniometer analysis.

More recently, Machine Learning (ML)-based techniques for disentangling spatial information from audio content have been demonstrated. With a primary use case of upmixing stereo to multichannel, [19] demonstrates that a latent vector formed by running multichannel audio through one half of the learned neural network represents the spatial information contained in the original signal. Similarly, [20] takes a stereo microphone input and regenerates one channel from the other by learning an inter-channel spatial representation. Mirroring our problem statement, [20] specifically mentions a lack of labelled real-world data as motivation for the use of unsupervised learning.

2.1. Selected algorithms

This paper will focus on four algorithms that estimate panning features from stereophonic commercial music recordings and do not require training on a dataset. Specifically, PH [14] PC [9], ASF [5], and SPF [1]. For mathematical definitions the reader is referred to the original papers, but a summary of each is given here.

2.1.1. Panning Histogram (PH) — Sarroff, 2008 [14]

The left and right channels are split into chunks and transformed into the frequency domain using a Fast Fourier Transform (FFT). Each FFT frame is converted into a frequency-azimuth (FA) plane through phase cancellation techniques. The left and right FA planes are concatenated, and a histogram is calculated across all frames and all frequencies. The result is a histogram of energy at each azimuth bin.

Note: The paper continues to collapse the histogram into a single scalar representing overall width, but we will not consider that here since the intent is to compare the PH vector with other high-dimensional representations of panning features.

2.1.2. Panning Coefficients (PC) — Gomez, 2008 [9]

The left and right channels are split into chunks and transformed into magnitude spectrograms using a Short-time Fourier Transform (STFT). The ratio of power spectra is calculated for each frame, resulting in an azimuthal panning factor each frequency bin. These values are warped by a function that accounts for the human perception of the direction of arrival of sounds. Each frame is then converted to an energy-weighted histogram and the mean over time is taken. The histogram is then converted to a compact representation by performing cepstral analysis and truncating the results to length 20.

2.1.3. Amplitude Stereo Features (ASF) — Tardieu, 2011 [5]

The ASF are created from a stereo cochleagram. Left and right audio channels are split into frequency sub-bands using gammatone filters spaced equally on the Equivalent Rectangular Bandwidth (ERB) scale. The amplitude difference of the two cochleagram channels is taken to produce a Cochleagram Difference (CD). The ASFs are created by taking the mean and standard deviation over various axes of the CD. The mathematical definition of a cochleagram is not given in [5], so for this paper we have followed the steps in [21], creating a cochleagram directly from time domain audio as opposed to using an FFT. [5] also does not state the exact form of the resulting vector, so for this paper all features are simply concatenated into a single vector.

2.1.4. Stereo Panning Features (SPF) — Tzanetakis, 2007 [1]

The left and right channels are converted into complex TF regions by chunking, windowing, and the STFT. These two TF regions are converted into an SPS, which is then split into high, mid, and low frequency bands. Each sub-band (along with the full spectrum) are collapsed by calculating the Root Mean Squared (RMS) across the frequency dimension, and smoothed in time by calculating a running mean and standard deviation over M frames. The mean and standard deviation over time are then calculated, resulting in a 16-dimensional vector.

3. METHOD

To assess each algorithm they are run on a selection of musical excerpts exhibiting panning features that it would be useful to differentiate between. As a way of delineating the entire panning feature gamut, we use two 'anchor tracks' - one strongly mono and one strongly stereo. The other tracks each represent an interesting panning feature. For each algorithm, the panning features for all tracks are calculated and considered to be points in an N-dimensional feature space. The Euclidean distances (D) between pairs of tracks are calculated, and the set for each algorithm is normalized so that $D \in [0, 1]$. Specifically, it is hypothesized that

1. The anchor tracks should be located farthest apart in the feature space
2. The panning feature tracks should be well separated and evenly distributed between the anchors

Any pair with low separation (i.e. the algorithm cannot differentiate between them) is further examined to find out why. We define a low separation to be $D < 0.1$.

3.1. Musical excerpts

All excerpts are defined by their title, artist, release year, and timing information. For the rest of the paper they will be referred to by abbreviated title. The panning features they exhibit are also defined here. Anchor tracks are denoted by '*' for mono and '**' for stereo.

- *'Round Midnight* (RM*)*, Miles Davis, 1957, 0:00–0:30. A jazz piece recorded in mono.
- *This Is The Thing (TITT)*, Fink, 2007, 1:00–1:30. An acoustic song with studio reverb added. The vocal switches from single-tracked to multi-tracked with moderate amplitude panning approximately halfway though the excerpt.

- *Laser Beam (LB)*, Low, 2001, 0:00–0:30. Electric guitar, bass, and vocals with an enveloping sound created mostly by reverberation and multi-miking, but only subtle amplitude panning.
- *Manifesto (M)*, Chilly Gonzales, 2025, 1:00–1:30. Solo upright piano recorded live on stage with close stereo microphones. Individual notes are audibly separated in space even though no studio panning has been applied.
- *Cut And Clicks (CAC)*, Tetsu Inoue, 2000, 0:00–0:30. An electronic piece that is characterized by heavy use of extreme panning, unnatural sounds, and singular audio events with very short durations.
- *Fly Me To The Moon (FMTTM)*, Frank Sinatra, 2008, 1:00–1:30. A full big band recorded live. Most instruments are statically panned to a fairly strong degree.
- *I Saw Her Standing There*** *(ISHST**)*, The Beatles, 1963, 0:00–0:30. A pop song with strong, rudimentary panning. Drums, bass, and lead guitar are panned hard left, while vocals and rhythm guitar are panned hard right.

We define 'natural panning' to mean that recorded instruments that have been amplitude panned to a static position in the final recording. Examples of natural panning are *TITT*, *FMTTM*, and *ISHST***. The counterpart is 'unnatural panning', which we use to refer to sounds that move very fast between left and right. *CAC* is an example of unnatural panning.

All recordings have a sample rate $f_s = 44100$ Hz and were normalized to an average loudness of -23 Loudness Units Full Scale (LUFS) before processing according to [22, 23].

3.2. Algorithm parameters

Each algorithm is defined by a set of input parameters, and the values used for this paper are shown in Table 1. For precise meanings of each one, the reader is directed to the original papers.

Algorithm	Parameters	Dimensions
PH [14]	FFT size = 1024 hop size = 512 azimuth bins = 129	129
PC [9]	FFT size = 1024 hop size = 512 azimuth bins = 129 cepstrum coefficients = 20	20
ASF [5]	Low f_c = 30Hz High f_c = 11050Hz number of bands = 70 chunk length = 20ms hop length = 10ms	272
SPF [1]	FFT size = 1024 hop size = 512 M = 40 low band = 0-250 Hz mid band = 250-2500 Hz high band = 2500-22050 Hz	16

Table 1: *Parameters for panning algorithms*

4. RESULTS

The Euclidean distances between pairs of tracks in each feature space are shown in Fig. 1. Within each feature space, distances have been normalized to the range $D \in [0, 1]$. Cells are coloured in greyscale with white and black representing 0 and 1 respectively, and track titles are abreviated.

Fig. 1 shows that some panning feature algorithms have a clear bias towards one of the anchor tracks. (visible as a much darker section in otherwise pale matrix). A bias towards the mono anchor (i.e. a dark horizontal line on the bottom row) means that all panning feature tracks appear 'very mono' to the algorithm. Conversely, a bias towards the stereo anchor (i.e. a dark vertical line in the left column) means that all panning feature tracks appear as 'very stereo' to the algorithm.

To quantify this, we define a 'bias score' (B) according to Eq. (1). Firstly the mean distances from each anchor track to all panning feature tracks are calculated, then B is defined as the natural logarithm of the ratio of mean mono distance ($\bar{D}_m$) to mean stereo distance ($\bar{D}_s$). A score of $B = 0$ means no bias, a positive score ($B > 0$) means a bias towards the stereo anchor and negative score ($B < 0$) means a bias towards the mono anchor. This is calculated by Eq. (1) where $T = \{t : t \in T\}$ is the set of panning feature tracks, $D_{m,t}$ is the distance between the mono anchor and a panning feature track, and $D_{s,t}$ is the distance between the stereo anchor and a panning feature track.

$$\begin{aligned} \bar{D}_m &= \frac{1}{|T|} \sum_{t \in T} D_{m,t} \\ \bar{D}_s &= \frac{1}{|T|} \sum_{t \in T} D_{s,t} \\ B &= \ln \frac{\bar{D}_m}{\bar{D}_s} \end{aligned} \tag{1}$$

The scores for each algorithm are shown in Table 2 alongside their interpretations.

Algorithm	$\bar{D}_m$	$\bar{D}_s$	B	Interpretation
PH [14]	0.72	0.26	1.02	strong stereo bias
PC [9]	0.91	0.16	1.73	very strong stereo bias
ASF [5]	0.31	0.87	-1.04	strong mono bias
SPF [1]	0.50	0.64	-0.25	slight mono bias

Table 2: *Bias scores for all panning feature algorithms*

From Fig. 1 it is simple to test hypothesis 1 (the anchor tracks should be the furthest apart) by finding the cell with a value of 1. It is also possible to assess hypothesis 2 (the other tracks should be well separated) by finding cells with low values. Each panning feature algorithm will be assessed separately.

4.1. Panning Histogram [24]

Fig. 1 shows that this is the only algorithm for which hypothesis 1 is not satisfied. Instead of the two anchor tracks (*RM** and *ISHST***) having the largest distance, *RM** and *CAC* are in fact the two most

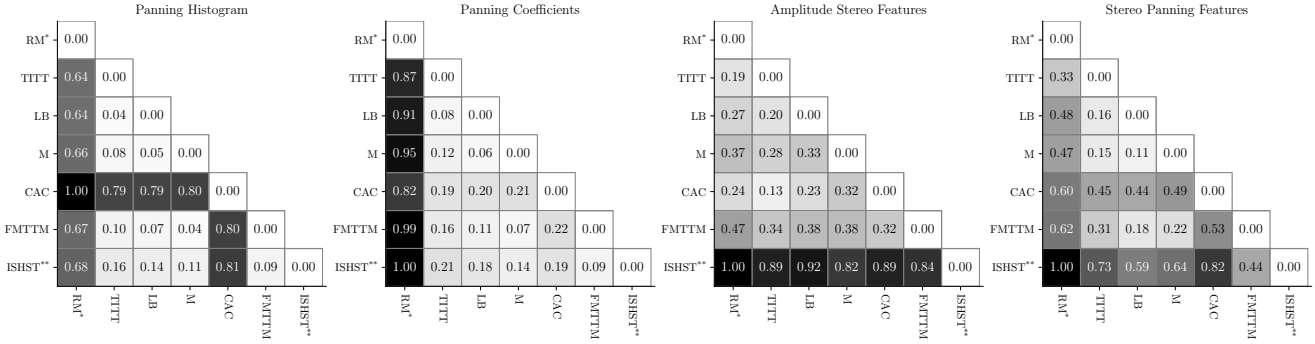

Figure 1: *Distance between pairs of tracks in the feature spaces of the four panning feature algorithms.*

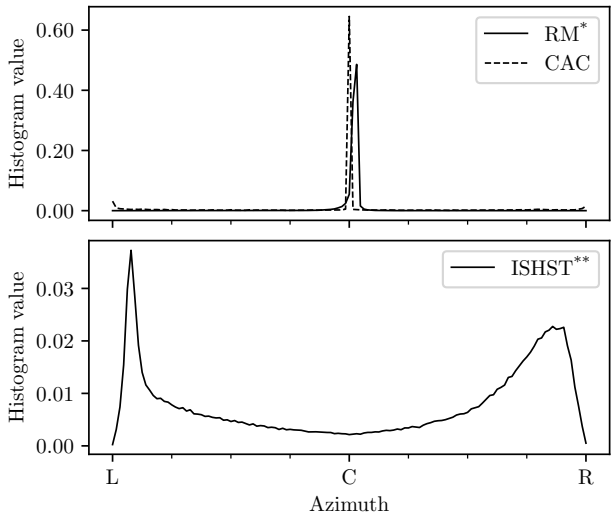

Figure 2: *PHs of RM*, CAC and ISHST**, demonstrating that CAC has been wrongly detected as monophonic.*

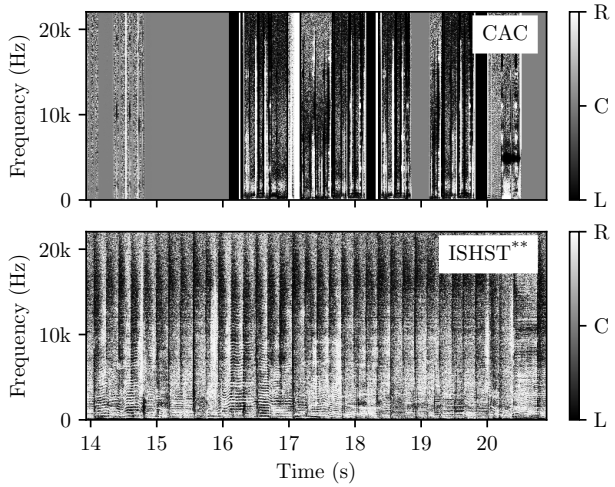

Figure 3: *Stereo Panning Spectrum of a short section of CAC and ISHST**.

separated tracks. To determine why this was the case, the PHs of all three tracks are plotted in Fig. 2.

From this diagram we can observe two important phenomena. Firstly, *CAC* (which is characterized by extreme and unnatural panning) actually appears close to mono in its PH. By inspecting the panning spectrogram (a section of which is shown in Fig. 3) we see that when hard panning occurs, time-frequency frames panned hard left and right are roughly evenly distributed. Hence, when the mean over time is taken, the hard-panned frames cancel each other out and all energy appears to be located in the centre. This differs from e.g. *ISHST*** which consistently has energy panned to both channels simultaneously and hence has two histogram peaks on the left and right.

Secondly, even though *RM** and *CAC* have similarly-shaped PHs, their Euclidean distance is very large. This is because the central peaks of each track appear in different azimuth bins. Since each bin is a separate dimension in the feature space, these two tracks effectively appear as two vectors with large amplitudes pointing in different directions. From this we can determine that the Euclidean distance is not a good measure of similarity for Panning Histograms, and we will therefore not assess hypothesis 2.

4.2. Panning Coefficients [9]

Inspection of Fig. 1 reveals that, as hypothesized, *RM** and *ISHST*** are the two tracks with the most separation. However, as Table 2 shows, it also has a strong bias towards the stereo anchor. The tracks *TITT*, *LB*, *M*, and *FMTTM* all appear in close pairs, implying that PC might have trouble differentiating between tracks that use moderate, but different, amplitude panning techniques. Reassuringly though, tracks that include extreme panning are separated from moderately-panned tracks, with $D \approx 0.2$.

The PC algorithm is based on a Panning Histogram that is quantized by truncation in the cepstral domain. To assess the effect of this, we can reverse the cepstral process using Eq. (2) where $\boldsymbol{c}$ is the vector of cepstral coefficients that has been truncated to length L. First, a pseudo-spectrum $\boldsymbol{s}$ is created by taking the exponential of the real part of the Discrete Fourier Transform (DFT) of $\boldsymbol{c}$. $\boldsymbol{s}$ is then truncated to half its length to recreate a histogram, and normalized so that the total area is 1. The result is a Panning Histogram $\boldsymbol{h}$ with $L/2$ azimuth bins.

$$\begin{aligned} \boldsymbol{c} &= (c_1, c_2, \ldots, c_L) \\ \boldsymbol{s} &= \exp(\mathrm{Re}(\mathrm{DFT}(\boldsymbol{c}))) \\ \boldsymbol{h} &= \frac{\boldsymbol{s}_{1:L/2}}{\frac{L}{2}\sum \boldsymbol{s}_{1:L/2}} \end{aligned} \tag{2}$$

Fig. 4 shows the quantized and original PHs for *ISHST*** and its three closest neighbours. From these diagrams it is clear to see why the PC algorithm is biased towards the stereo anchor - the anchor itself has had its panning energy moved from the side lobes towards the centre, and hence the detected panning features are more akin to the moderately-panned tracks (which are not so affected by the quantization).

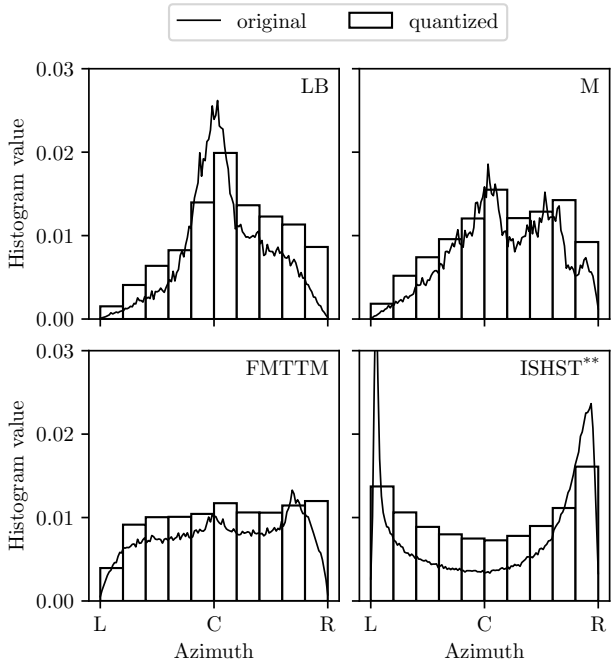

Figure 4: *PHs of ISHST** and its three closest neighbours before and after cepstral quantization.*

4.3. Amplitude Stereo Features [5]

Similarly to PC, the ASF distance matrix (Fig. 1) shows that the two anchor tracks are indeed the furthest separated. Conversely though, Table 2 shows that ASF has a bias towards the mono anchor. The separation of all tracks except *ISHST** is, however, quite adequate, with a range of $D \in [0.13, 0.47]$.

Curiously, *FMTTM* and *ISHST** have a high separation ($D = 0.84$) despite both exhibiting strong natural panning. Fig. 5 shows a visual representation of ASF for these two tracks. The panning features 'mean over time' and 'standard deviation over time' (see [5]) for each cochlea band are depicted as dots with error bars. From this diagram it is clear the two tracks have roughly opposite panning along the frequency axis. For example, *FMTTM* has a double bass panned to the right, while *ISHST** has a bass guitar panned to the left. This difference in panning direction is certainly

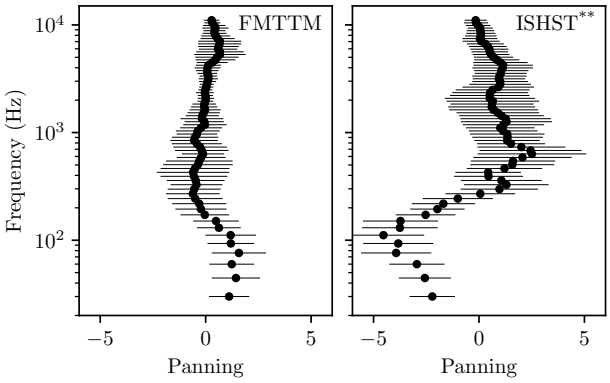

Figure 5: *ASF for FMTTM and ISHST** showing similar frequency bands panned in opposite directions.*

a contributor to the high separation between these two tracks, and hence the overall mono bias of ASF.

It is also notable that the panning variation for both tracks shown in Fig. 5 is quite high, when we can hear in the original audio that instruments are panned statically. This is most likely caused by overlapping frequency components in the left and right spectra – a common issue discussed in previous literature [8, 6, 25]. Certainly the phenomenon is not unique to the ASF algorithm but nevertheless is illustrated particularly well by this diagram.

While not too close by our standards, *CAC* and *TITT* ($D = 0.13$) are the closest pair. This is surprising since one would imagine that the moderate, natural panning of *TITT* would be well separated from the extreme, unnatural panning of *CAC*. Examining their ASF plots (see Fig. 6) reveals that while the panning variation of each track is quite different, they both exhibit mean panning that is roughly centred, mirroring the PH error described in Section 4.1. One possible explanation is that the fast panning of *CAC* manifests more as phase difference than amplitude difference, which would match the dispersion introduced by the reverberation in *TITT*. While panning variation for *CAC* is much higher in the lower frequencies, it could be the case that the high number of dimensions in ASF is acting to quash those differences.

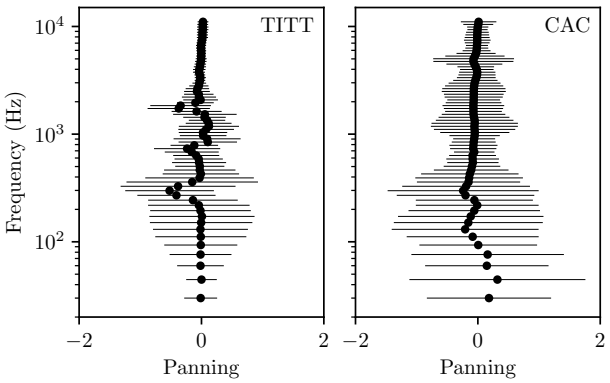

Figure 6: *ASF for TITT and CAC, both exhibiting average central panning.*

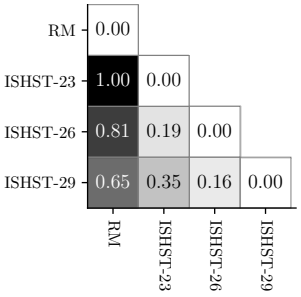

Figure 7: *Distance matrix comparing RM* (at -23 LUFS) to ISHST** at -23, -26, and -29 LUFS.*

One important factor in the use of ASF is that its calculation (defined in [5]) does not include any step to account for the overall loudness of the audio. To demonstrate the effect of this, we can calculate another distance matrix comparing *RM** to several versions of *ISHST*** that are normalized to different loudness levels. Fig. 7 visualizes this and clearly shows that the relative loudness of two tracks affects their distance in the feature space. All tracks analysed in this paper were normalized to equal loudness to account for this, but that might not be possible in a real-world use case, e.g. analysing streaming audio in real time.

4.4. Stereo Panning Features [1]

The distance matrix (Fig. 1) for SPF reveals that the two anchor tracks are indeed the furthest separated, and Table 2 shows the lowest bias. There are also no track pairs with a low separation. However, the distance matrix does show a generally high separation for *CAC*. This implies that SPF is able to differentiate between natural and unnatural panning quite well. In particular, *CAC* is maximally separated from *ISHST*** even though both contain extreme panning.

The definition of SPF includes estimations of both 'short-term panning' (the running standard deviation over M frames) and 'long-term panning' (the running mean over M frames). These are calculated over four frequency bands (fullband, lows, mids, and highs), and the global means and standard deviations of these comprise the final feature vector. We can therefore visualize the SPF as a scatter plot with the vertical position of a point denoting the amount of panning in each frequency band and error bars denoting panning variation, with short- and long-term values separated.

Fig. 8 shows the SPF plots for *CAC* alongside the two anchor tracks. It reveals that indeed, the unnatural panning of *CAC* is visible as large error bars in all dimensions whereas *ISHST*** has much higher average panning and relatively low variation, thus demonstrating how SPF is able to differentiate between natural and unnatural panning.

*RM** has low, but non-negligible values in all dimensions. Inspection of the original formula [1] shows that a truly mono signal would produce an SPF vector of all 0s, revealing that *RM** is, in fact, not entirely mono. By listening to the difference between the left and right channels, we can hear that the original mono recording has had a decorrelation effect applied at the mastering stage. We can therefore conclude that SPF is sensitive to this kind of effect.

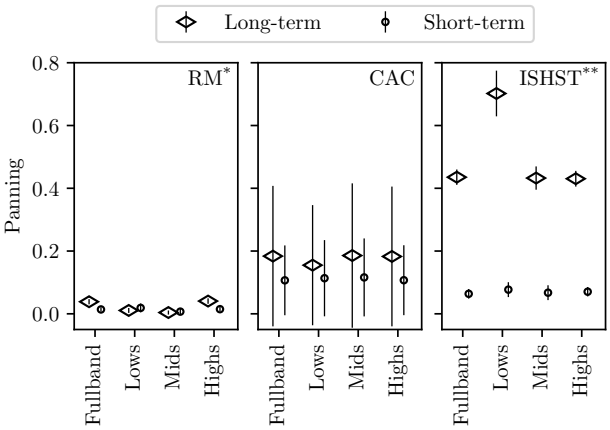

Figure 8: *SPF plots for RM*, CAC and ISHST** showing clear separation between all three.*

5. DISCUSSION

Table 3 shows a summary of each algorithm. For the purposes of unsupervised learning on commercial music, SPF has shown the most promise as it has the lowest bias score and good separation between all the panning feature tracks. The features related to panning variation over time appear to be important as they can help avoid issues such as the misidentification of unnatural panning as monophonic audio (see Section 4.1).

Each algorithm explicitly chooses whether to represent the panning *direction* or just the panning *amount*. ASF even includes features for both. Likewise, some algorithms act on frequency sub-bands whereas others process the entire spectrum. When the intention is to use panning features for unsupervised learning, the importance of both panning direction and frequency sub-bands is application-specific; any of the algorithms could be modified to include or exclude these two features.

The computational complexity of the algorithms was not addressed in this paper, but varied significantly among them. SPS-based descriptors ([1, 9]) are reliant on an STFT for their TF representation and hence are efficient but sometimes difficult to reconcile with human perception, whereas algorithms based on cochleagrams ([5]) produce more perceptually relevant results but require very large banks of time-domain filters.

An additional variable not covered by this paper is the duration of analysis. For the purposes of this experiment, 30s was chosen to roughly match [1], although [9] mentions that any duration between 2s and the entire recording is acceptable and [5] shows examples of entire songs.

All algorithms differ greatly in the number of dimensions represented in the final panning features. When calculating Euclidean distances in N-dimensional space, it has been proven that the inclusion of variables which provide no additional accuracy both increases the distance between points and raises the likelihood of attributing significance to spurious measurements [26]. A balance must therefore be struck between the resolution of the panning features and the ease of differentiating between tracks. Again, each algorithm can be modified to increase or decrease the dimensionality.

Algorithm	Frequency bands	Variation over time	Panning direction	Azimuthal resolution	Loudness invariant	Notes
PH [14]	Fullband	No	Yes	High	Yes	Cannot use Euclidean distance. Azimuthal resolution configurable.
PC [9]	Fullband	No	Yes	Low	Yes	Strong stereo details are lost. Azimuthal resolution configurable.
ASF [5]	ERB, 70 bands	Yes	Yes	High	No	Strong mono bias. Very large feature space.
SPF [1]	Lows, Mids, Highs	Yes	No	High	Yes	Relatively low bias. Separates natural/unnatural panning well. Sensitive to decorrelation effects.

Table 3: *Summary of properties for all panning feature algorithms.*

6. CONCLUSION

Four panning feature extraction algorithms were tested with respect to their use for unsupervised learning. They were run on five tracks that exhibit interesting panning features, plus two 'strongly mono' and 'strongly stereo' anchor tracks. The algorithms' results for individual tracks were analysed to assess what panning features each algorithm is able to differentiate between. Each algorithm was assigned a 'bias score' based on how the panning feature tracks were distributed within the anchor tracks. SPF from [1] showed the most promise, although it lacks differentiation between left and right and has relatively low frequency resolution.

In future work using panning features for unsupervised learning we recommend setting precise requirements for relevant panning features and choosing an algorithm accordingly based on the results given here. Alternatively, algorithms may be modified or combined to suit a specific need. The inclusion of other spatial descriptors such as those based on phase differences or generated via ML might also prove fruitful. To that end, a much larger survey of spatial analysis techniques would be beneficial.

In the cases where the original literature performed experiments on a commercial music collection [1, 5], it is not possible to fully reproduce their results since the exact tracks are unspecified. Analysis of modern, open datasets (e.g. [12, 13]) would therefore be advantageous and would strengthen the results obtained in the current experiments.

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Perceptual Evaluation of Binaurally Panned Speech and Music Mixtures

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The current practice for mixing music and speech is to decrease the volume of of the music in order to increase the intelligibility of the speech. However, it is sometime desirable to keep the music in focus, while mixing speech in another way. In VR it is common to separate sounds in space rather than changing the relative levels, but there is a lack of research on how to mix music and speech using similar techniques while retaining a pleasant listening experience. This paper describes a subjective test in which volunteering participants were asked to rate the pleasantness of music mixed with relevant speech. The speech signal was panned binaurally between 45° left and 45° right, in 15° increments, and mixed with various music tracks at equal loudness, with the hypothesis that this would improve the pleasantness ratings compared to centrally panned speech. The results indicate that the centre is not always the optimum panning position and that participants had strong individual preferences. Future work should concentrate on controlling and expanding the audio stimuli, and investigating the reasons for participants' preferences.

Keywords: Music, Speech, Music Mixtures, Binaural Panning, User Study, Subjective Test, Perceptual Evaluation

1 INTRODUCTION

When mixing a speech recording with a music track, the accepted norm in industry is to reduce the volume of the music to make the speech intelligible. This paradigm is called "ducking", and is so widespread that literature on the subject generally does not acknowledge that other mixing techniques are available [1, 2, 3, 4]. In the radio and TV industries in particular, there has been research into optimal ducking parameters such as speech-to-background level differences, and speech timing [1].

Another option is to intersperse speech and music so that they never mask each other [5]. Alternatively, in the realm of VR or AR, it is standard practice to adjust the physical location of sound sources to achieve perceptual separation [6, 7].

Previous literature generally forms the problem as one of speech intelligibility, with any music being classified as a "background" or "ambient" sound [1, 7, 8]. In this paper we posit that the inverse is also valid - music is the dominant and most desirable sound, whereas speech is an addition. The question then becomes how to mix the two in such a way that the listener's enjoyment of the music is minimally affected. Consider the use case of a radio Disc Jockey (DJ) providing commentary (which is typically done at the transition or mixing point between two tracks) or exercise instructions set over music. In both cases, music is the dominant auditory content and the speech consists of short and sparse interjections.

During the rise of music streaming services in recent years, personalized experiences have been identified as a strong factor in their adoption by consumers [9]. When personalization is mentioned, it generally refers to recommending tracks or playlists that a user might like. Since personalization of the music choice serves to improve a listener's experience, it logically follows that enjoyment of the music is paramount whereas any additional speech might be treated as secondary.

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More recently, other personalization options have emerged, such as personalized user interfaces [10]. In parallel, high-quality text-to-speech services have become much more readily available, to the point where they can be used to generate personalized informational audio about the music [5].

Although these new technologies are already established and being put into production, the usual practice of ducking the music when speech is present remains. There is currently little research on other methods to mix music and speech acceptably and pleasantly. For example, it is known that mixing music or acoustic noise with speech results in a drop in intelligibility and perceived quality *of the speech itself* [11], but further research is required to determine how to minimally affect the perceived quality of the *music*.

This paper describes a perceptual evaluation designed to investigate how the panning position of speech audio mixed with music influences the perceived pleasantness of the mixture as a whole. Speech audio was binaurally panned between 45° left and 45° right in 15° increments, and overlaid on various music tracks.

The remainder of the paper is organized as follows. Sec. 2 describes the relevant concepts for the study and presents related work. The experimental method is presented in Sec. 3, and the results are analysed and discussed in Sec. 4. Finally, the work is concluded, and future work is suggested in Sec. 5.

2 BACKGROUND AND RELATED WORK

Binaural panning is an advanced audio spatialization technique in which a monophonic signal is modelled as a point source at a specific angle from the listener's ears. It can be achieved by convolving the mono signal with the left and right Head-Related Impulse Response (HRIR) recorded from binaural microphones. Headphones are typically required to hear the correct effects of binaural panning. It differs from amplitude panning, in which the amplitude of a signal is adjusted in the left and right channels to make the audio source appear to come from a different direction. Amplitude panning is common in music production, whereas binaural panning is typically used more in VR, AR, and Extended Reality (XR).

Spatial Release from Masking (SRM) is a well-studied phenomenon in which humans can perceive more detail in two sounds that come from different directions than if they come from the same direction. The associated research problem is "The cocktail party effect", which refers to the problem of perceiving speech in the presence of other interfering noises [12]. SRM can be exploited to improve the clarity of recorded voices and to create a deeper sense of immersion in the final mix [13].

The clarity and intelligibility of speech when masked with another sound source have been major research topics for decades. One particular area of interest is speech in reverberant environments, with research dating back to the mid 20th century and continuing to this day [14, 15]. A second broad area of speech intelligibility research is related to the effects of interference noise. This can take the form of other speakers, environmental noise, artificial noise, and (more recently) music. For example, [1] investigates the ducking parameters necessary for clear speech in TV audio mixes, concluding firstly that music needs to be attenuated by 10 dB in order to make speech intelligible, and secondly that listeners have highly personal preferences. In the VR space, music has also been used as an interfering source with the express intention of measuring or improving the intelligibility of speech [16, 11]. Until recently the perceptual quality of *music* that has been distorted or manipulated in a VR or AR setting has been historically neglected [17, 18, 8]. This perceptual evaluation aims to investigate how music and speech can be mixed in a suitable way, with the previously mentioned use cases in mind.

3 METHOD

This experiment aims to explore the subjective evaluation of spatialized speech in music. Data were collected using rankings and questionnaires. The test procedure follows a MUSHRA-like structure [19].

3.1 Setup

The experiment was conducted in a controlled laboratory environment, free from external disturbances. During the experiment, participants were comfortably seated on a soft chair. A Mac laptop was used to display the experiment stimuli, which consisted of a graphical interface, as shown in Fig. 1. A pair of Sennheiser HD 650 headphones was used together with a Focusrite Scarlett Solo 3rd Gen, to allow each participant to modify the volume to a suitable setting, as permitted by [19].

3.2 Participants

A subjective test was conducted with 29 participants, comprising 22 males, six females, and one participant who preferred not to disclose their gender. Age data were available for 28 participants, ranging from 20 to 60 years, with a mean age of 36.6 years and a standard deviation of 10.0. Generally, participants who were over 18 years of age and had no hearing impairment were invited to participate in the study. Before participating, they received an information letter with detailed instructions on the experiment. In the laboratory, the participants were always accompanied by a researcher conducting the study. If necessary, further details were explained to them upon their arrival at the laboratory.

During the experiment, the participant would listen to audio on commercial headphones. No risk was foreseen to the participant, and they were informed in advance that the experiment could be terminated at any point without giving a justification. If they volunteered to participate, they signed a consent form that was kept separate from any data collected through rankings and questionnaires. There was no reward for participating in the experiment. After informed consent, demographic information and experience with streaming services were collected using a questionnaire with a randomized identification number.

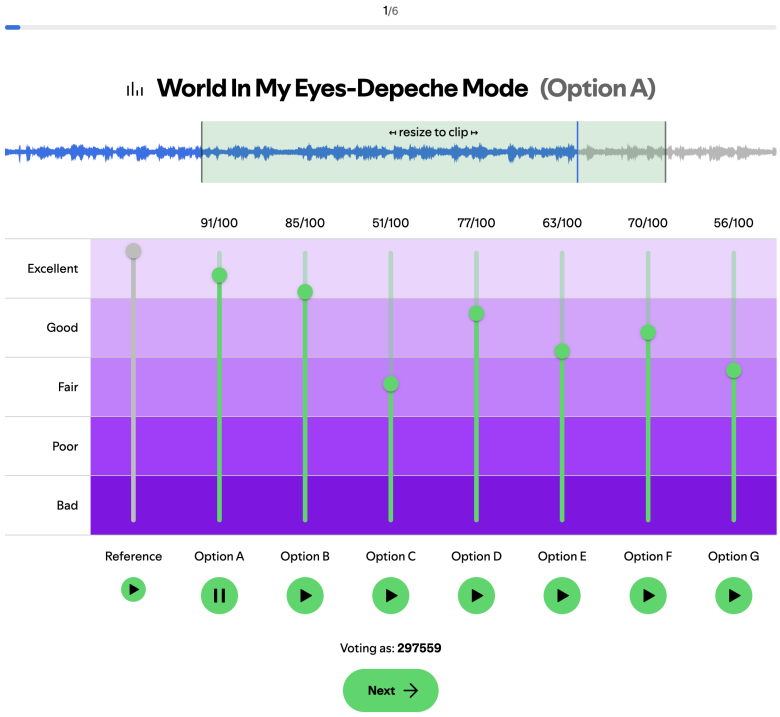

Fig. 1. The experiment interface used in the perceptual evaluation.

3.3 Audio Stimuli

All audio stimuli consisted of a commercially recorded music track combined with a speech recording. The speech recordings were generated using a commercial Text-to-Speech (TTS) service, with the text being extracted from the Wikipedia[1] page of the track's primary artist. To balance the gender of the speech voice, 50% of the tracks were assigned a male TTS voice and 50% a female TTS voice. The mono speech recording with no music was used as the reference signal for each track. Full details can be found in Appendix A.1.

Each variant of a track was created by applying binaural panning of the speech signal at angles {45L, 30L, 15L, 0, 15R, 30R, 45R} (where L and R denote left and right, and the number denotes the panning azimuth in degrees from the centre line), as shown in Fig. 2. Binaural panning was achieved by convolving the mono speech signal with the left and right HRIRs recorded by binaural microphones on the KEMAR dummy head subject of the SADIE II database [20]. This database of HRIRs contains a set in which the excitation sound came from the exact angles specified; no interpolation of the HRIR coefficients was performed. A block diagram for a single azimuth is shown in Fig. 3, in which solid lines represent audio signals, and the dashed line represents duration metadata. The "Upmix" block represents a simple mono-to-stereo process whereby a mono recording is copied into both the left and right channels. The "Trim" block represents a process where the music excerpt is truncated to match the length of the binaurally panned speech. The "Normalize loudness" blocks represent changes in audio gain to ensure that the resulting loudnesses of both speech and music were equal.

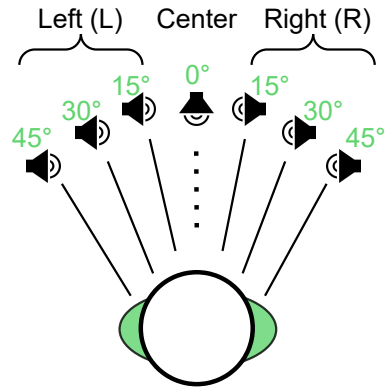

Fig. 2. The binaural panning of the speech signal at the different angles.

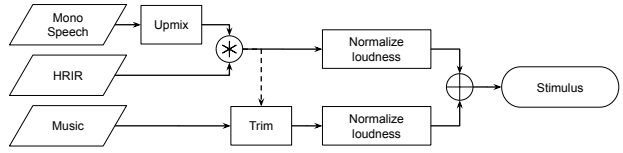

Fig. 3. Block diagram of the audio stimulus creation process for a single azimuth. Solid lines represent audio signals, the dashed line represents duration metadata, the $*$ operator represents channel-wise convolution, and the $+$ operator represents channel-wise addition.

[1] https://en.wikipedia.org/

Once the binaurally panned speech signal was created, an excerpt of the music track was selected to be (1) musically representative of the track as a whole and (2) precisely the same length as the speech signal. The method of finding a representative section is detailed in [21]. The speech and music signals were then normalized to the same loudness using the pyloudnorm library [22] (which is an implementation of the ITU-R BS.1770 standard [23]) and summed to produce the final audio stimuli. In total, six different music tracks were selected, each combined with a different speech recording. The speech recordings were generated using both male and female voices, with three recordings featuring male voices and three featuring female voices. Full details are given in Appendix A.1.

3.4 Procedure and Experiment Interface

In the experiment, participants could first adjust the volume to a comfortable level for themselves [19]. The application then presented the interface shown in Fig. 1 to familiarize them with the experimental procedures and the different equipment. This was done before the formal experiment, which involved listening to and rating various audio clips. Each session lasted up to 30 minutes.

Before the start of the experiment, participants were asked four questions: (1) How old are you?, (2) What is your gender? (Male/Female/Other/Prefer not to say), (3) How often do you use streaming services to listen to the following? Music streaming (A. Never use/B. Daily/C. Weekly/D. Monthly), How many hours do you listen to B/C/D?, Podcast (A. Never use/B. Daily/C. Weekly/D. Monthly), How many hours do you listen to B/C/D?, and (4) Do you consider yourself an expert listener? (A. Yes/B. No/C. I do not know).

Participants are presented with a reference signal along with the seven variants described in Sec. 3.3, and are asked to rate each based on how pleasant it is to listen to compared to the reference. The ranking scale follows the MUSHRA standard in that the opinion is a numeric value between 0 and 100, with the scale split into five equal sections marked by the words “Excellent”, “Good”, “Fair”, “Poor”, and “Bad”. Both the order of the variants and the order of the tracks are randomized for each subject, and the names of the variants are replaced with anonymized versions (see Fig. 1). The test procedure differs from MUSHRA in that the variants for each track do not contain a hidden reference or hidden anchor. This is because the subjects were not asked to evaluate a signal degradation but rather a new signal assumed to be more pleasant than the reference. The information collected by the test interface comprised the ratings given to each variant.

After the experiment, participants were asked to complete a questionnaire about their listening experience. In particular, they were asked: 1. Can you justify what influenced your ranking of spatialized voice in music? (with a free-text answer) 2. Do you prefer female or male voice in music? (A. Female voice, B. Male voice, C. No difference, or D. I do not know) 3. What did you think about the relationship between the volume of voice and music? (A. The voice was: too quiet, good, too loud, B. The music was: too quiet, good, too loud). 4. Do you have any further feedback on the experiment? (with a free-text answer).

The experiment interface, in Fig. 1, presented the participant with (for each track) a waveform display depicting the track, the track title, the artist name, a set of vertical sliders to give their rating of each variant, and a play/pause button for each variant in addition to the reference. When audio is playing, it is looped, and the waveform display can be used to set the loop points.

4 RESULTS AND ANALYSIS

The test participants were informed that a higher score indicated a greater pleasantness, but were given only basic instructions regarding the exact value of the score. Therefore, in the analysis, we assume that the scores are ordinal but with unequal intervals, and hence we only use statistical tests on the ranks of the scores. Ranks are calculated per-person-per-track.

4.1 Comparison of Centre to Non-Centre Preference

In Sec. 3.3, it was established that the reference signal was a monophonic speech recording. It is standard practice that, by default, a monophonic audio signal is placed exactly in the centre of a stereo rendering. The perceptual evaluation, therefore, assesses whether the test participants gave any preference to the non-centred variants of panned speech. To do this, a Boolean variable was assigned, denoting whether the top choice was centre or non-centre. If there was a tie, the centre won. This new dataset comprises a list of 174 preferences (29 participants $\times$ 6 tracks).

A ratio R representing the proportion of non-centre preference was calculated by S/N where S is the number of results indicating a non-centre preference and N is the total number of results. Hence, R is in the range $[0, 1]$ where 0 means that the voice overlay was always rated best in the central position, and 1 means that the voice overlay was always rated best in a non-centre position. These values are presented in Table 1, both for the entire dataset and separated by music track. Under the previously mentioned assumption that the default location is centrally panned, the given values of R can be interpreted as the proportion of instances in which the default option (centre-panned) was suboptimal. As can be seen from the table, across all tests (i.e., the group labelled “All”) the default position was suboptimal in 59% of cases.

Table 1. Ratios (R) of centre to non-centre preference for all test results and each individual track.

Group	S	N	R
Brown Eyed Girl	18	29	0.62
Everyday Is A Winding Road	18	29	0.62
Fake Plastic Trees	22	29	0.76
Fly Me To The Moon	16	29	0.55
World In My Eyes	14	29	0.48
bad guy	15	29	0.52
All	103	174	0.59

To further analyse these results, we used a series of χ^2 tests to assess whether any of the categorical data collected in the questionnaire explained the participants' preferences. Table 2 shows the results of these tests. They show that the only variable that is significantly correlated to non-centre preference is the assessor themselves. Essentially, the participants had preferences as to in which position the voice sounds most pleasant, but the demographic data collected in the survey did not explain those preferences. This matches the conclusion of Torcoli et. al regarding the importance of personalized loudness differences between speech and music [1]. Although that study found statistical links between loudness preferences and factors such as assessor age and listening experience, and the audio material itself, conversely, this study did not find such links for the voice panning position. This indicates that the reasons for the participants' preferences remain unknown.

Table 2. Results of χ^2 tests between independent variables and non-centre preference. The column "CV (DOF)" shows the critical values in the χ^2 distribution for 95% confidence with the given degrees of freedom. Significant variables are shown in bold.

Variable	CV (DOF)	χ^2	P-value
Participant	**41.34 (28)**	**75.33**	$\mathbf{3.24 \times 10^{-6}}$
Gender	5.99 (2)	2.82	0.24
Expert listener	5.99 (2)	2.27	0.32
Track	11.07 (5)	5.83	0.32
Voiceover gender	3.84 (1)	0.86	0.35
Age range	5.99 (2)	1.48	0.48
Podcast usage	7.82 (3)	2.49	0.48
Music streaming usage	5.99 (2)	0.31	0.86

4.2 Comparison of Ranked Azimuths

In Sec. 4.1 it was established that the central panning position is sometimes suboptimal. In this section, the scores given to individual panning positions are assessed to establish whether any particular position is more desirable. To that end, Fig. 4 shows box plots of the ranks given to each position.

While the confidence intervals of some adjacent panning positions do overlap, there is a clear general trend that the more extreme panning received worse ranks. However, all positions received the full range of the ranks, indicating that there was clear disagreement between the test participants. This aligns with the observation from Sec. 4.1 that the participants' choices were highly personal.

To test for statistically significant differences between azimuths, a Kruskal-Wallis-H test is first conducted with the null hypothesis that all azimuths are ranked equally. With a confidence value of 95% and seven azimuth classes, a critical value of 12.59 is found in the χ^2 distribution with six degrees of freedom. The result of the test is an H-statistic of 18.0 with a p-value of 0.00636. Therefore, the null hypothesis is rejected and it can be concluded that at least one of the azimuths is ranked differently from the others. To determine which one, a Dunn's post-hoc test with Bonferroni correction is performed to compare pairwise p-values. The results

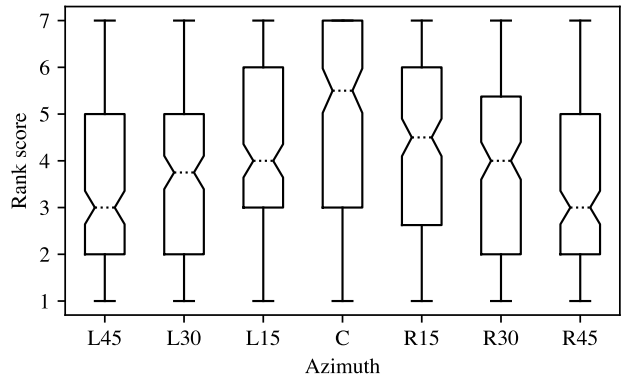

Fig. 4. Box plots of panning position ranks across all tracks. Dotted lines show the median and notches show the 95% confidence interval calculated with Gaussian-based asymptotic approximation [24].

of all pairs that included "Centre" are shown in Table 3. All other pairs had p-values of 1 and are omitted from the table. There are two azimuths with a significant difference from "Centre" - "Left 45" and "Right 45". Looking at the median scores of these two classes, it can be concluded that the extreme-panned speech was indeed less highly rated than the more modest panned or centred speech. Interestingly, even though there are no other statistically significant differences, the *confidence* of a difference from "Centre" is lower on the right side. We speculate that this might be due to the "right-ear advantage" – a phenomenon in which a person is more easily able to interpret a voice presented to the right ear than the left [25]. More data would need to be collected to test this speculation.

Similar Kruskal-Wallis-H tests were performed with data for single tracks, but the null hypothesis could not be rejected for any of them.

Table 3. Medians and p-values for pairwise differences between the centre-panned voice and all other azimuths. Significant differences are shown in bold. Centre median value is shown for comparison.

	L45	L30	L15	C	R15	R30	**R45**
Median	**53.0**	54.5	55.0	63.0	59.0	57.0	**55.0**
P-value	**0.006**	0.064	0.341	-	1.000	0.267	**0.015**

4.3 Analysis of Free-Text Responses

To reiterate, the questions with free-text responses given to test participants were "Can you justify what influenced your ranking of spatialized voice in music?" and "Do you have any further feedback on the experiment?". The participants answered both questions with similar information and therefore the responses were combined. Subsequently, they were thematically analysed and classified by the authors. The themes mentioned are shown in Table 4 alongside the number and percentage of participants who mentioned them.

The most prominent overarching theme was that difficulty in hearing the panned speech or its lack of clarity was

the primary factor in the participants' ratings of pleasantness. Some participants provided more detailed feedback, specifying that the music was too loud or that the type of music affected their scores. Eight participants stated that they felt the voice was always clearer on either the left or the right, and seven stated that they always wanted the voice to be in the centre. Smaller sets of participants mentioned that clashes between any singing voice in the music and the speech led to lower ratings. They also noted that the pitch of the voice affected their ratings, either positively or negatively, depending on the music. A similarly small group mentioned that the spatial factors affected their scores. Some comments also specifically mentioned that speech with more extreme panning was less pleasant, which is also reflected in the data (see Sec. 4.2).

Table 4. Counts of test participants who mentioned that specific factors affected their ratings.

Theme	Count	Percentage
Clarity of voice / listening effort	17	59%
Music too loud	10	34%
Left/right preference	8	28 %
Centre preference	7	24%
Clash between singing and speech	5	17%
Speech pitch	5	17%
Speech location / spatial factors	4	14%
Music type / genre	4	14%

Based on these findings, it is suggested that further experiments stratify the music tracks into those with and without strong vocals, include a more varied range of music genres, and apply less extreme panning to the voiceover.

4.4 Analysis of Loudness Perception

Recall that participants were asked if the music and voice were too loud, too quiet, or good. A tabulation of their responses can be found in Table 5. The raw results are skewed towards the music being too loud or good, and the voice being good or too quiet. Bearing in mind that, in actuality, both were normalized to equal loudness, it can be speculated that participants considered the music to be an interfering sound source and the voice to be the desired sound source. This was contrary to the original assumptions of the authors and needs to be considered in future iterations of the experiment. It does, however, mirror the findings of the free-text responses - voice clarity was paramount.

Table 5. Test participants' opinions of the loudness of music and voice.

	Voice	Music
Too loud	2	18
Good	18	10
Too quiet	9	1

4.5 Discussion

A constant source of uncertainty during data analysis was the relatively small number of both tracks and participants. In particular, only six tracks were used to limit the duration of each test session. In reality, it was not enough to draw strong statistical conclusions about differences between tracks.

Furthermore, several of the statistical tests used are based on the assumption that all data are independent. It could be argued that the data collected are not strictly independent since each participant scored multiple tracks. This was a practical compromise made to gather enough data for any analysis. It is a known problem that subjective tests of this variety are slow and costly, and recent efforts in crowdsourcing MUSHRA tests have proven fruitful [26] (although this work focused on assessing solo speech signals rather than music).

5 CONCLUSIONS AND FUTURE WORK

A perceptual evaluation was performed to determine the pleasantness of music mixed with voice that has been binaurally panned to various angles by convolving the signal with an HRIR. Participants were presented with a MUSHRA-like test in which they moved sliders to give their opinions of the different panning azimuths. The raw scores were transformed into ranks on a per-person-per-track basis.

The non-centre positions were preferred over the centre in 59% of the test cases, and the ratings given by test participants were highly personal. However, the demographic data collected about the participants did not explain their choices.

Kruskal-Wallis tests revealed that the outermost azimuths were consistently ranked lower than the moderately panned and centre azimuths. There was also the implication that the voice panned to the right was ranked higher than that on the left, but as stated above, more data would be required for confirmation.

The free-text explanations given by the participants showed that "listening effort" or "voice clarity" were the factors that influenced the pleasantness of the audio for the highest number of people. They suggest that to increase overall pleasantness, a voiceover should be cleanly separated from any vocals in the music, although the best method of separation remains unclear. Alongside this, there were indications that other factors relating to listeners' personal preference should be investigated in future experiments.

Based on these findings, we suggest that further experiments be performed in which more tracks and more test participants are used. The musical excerpts chosen should be controlled so that they have either strong vocals or a lack thereof throughout, and the vocal overlays should be panned more towards the centre and in finer increments.

6 ACKNOWLEDGEMENTS

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A.1 DETAILS OF AUDIO STIMULI

The audio stimuli contain copyrighted material and therefore cannot be distributed. For reproducibility, the parameters to recreate them are provided in this appendix.

Track metadata

Title	Artist	Start Time	Duration	Voiceover Gender
Everyday Is A Winding Road	Sheryl Crow	136.19	21.04	male
Fly Me To The Moon	Frank Sinatra	64.00	20.06	male
Fake Plastic Trees	Radiohead	173.32	23.37	female
Brown Eyed Girl	Van Morrison	81.08	19.88	female
bad guy	Billie Eilish	114.67	16.77	female
World In My Eyes	Depeche Mode	169.49	13.84	male

Voiceover text

Sheryl Crow

Sheryl Suzanne Crow (born February 11, 1962) is an American singer, musician, songwriter, and actress. She is noted for her optimistic and idealistic subject matter, and incorporation of genres including rock, pop, country, folk, and blues. She has released twelve studio albums, five compilations, and three live albums, and contributed to several film soundtracks.

Frank Sinatra

Francis Albert Sinatra (born December 12, 1915) was an American singer and actor. Nicknamed the "Chairman of the Board" and "Ol' Blue Eyes", he is regarded as one of the most popular entertainers of the 20th century. Sinatra is among the world's best-selling music artists, with an estimated 150 million record sales globally.

Radiohead

Radiohead are an English rock band formed in Abingdon, Oxfordshire, in 1985. They comprise Thom Yorke; brothers Jonny Greenwood and Colin Greenwood; Ed O'Brien; and Philip Selway. They have worked with the producer Nigel Godrich and the cover artist Stanley Donwood since 1994. Radiohead's experimental approach is credited with advancing the sound of alternative rock.

Van Morrison

Sir George Ivan "Van" Morrison OBE (born 31 August 1945) is a Northern Irish singer-songwriter and musician whose recording career started in the 1960s. Morrison's albums have performed well in the UK and Ireland, with more than 40 reaching the UK top 40, as well as internationally, including in Germany, the Netherlands and Switzerland.

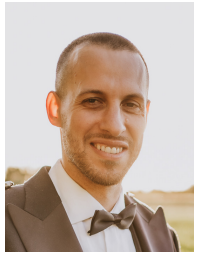
Billie Eilish

Billie Eilish Pirate Baird O'Connell (born December 18, 2001) is an American singer-songwriter. She first gained public attention in 2015 with her debut single "Ocean Eyes", written and produced by her brother Finneas O'Connell, with whom she collaborates on music and live shows.

Depeche Mode

Depeche Mode are an English electronic music band formed in Basildon, Essex in 1980. Originally formed with the lineup of Dave Gahan, Martin Gore, Andy Fletcher and Vince Clarke, the band currently consists of Gahan and Gore.

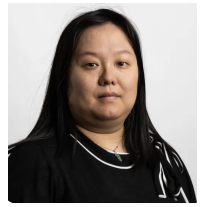
THE AUTHORS


Richard Mitic

Majed Elwardy

Veronica Sundstedt

Andreas Rossholm


Yan Hu

Richard Mitic is a doctoral candidate at Mälardalen University, currently researching spatial audio, digital signal processing, and large-scale music information retrieval. He holds a masters degree in Music and Electronics from the University of Glasgow where he specialised in digital electronics and audio processing. He has 14 years experience in industry, starting as a researcher of media streaming protocols and participating in MPEG standardization, later progressing into the music industry where he wrote software for music streaming, media transcoding, real-time audio, and embedded devices.

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Majed Elwardy is a Lecturer in Computer Science at Blekinge Institute of Technology (BTH), Sweden. He received his Ph.D. in Computer Science from BTH in 2025, where his research focused on 360-degree video quality assessment and user behavior in virtual reality. His research lies at the intersection of human–computer interaction and immersive technologies, with particular interests in immersive multimedia, quality of experience, signal processing, user experience evaluation, cybersickness, and pedagogy-driven approaches for XR education.

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Veronica Sundstedt is a Professor in Computer Science at BTH, bringing extensive international experience gained at institutions including Trinity College Dublin, the University of Bristol, and the University of Bath. She holds a Ph.D. in Computer Graphics from the University of Bristol. Her research expertise lies in the intersection of computer graphics and perception, focusing particularly on perceptually-based rendering, novel interaction techniques, and eye tracking. Her academic contributions include organizing conferences and serving on multiple ACM editorial boards. Currently, she develops and teaches specialized courses at BTH, and she is the project leader for a new research initiative focused on “Human-Centered Intelligent Realities (HINTS), funded externally from the Knowledge Foundation (20220068).”

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